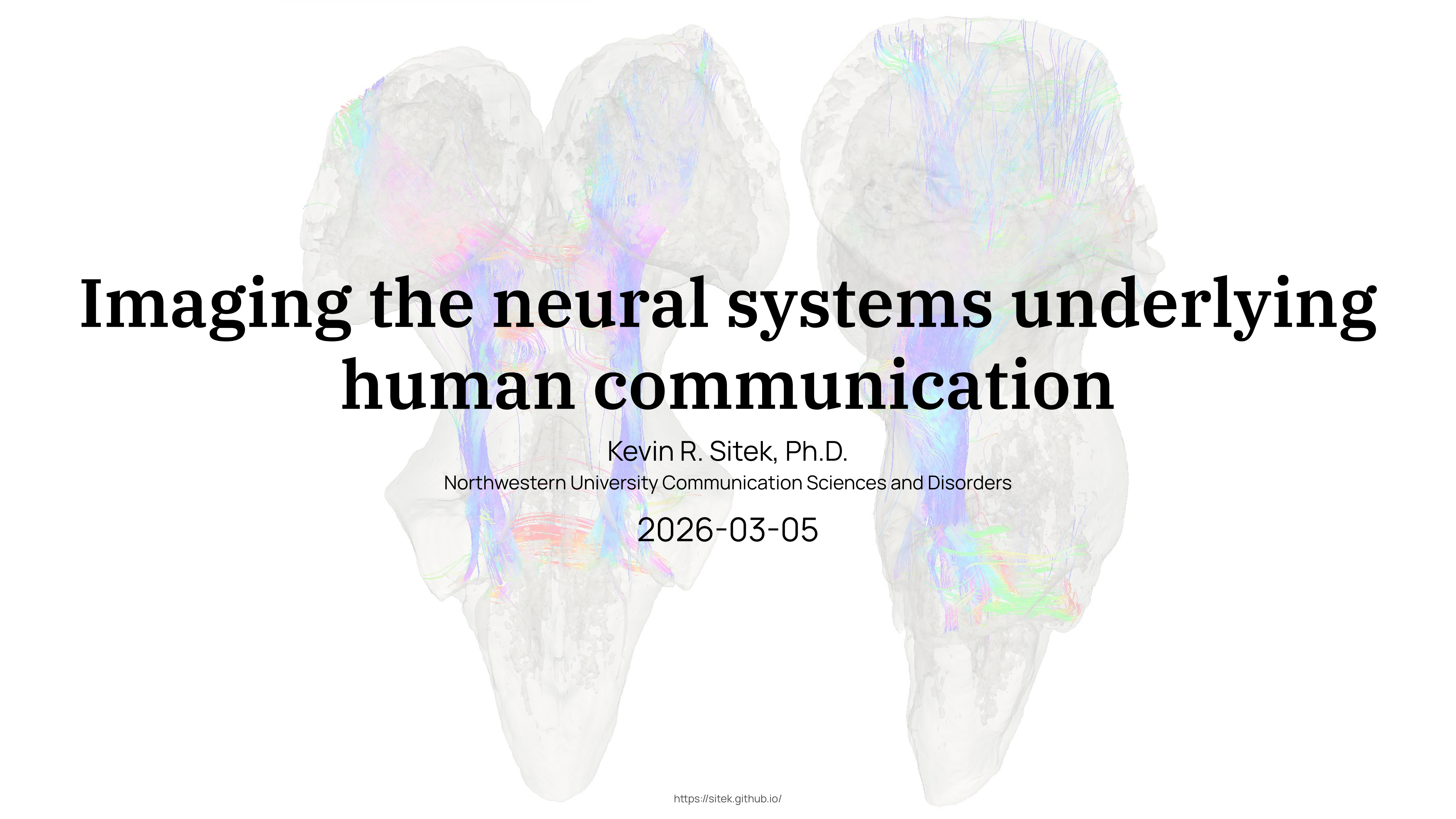




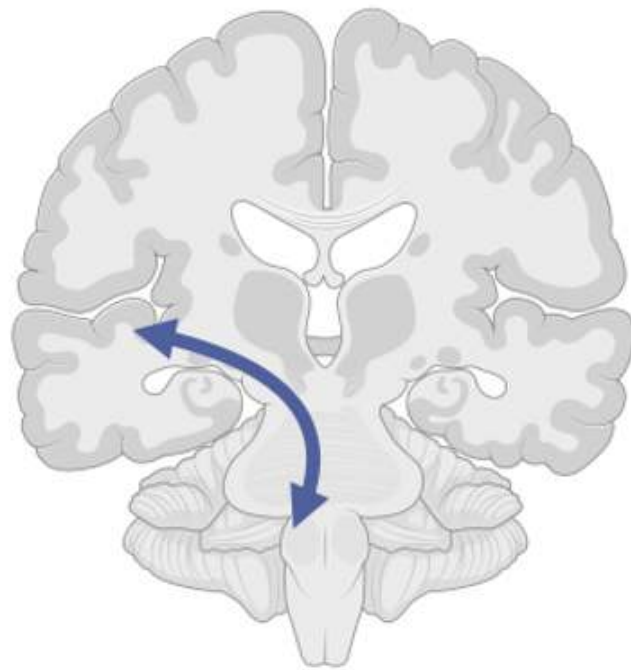
# Imaging the neural systems underlying human communication

Kevin R. Sitek, Ph.D.

Northwestern University Communication Sciences and Disorders

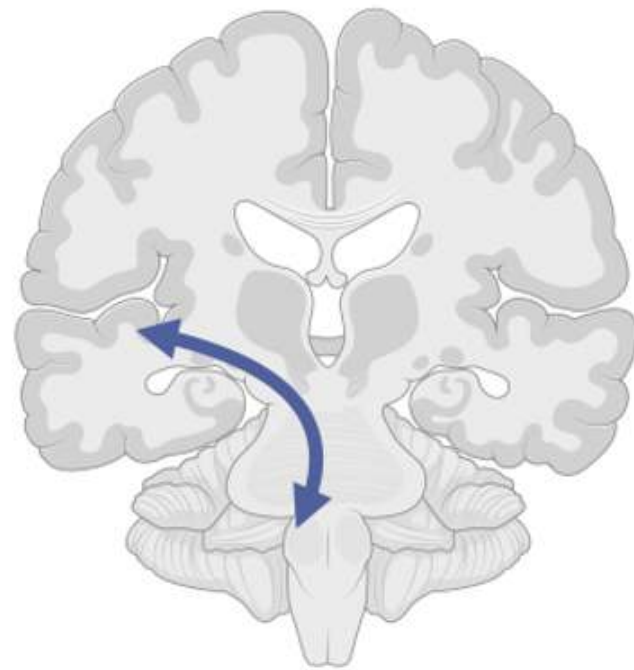
2026-03-05

# A systems-level approach to human auditory neuroscience

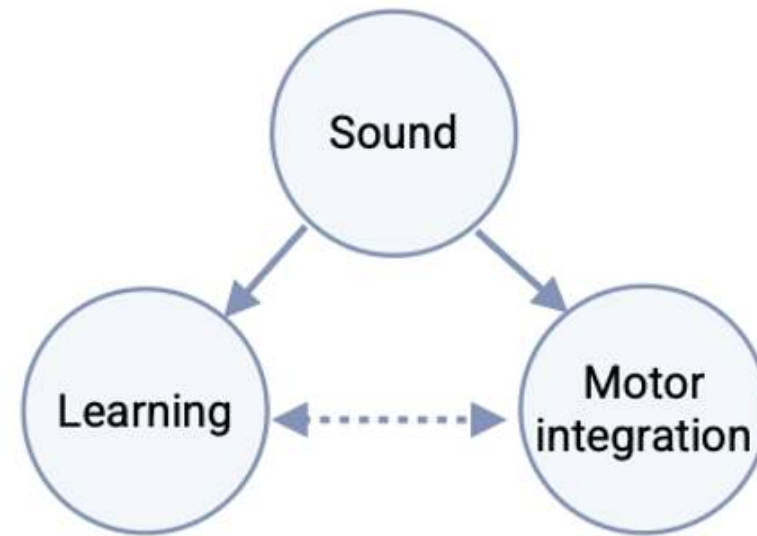


**Auditory  
circuits**

# A systems-level approach to human auditory neuroscience



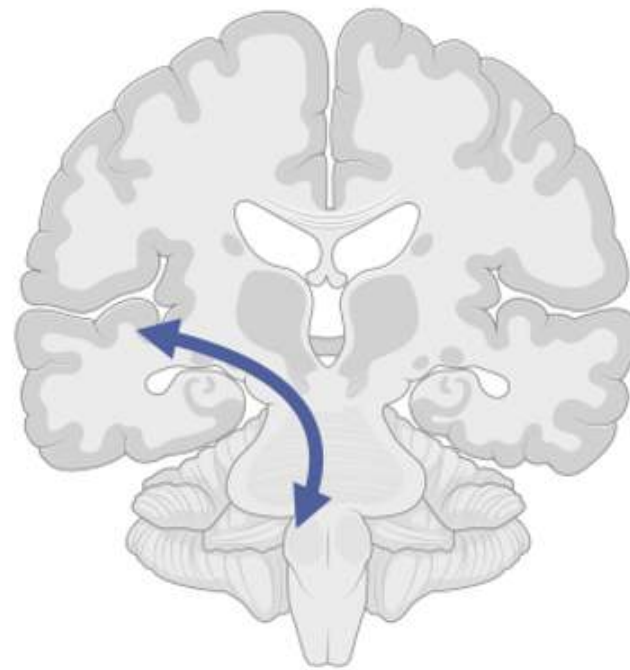
**Auditory  
circuits**



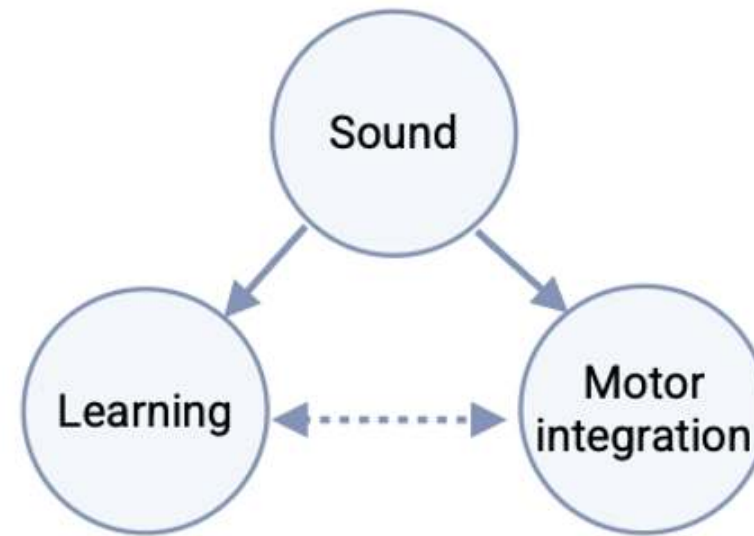
**Speech and  
learning**



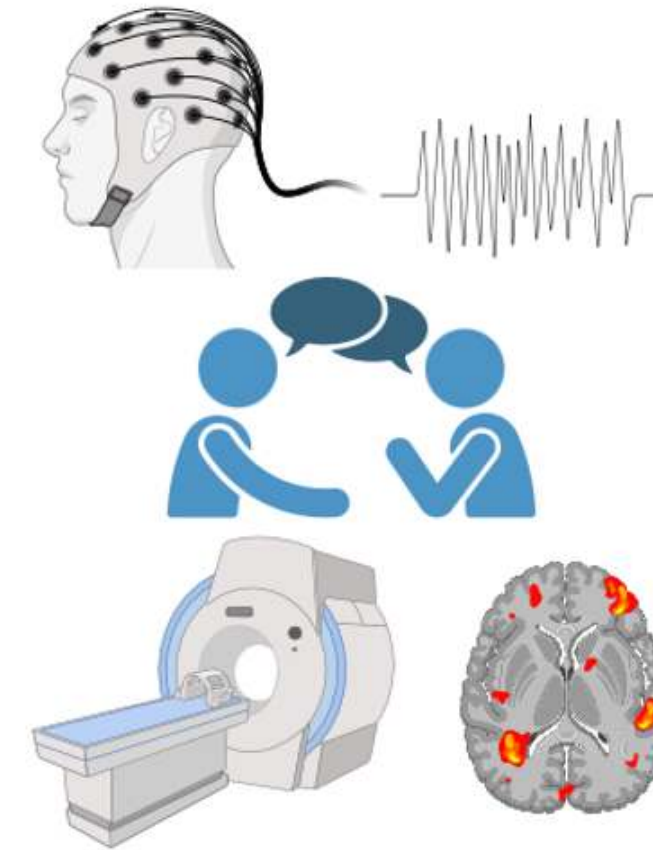
# A systems-level approach to human auditory neuroscience



**Auditory  
circuits**



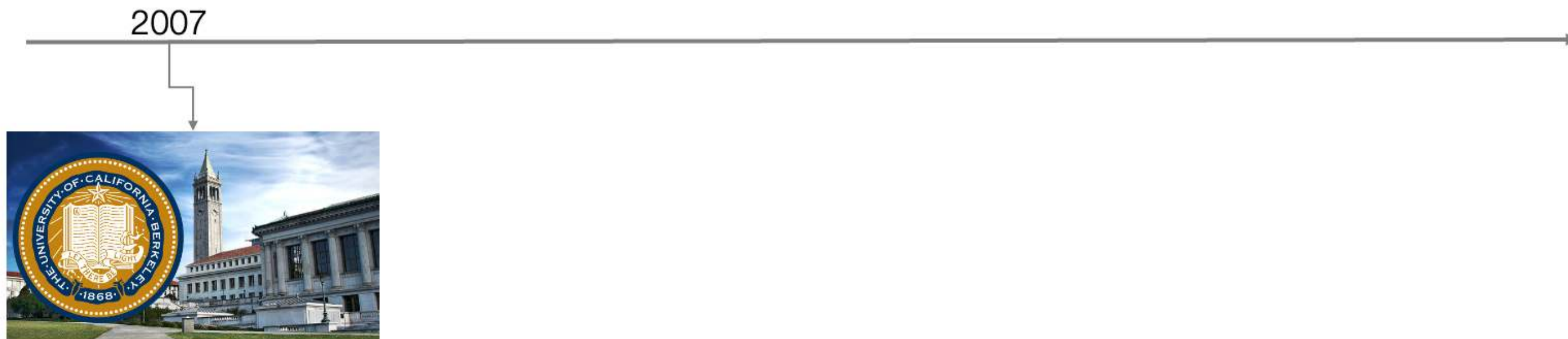
**Speech and  
learning**



**Communication  
biomarkers**

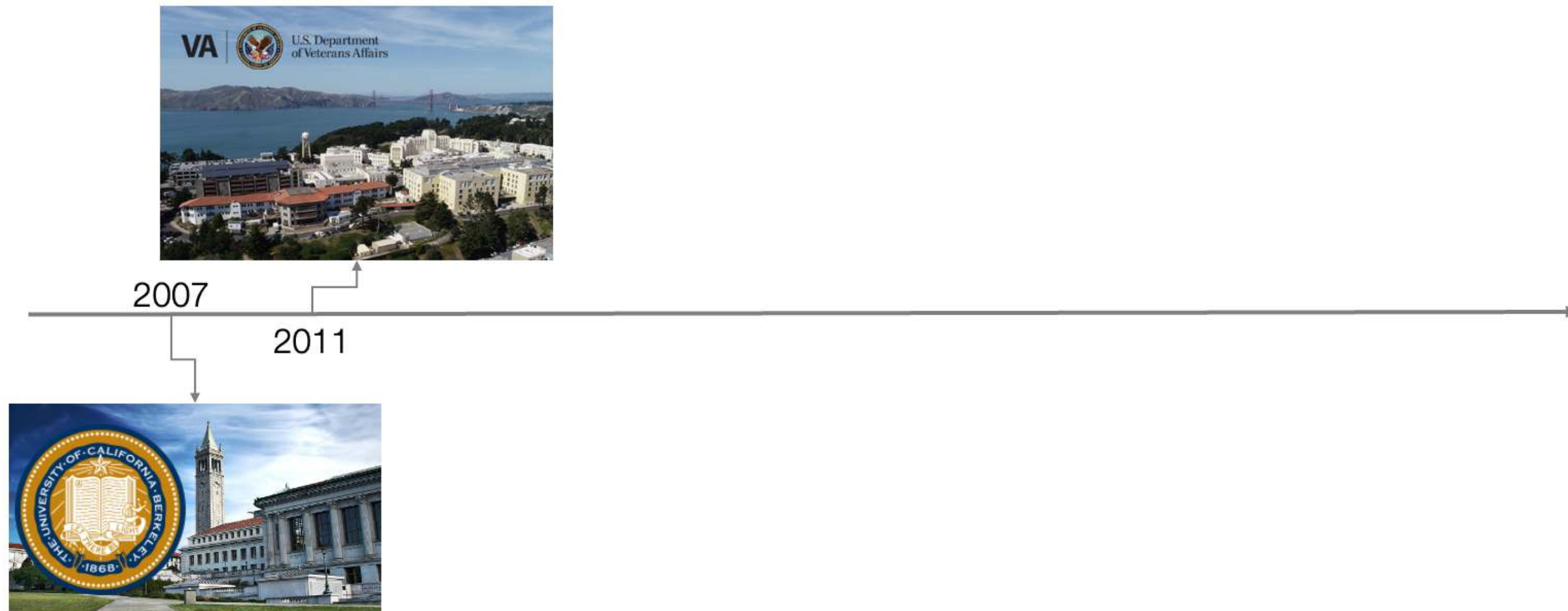
# **My path to hearing, speech, and the brain**

# First steps: Linguistics and Cognitive Science Undergraduate at UC Berkeley



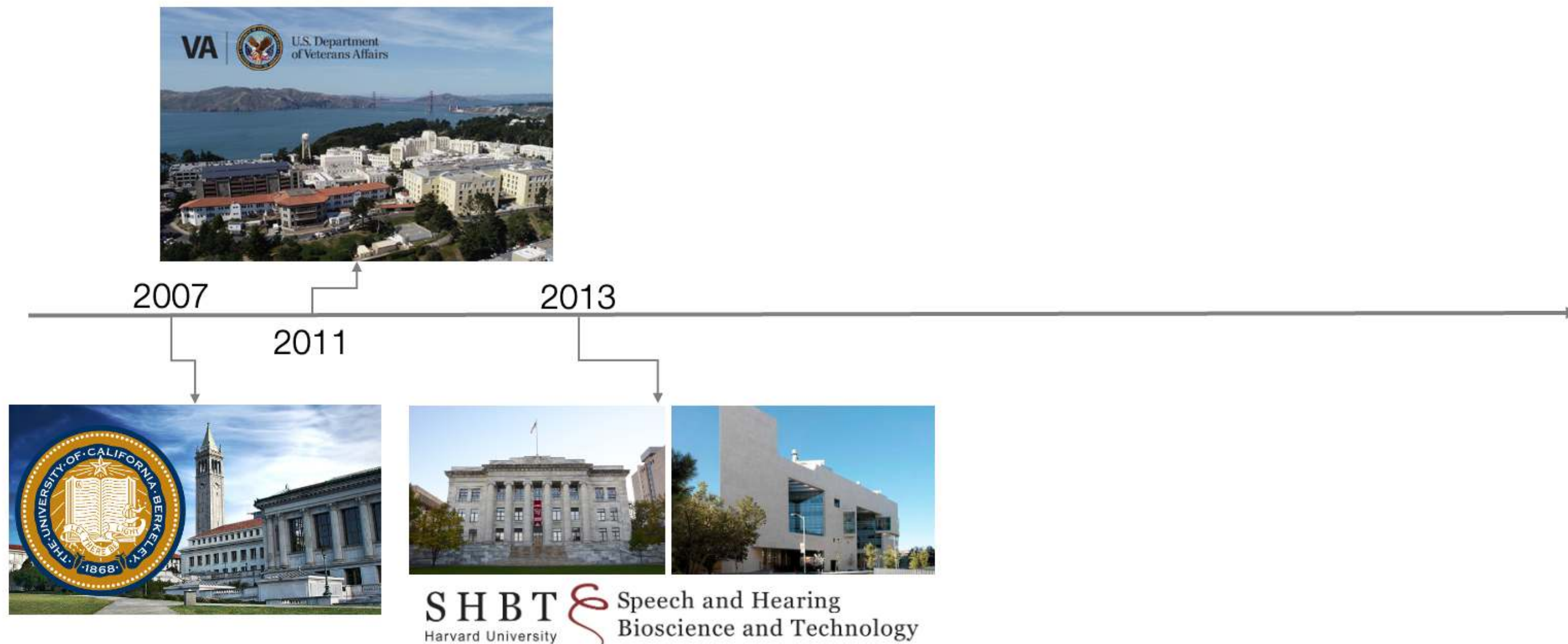
Sitek & Johnson (2012 Chicago Linguistic Society). “Ipsilateral and contralateral phonetic context effects”.

# Brain imaging research assistant at UCSF/San Francisco VA Medical Center



Sitek et al. (2013). “Auditory cortex processes variation in our own speech”.

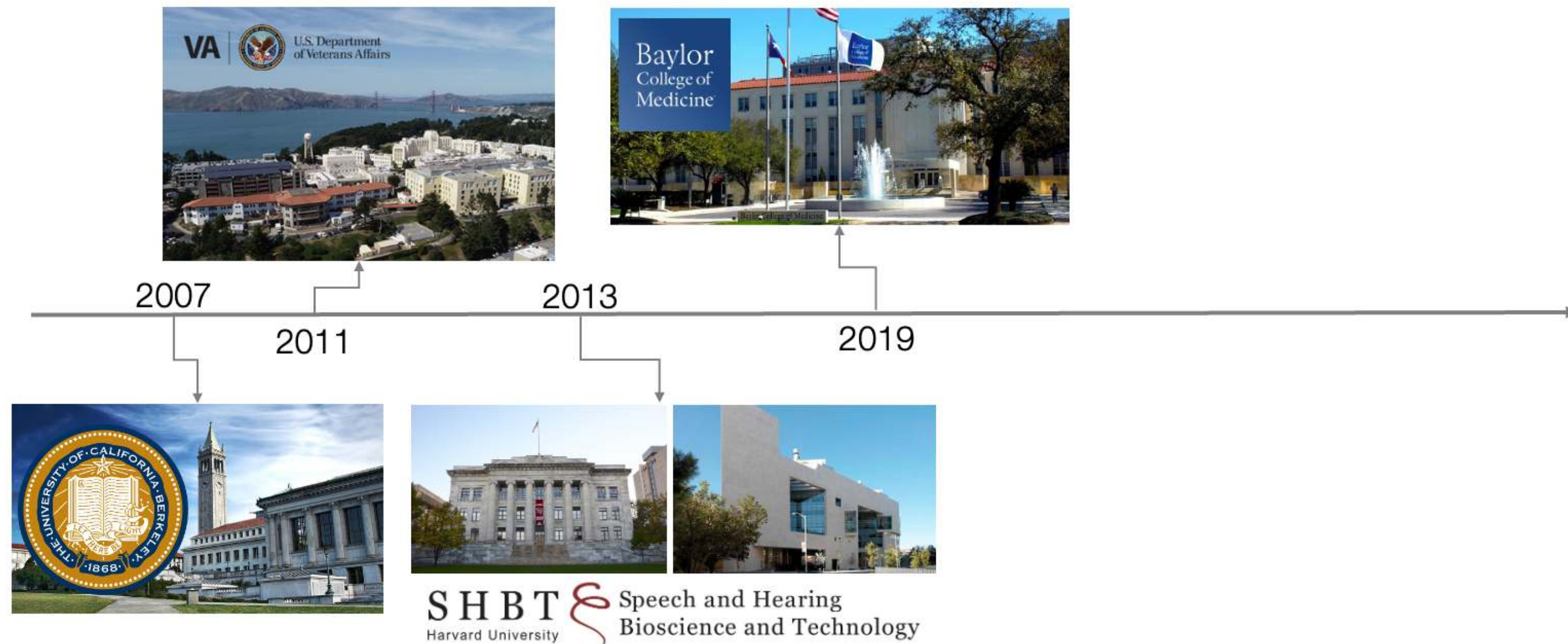
# PhD in Speech and Hearing Bioscience and Technology at Harvard/MIT



**NRSA F31** (2016–2018); Sitek et al. (2016), Connectivity and **stuttering** severity; Sitek and Gulban et al. (2019), “Mapping the human **subcortical auditory system**”.



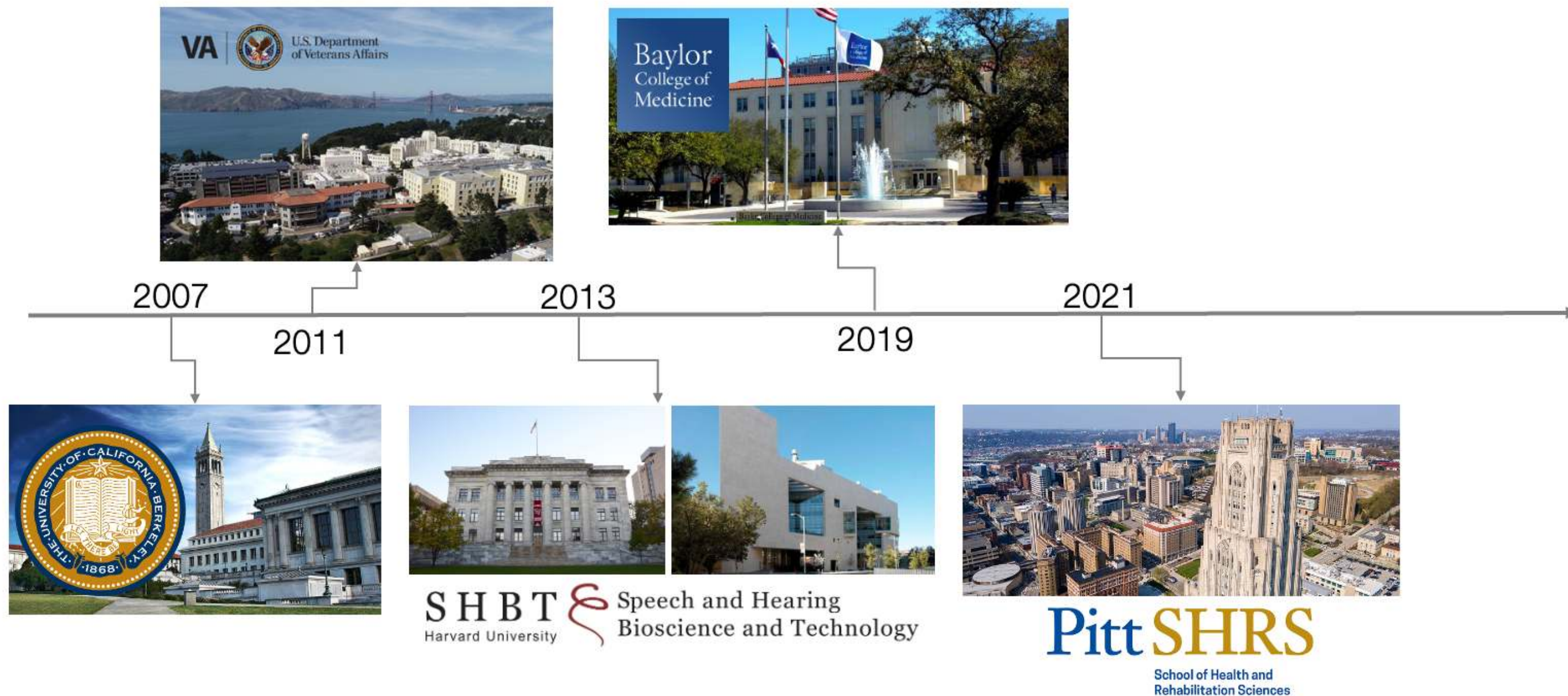
# Postdoctoral research in Neuroscience at Baylor College of Medicine



Ultra-high field 9.4 Tesla functional MRI of the human midbrain (collaborative CRCNS with Tübingen)



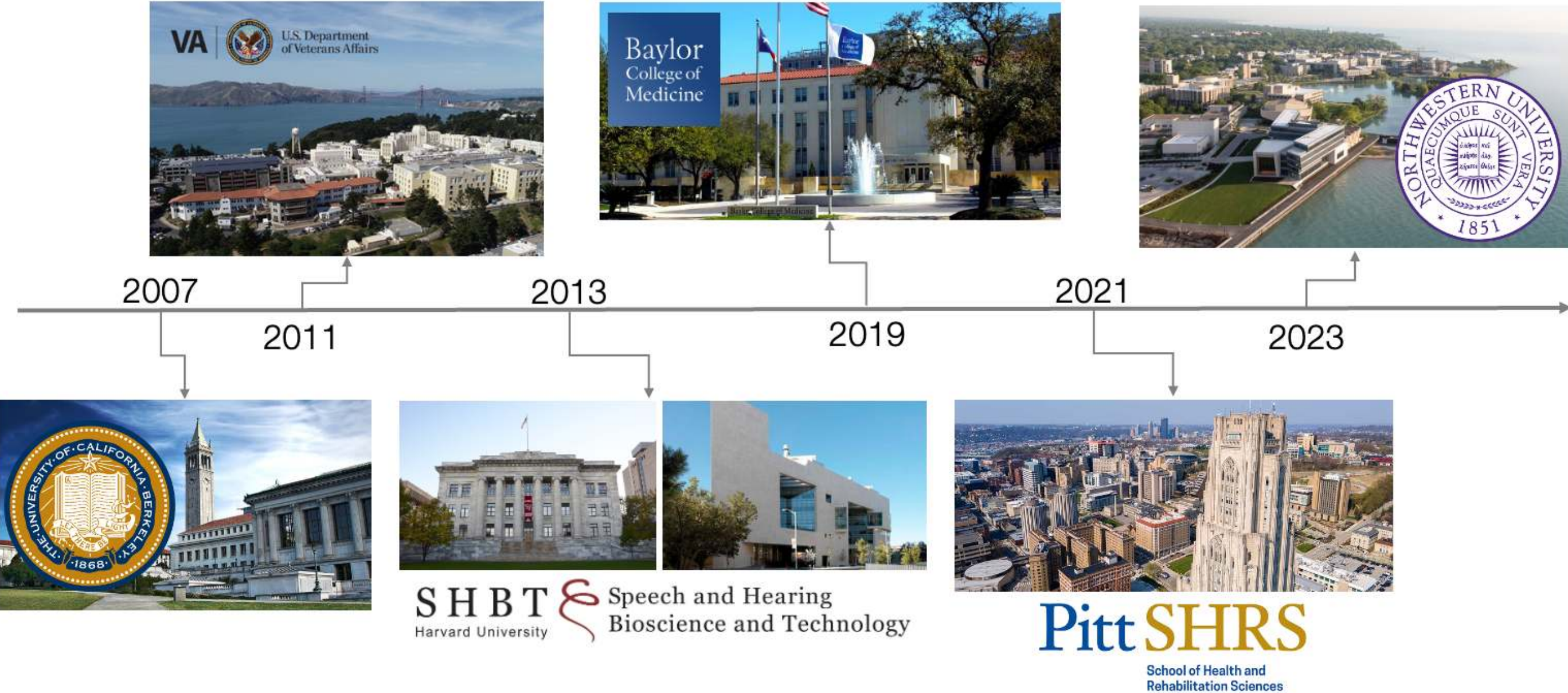
# Research Scientist in Communication Science and Disorders at the University of Pittsburgh



NIH K01 (2022–2027); Sitek et al. (2022), Structural connectivity of human inferior colliculus



# Now: Research Assistant Professor at Northwestern University

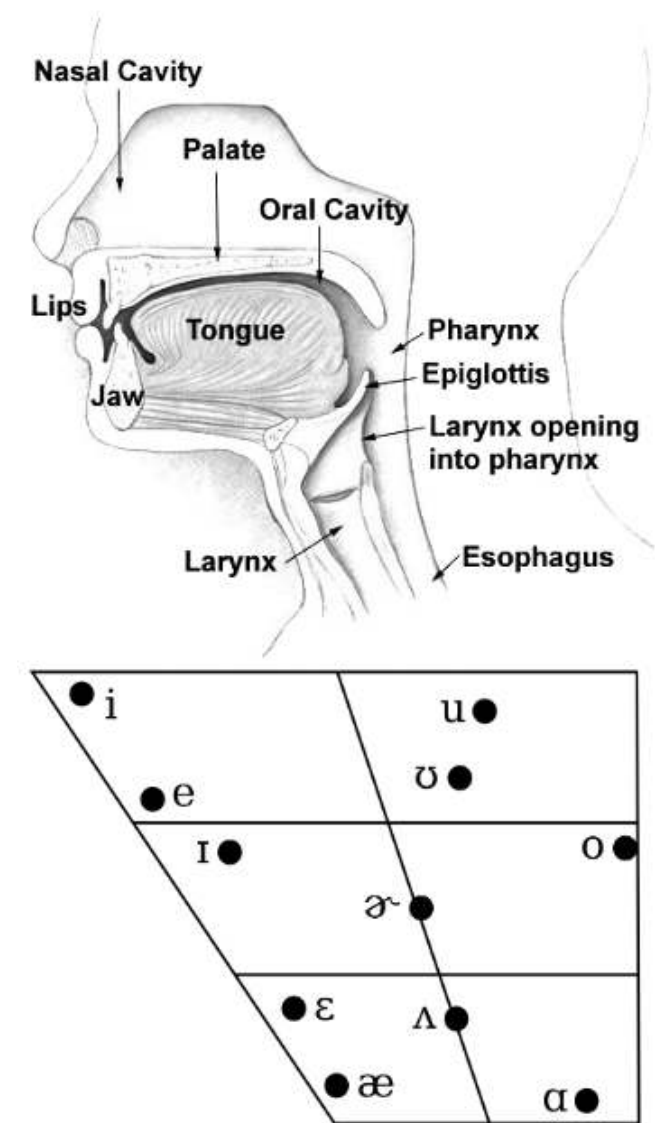


NIH R21 (2025–2028); auditory learning and auditory–motor interactions

# **Widening perspectives and expanding opportunities through teaching and mentorship**

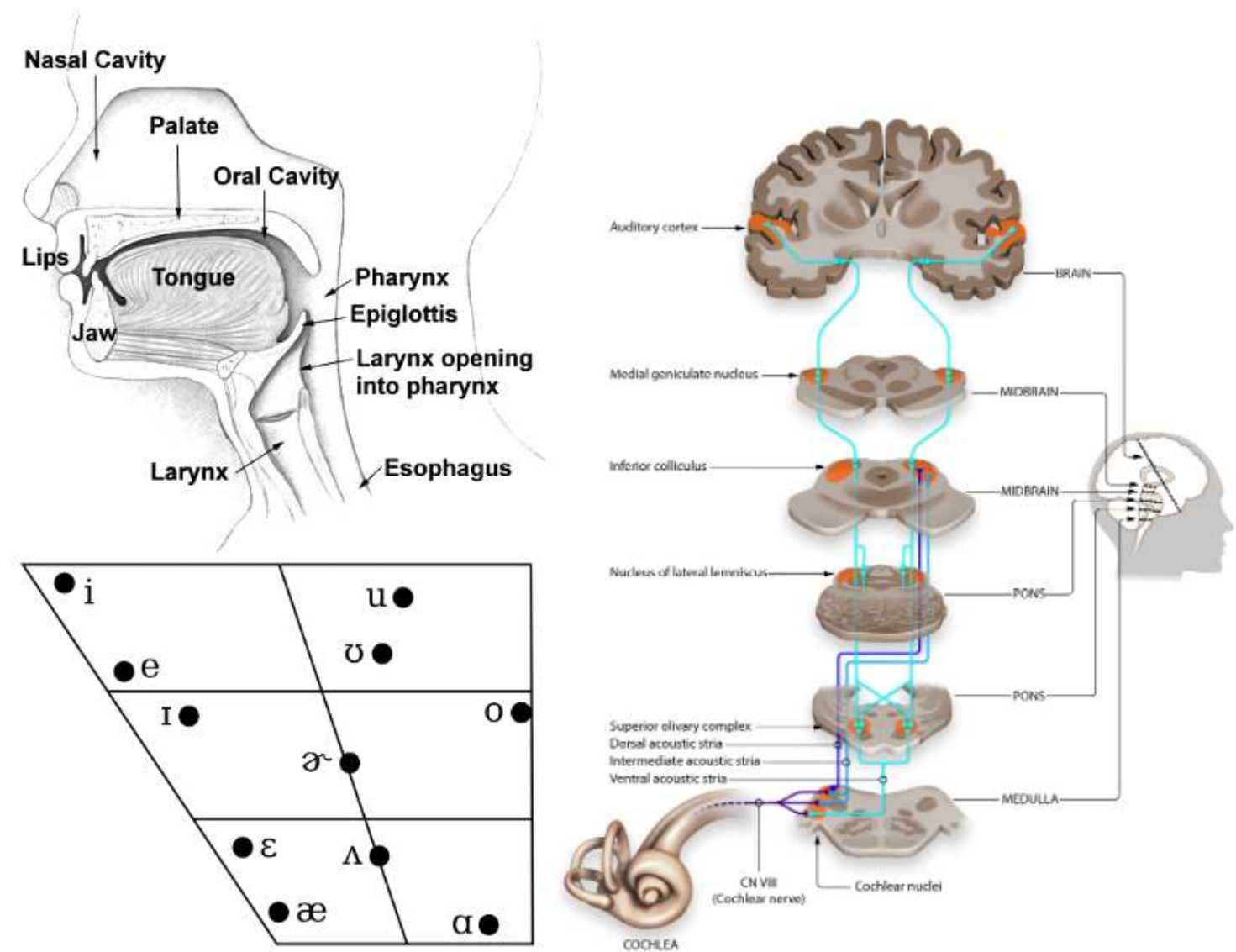


# Widening perspectives and expanding opportunities through teaching and mentorship



**Speech  
communication**

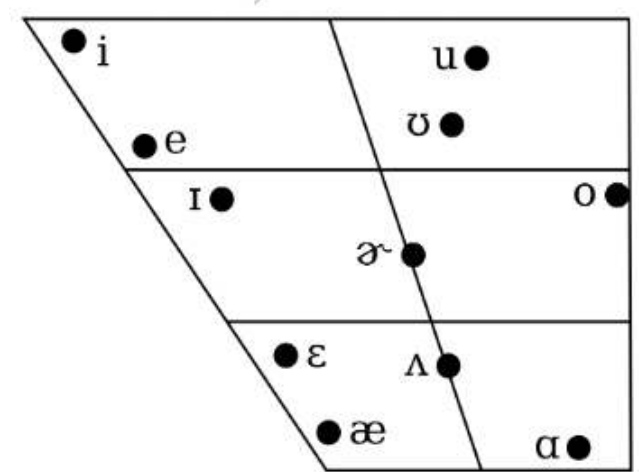
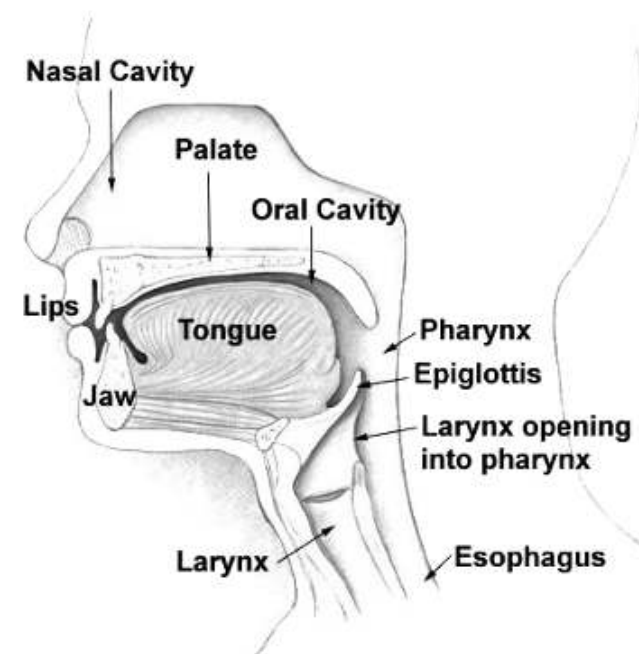
# Widening perspectives and expanding opportunities through teaching and mentorship



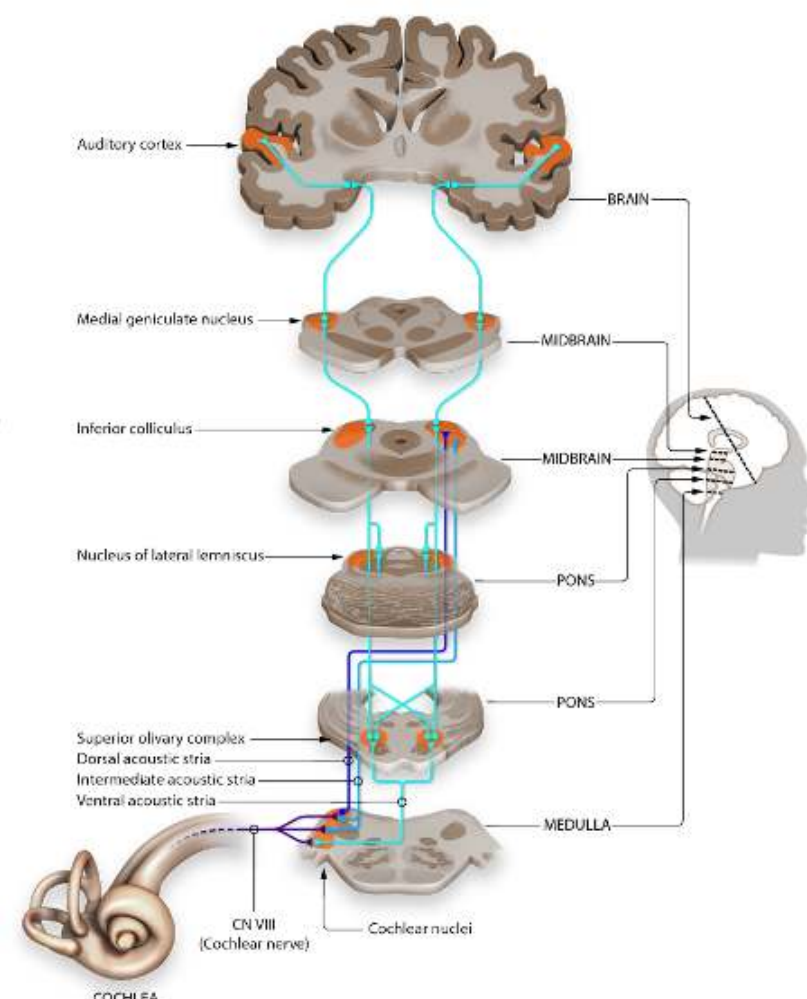
**Speech communication**

**Subcortical auditory circuits**

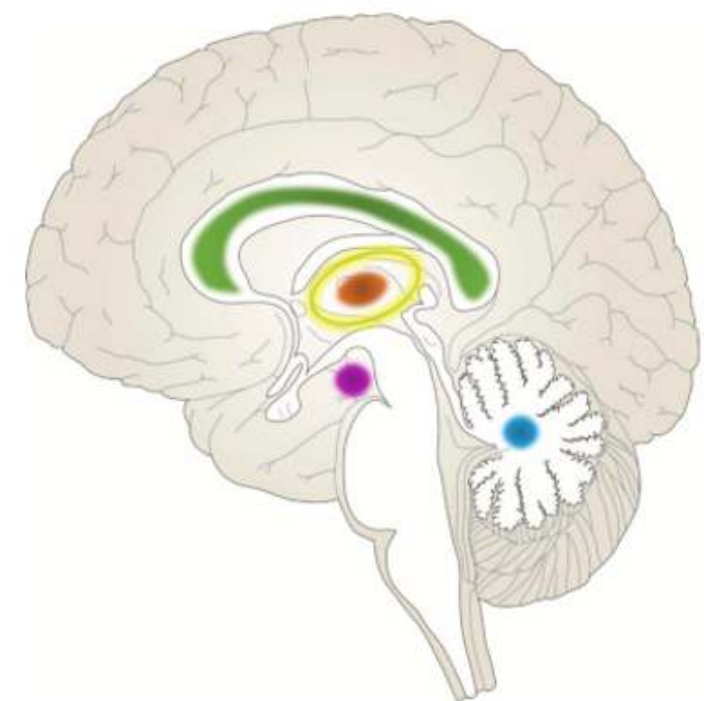
# Widening perspectives and expanding opportunities through teaching and mentorship



**Speech communication**



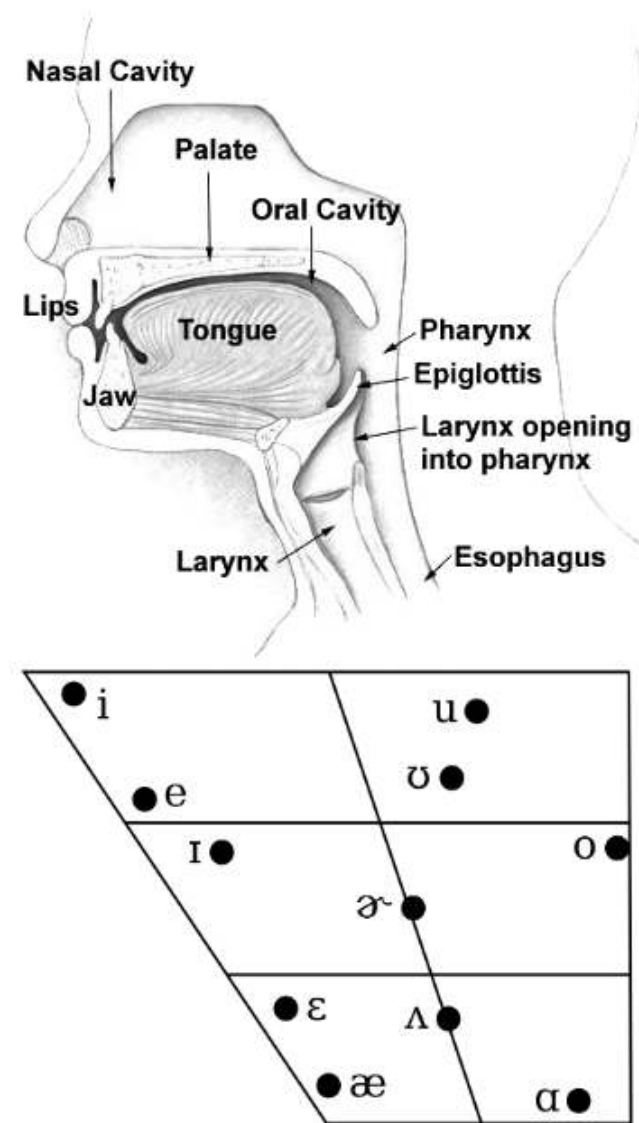
**Subcortical auditory circuits**



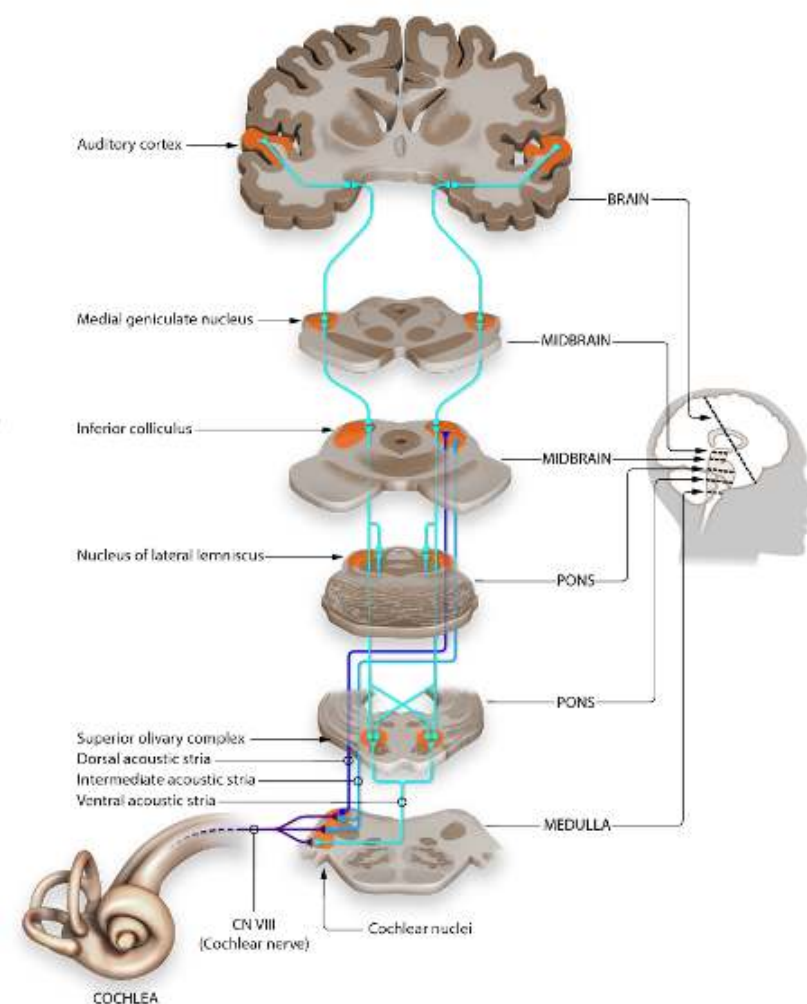
**Subcortical contributions to language**



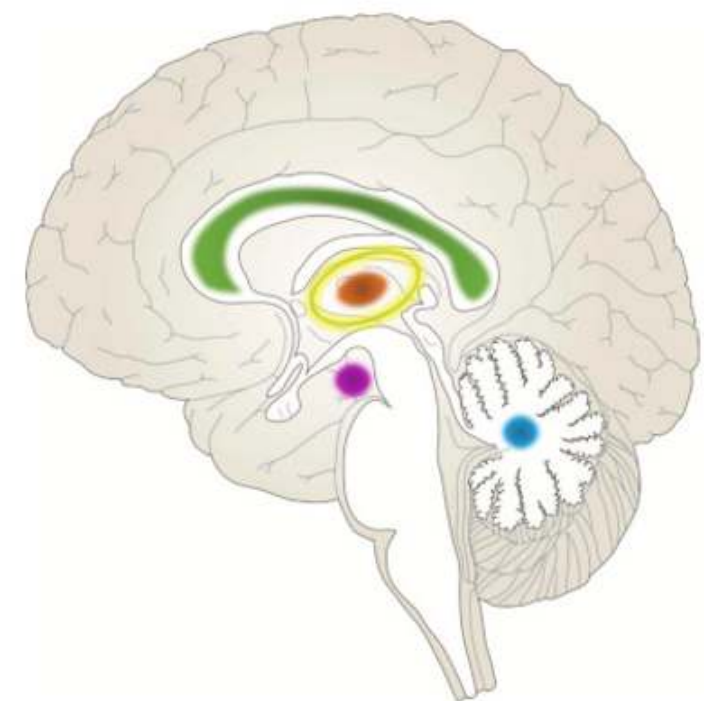
# Widening perspectives and expanding opportunities through teaching and mentorship



Speech communication



Subcortical auditory circuits



Subcortical contributions to language



Communicating science

# Research contributions to communication neuroscience

## Mapping the human subcortical auditory system

- **Anatomy** (Sitek & Gulban et al., 2019)
- **Function** (Sitek & Gulban et al., 2019; Medina, Sitek, et al., submitted preprint)
- **Structural and functional connectivity** (Sitek & Gulban et al., 2019; Sitek et al., 2022; Chandra & Sitek et al., 2024)
- **Neurovascular organization** (Xu et al., ISMRM 2025; Xu, ..., S. Bollmann, *Sitek*, in prep.)

## Behavior of the human auditory system

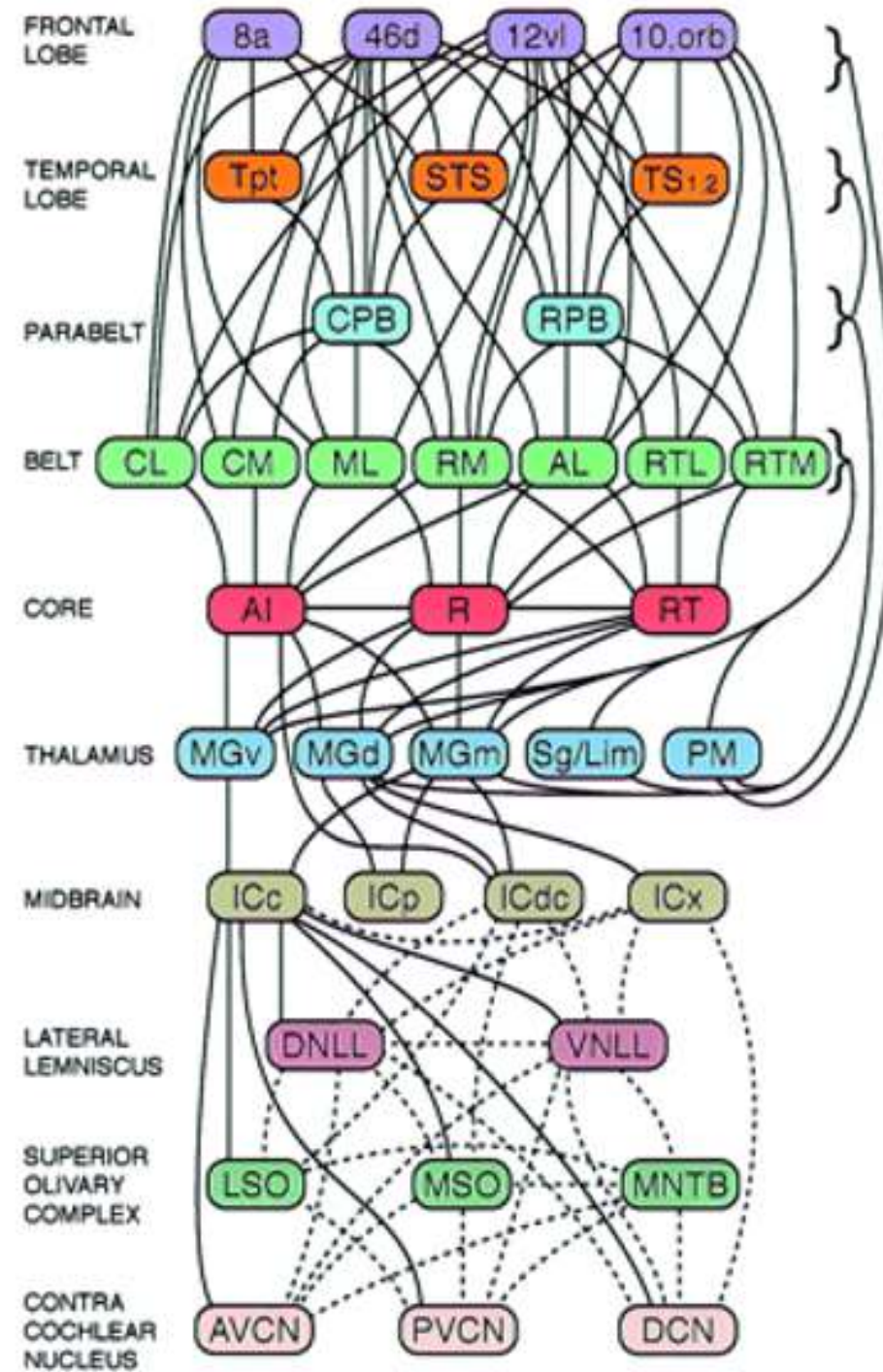
- **Auditory categorization** in the human brain (Sitek et al., in re-review; Llanos & Sitek et al., in re-review; Deschamps, Sitek, et al., in prep.)
- **Auditory-motor interactions** across the auditory pathway (Butler, ..., Sitek, in prep.)
- **Top-down modulation of predicted sound processing** across the auditory pathway (Ara et al., 2024)
- **Speech-in-noise processing** in the developing brain (Sitek et al., SNL 2025; in prep.)

This work is actively supported by K01-DC019421 and R21-DC022906 and was previously supported by F31-DC015695.

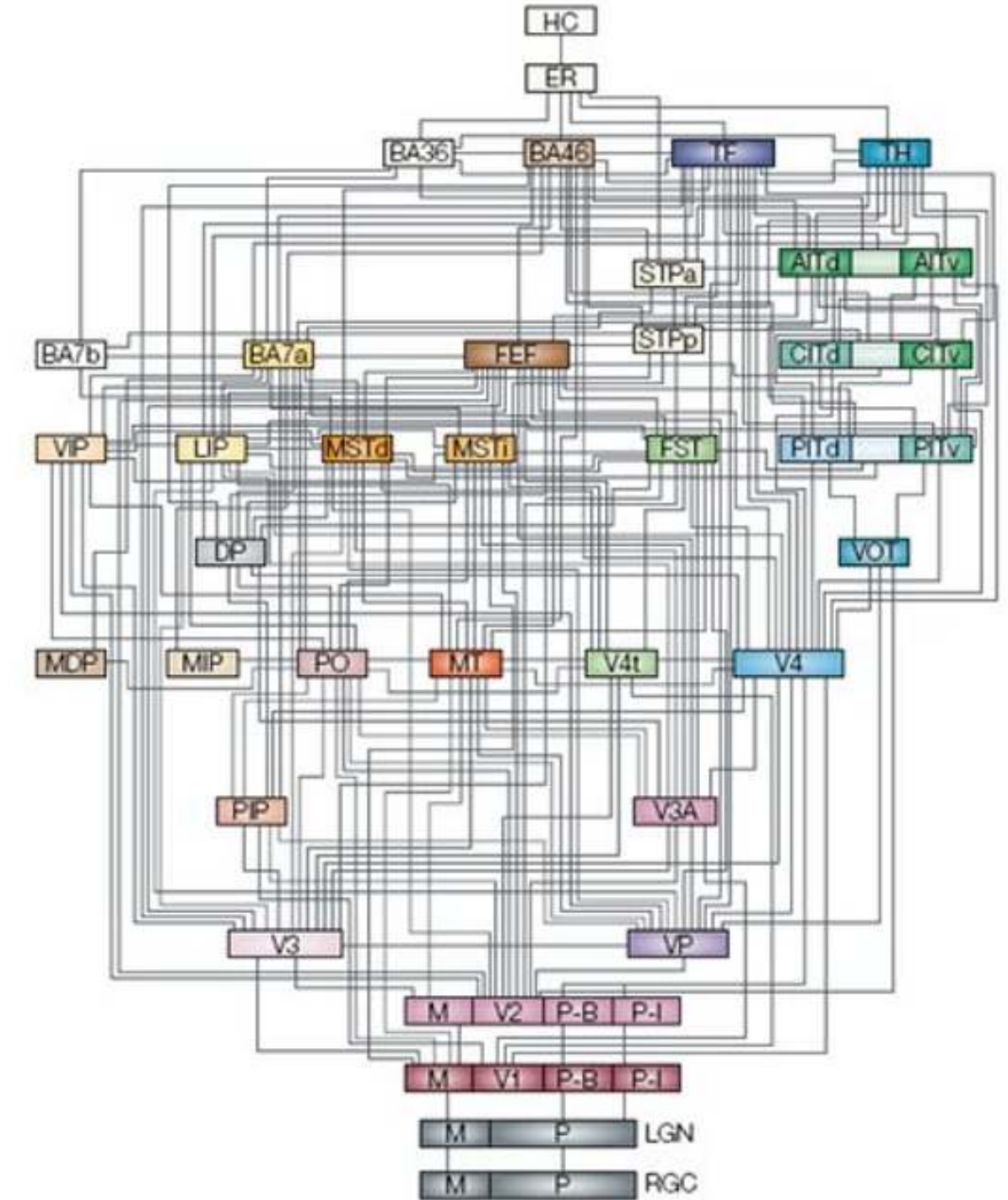
# **Research contributions: Auditory circuits throughout the human brain**



# What's special about the auditory system?



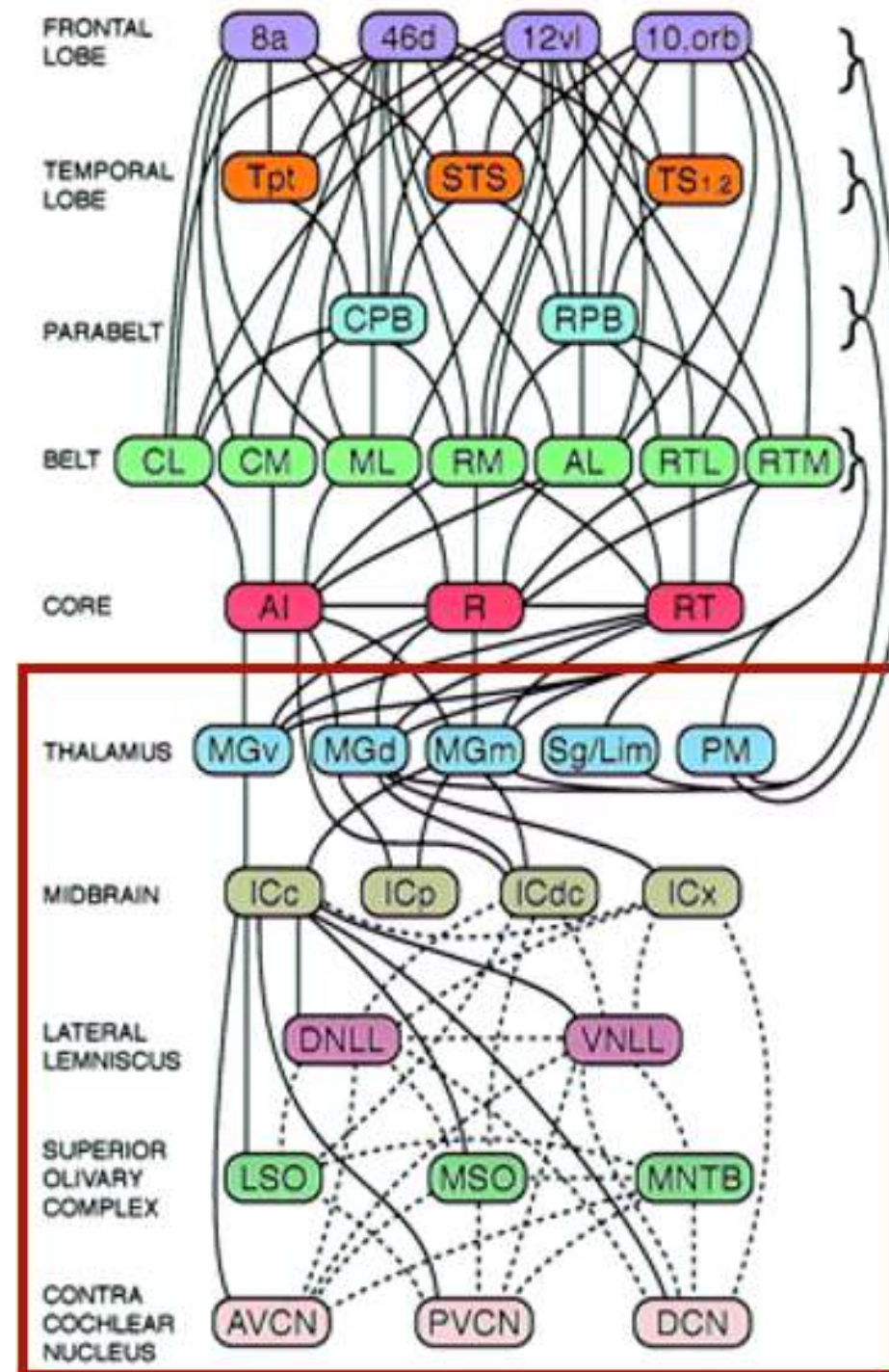
Kaas & Hackett (2000)



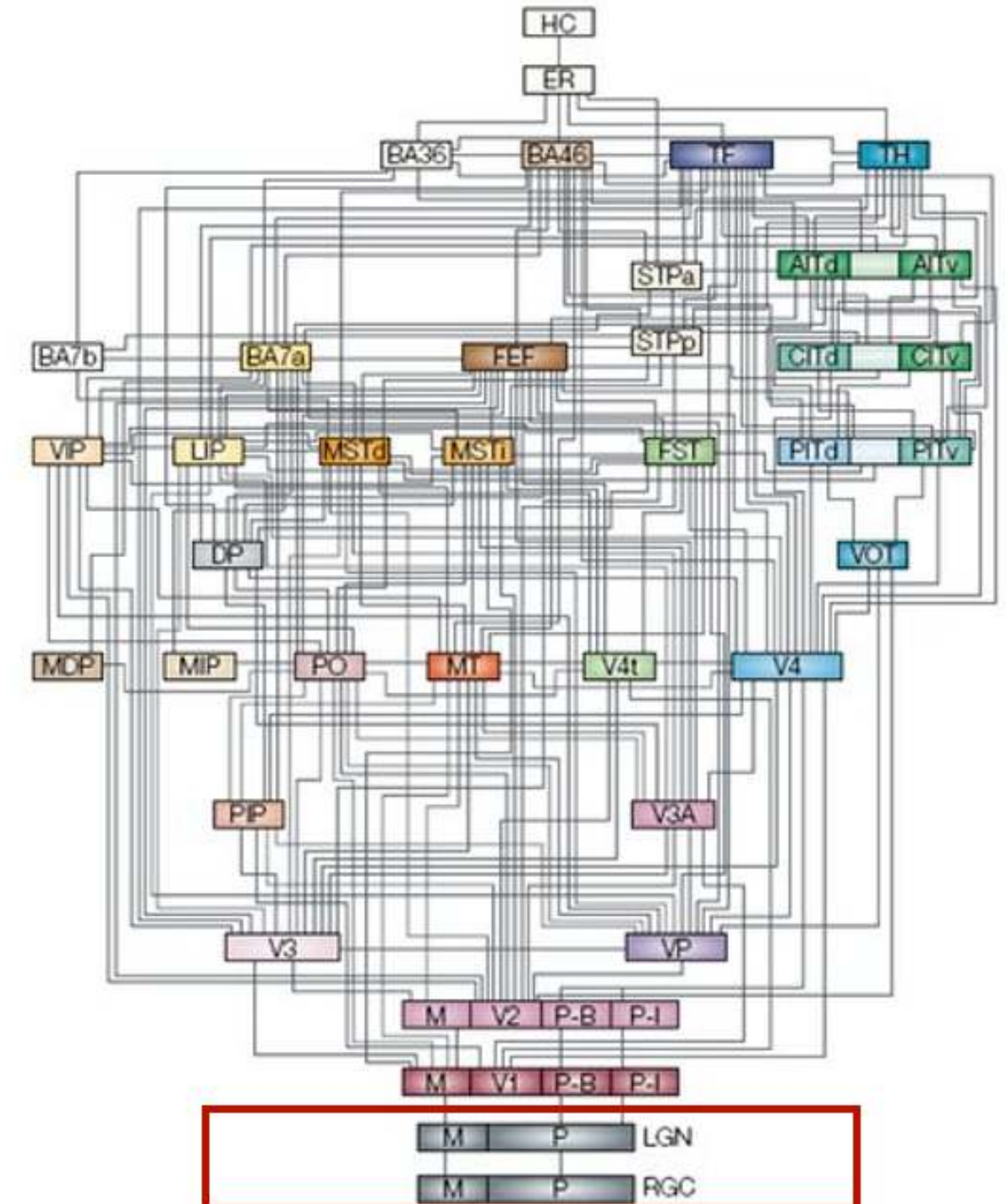
Van Essen et al. (1992)



# What's special about the auditory system?

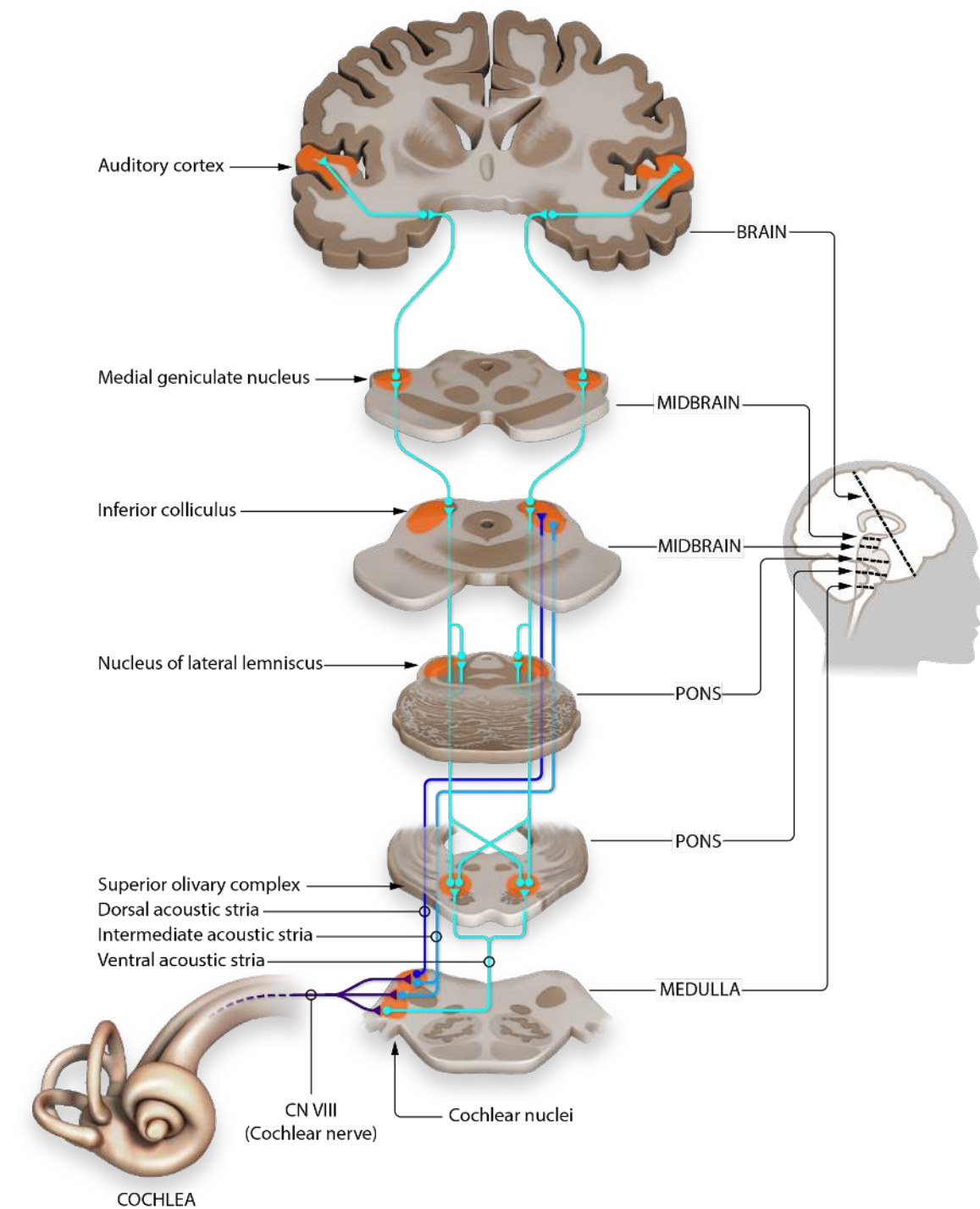
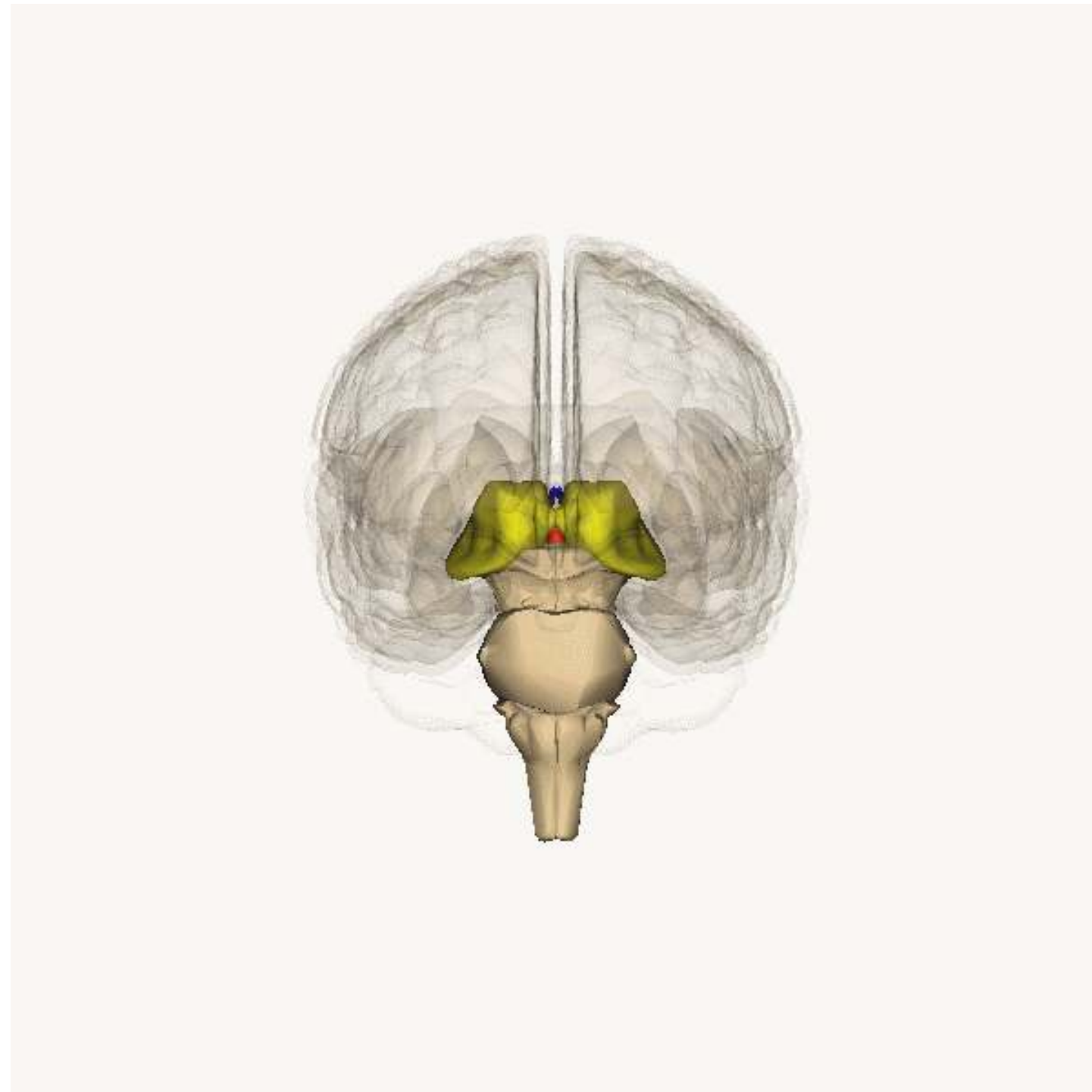


Kaas & Hackett (2000)



Van Essen et al. (1992)

# The human subcortical auditory system

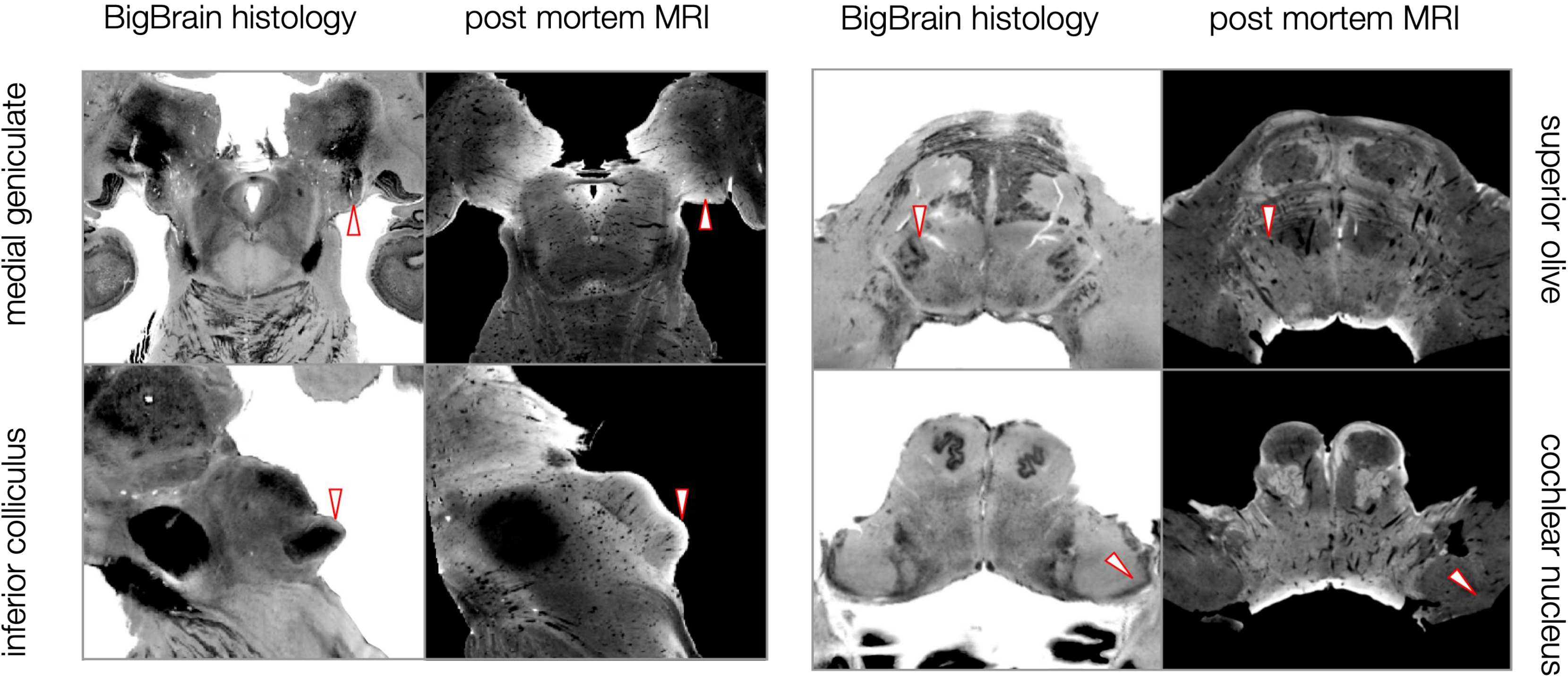


**But: small, deep brain structures are challenging to image.**



# Mapping the human subcortical auditory system

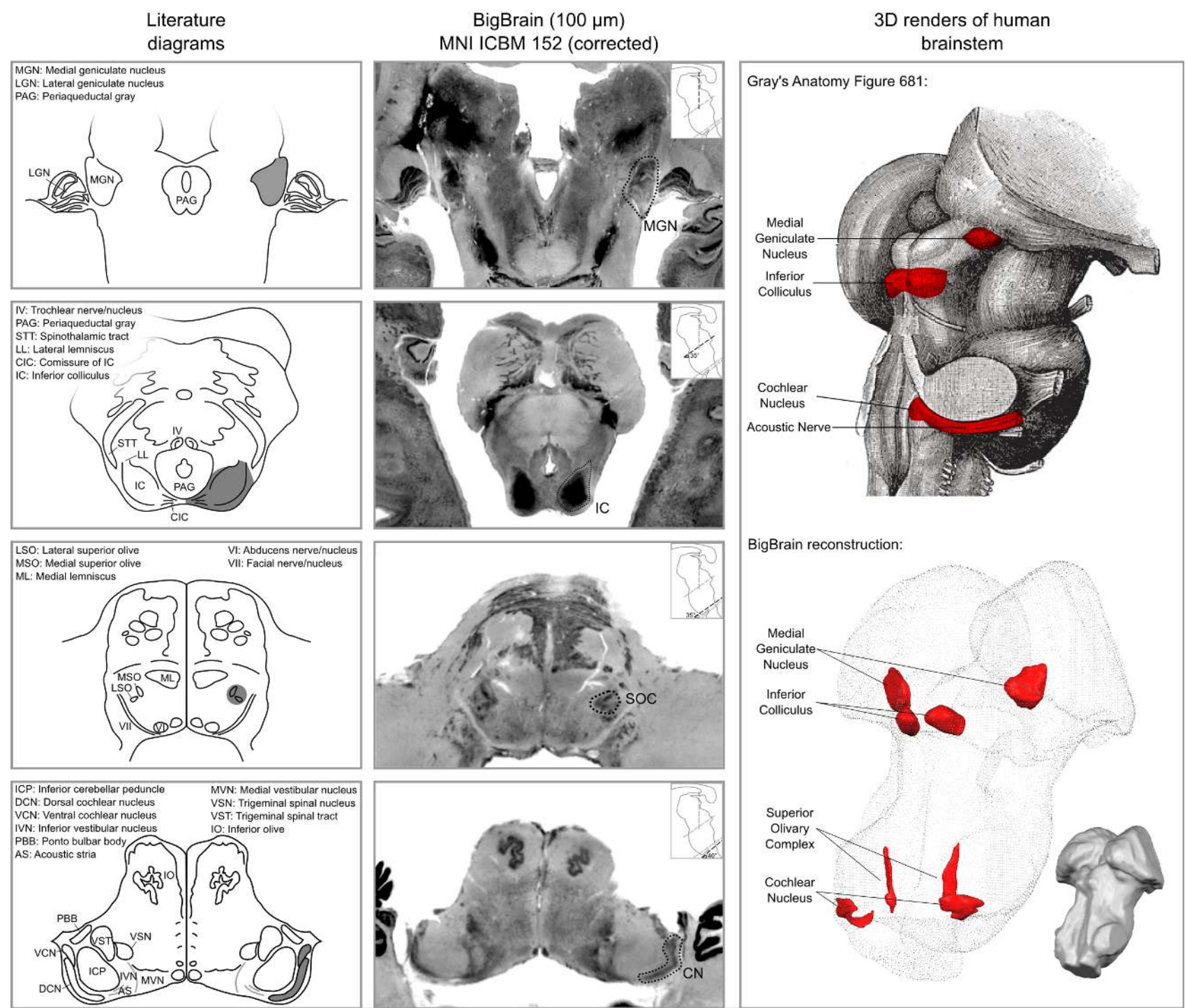
# Post mortem anatomical localization



Sitek and Gulban et al. (2019 *eLife*)



# 3D subcortical atlases



Sitek and Gulban et al. (2019 *eLife*)

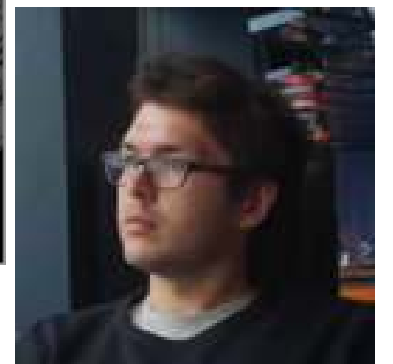
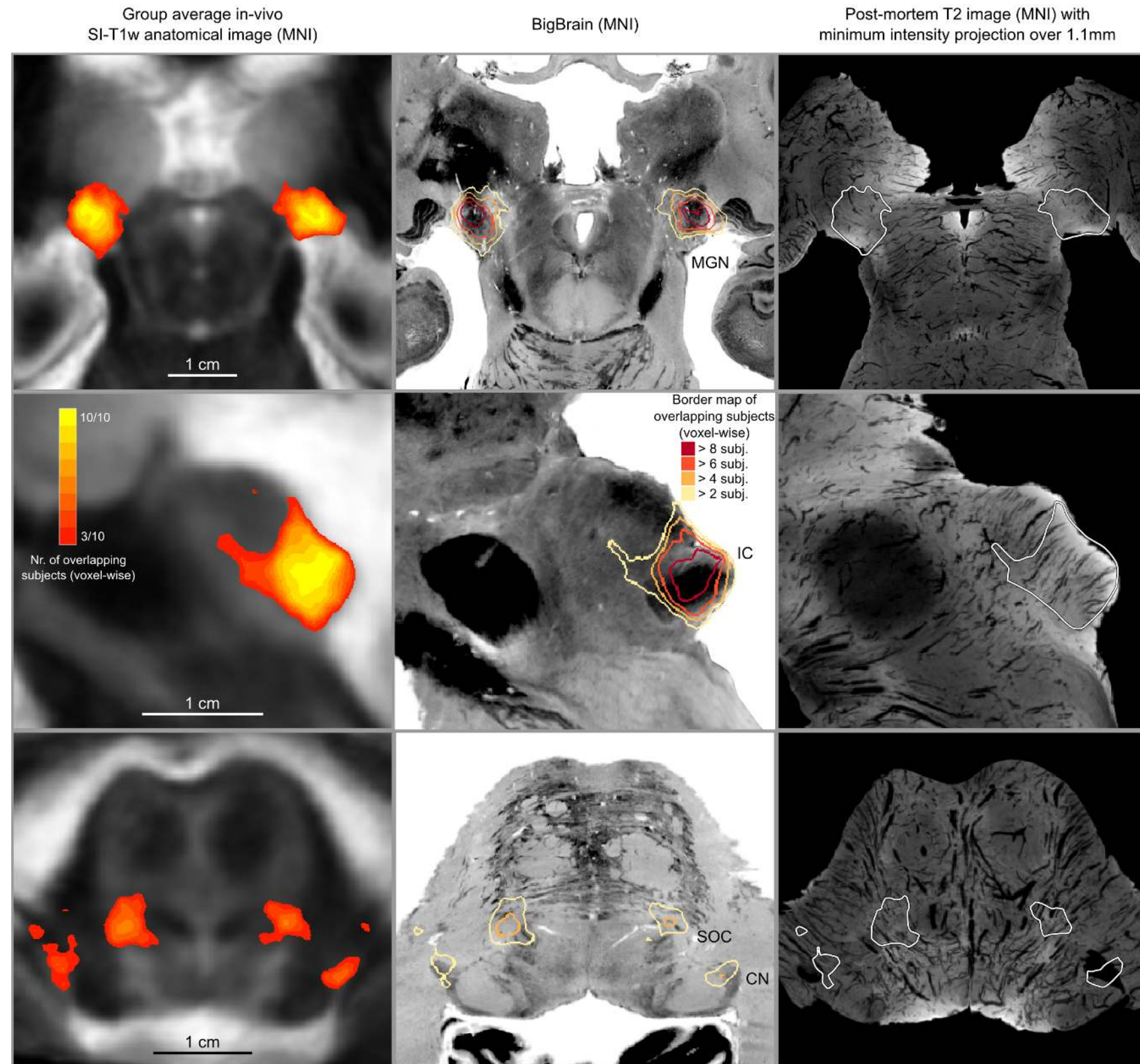


# In vivo functional MRI mapping

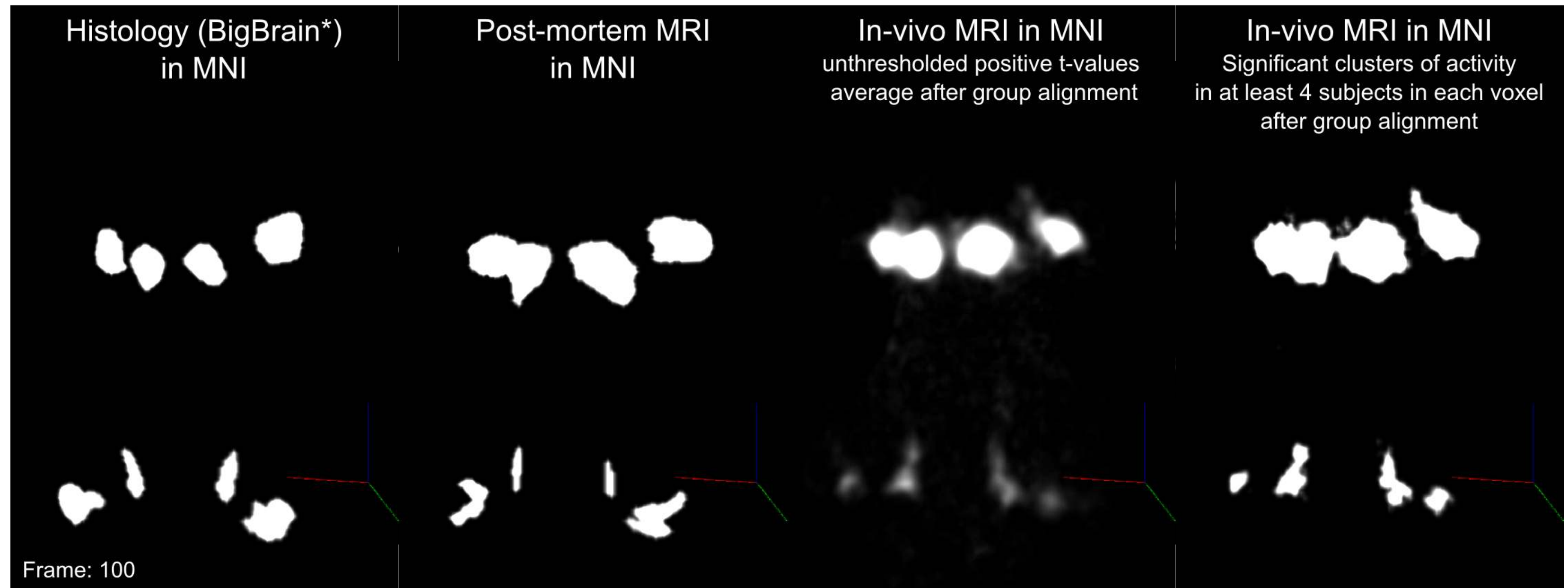
## 7 Tesla functional MRI (1.1 mm voxels)

- 168 natural sounds (1 s each): speech, voice, nature, tools, music, animals and monkey calls
- Sound > baseline contrast
- 10 healthy participants
- 2 hours per session
- Collected over 2 sessions

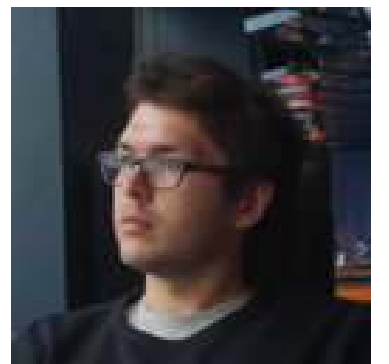
Sitek and Gulban et al. (2019 *eLife*)



# *Where* are the auditory structures: Three public atlases



Sitek and Gulban et al. (2019 *eLife*)

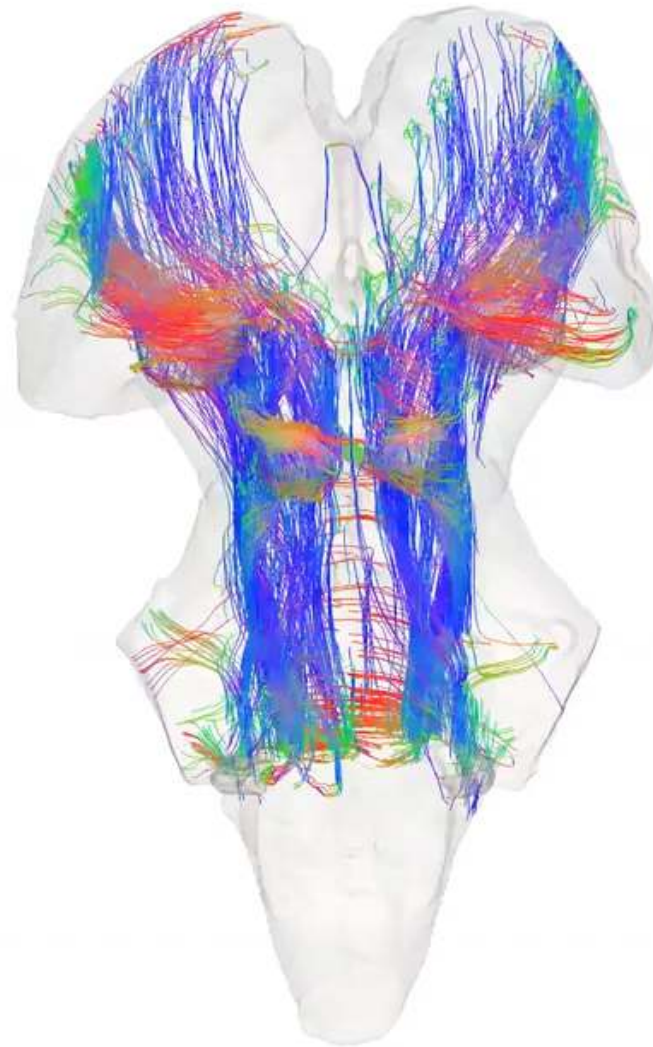


# Connectivity of the human subcortical auditory system

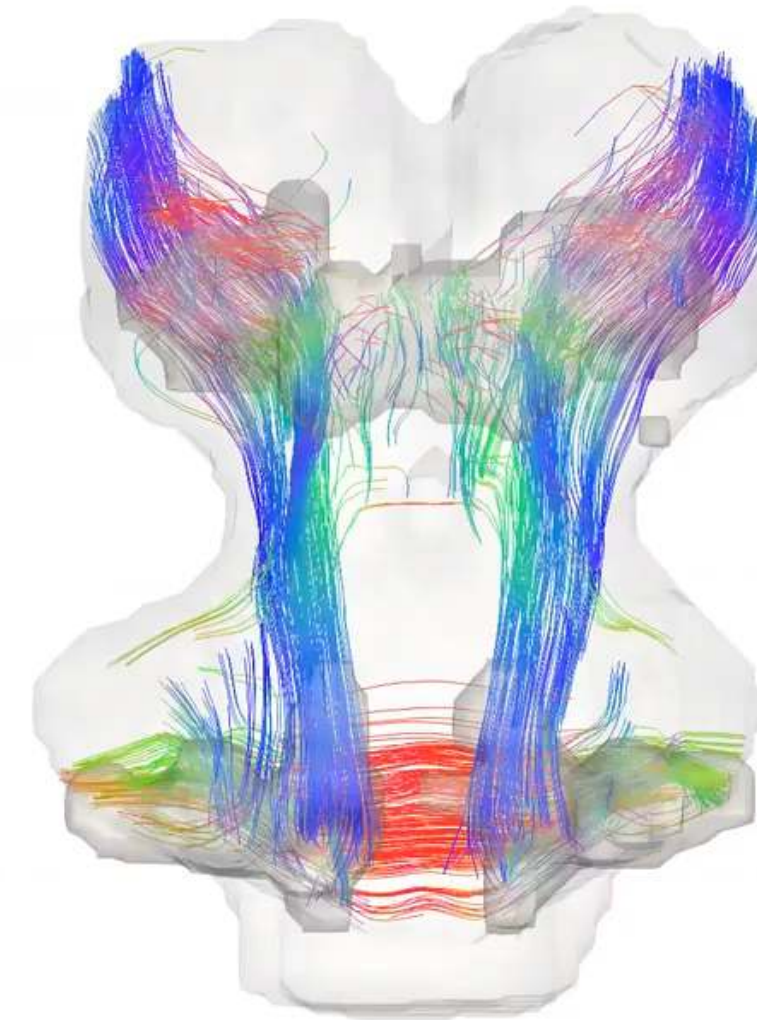


# How are the auditory structures connected anatomically?

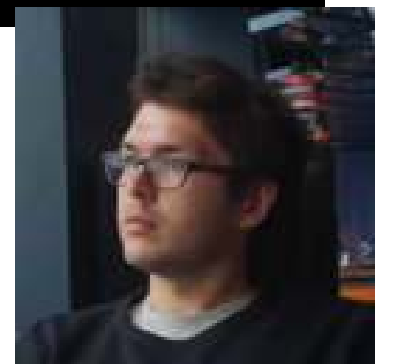
Post mortem



In vivo

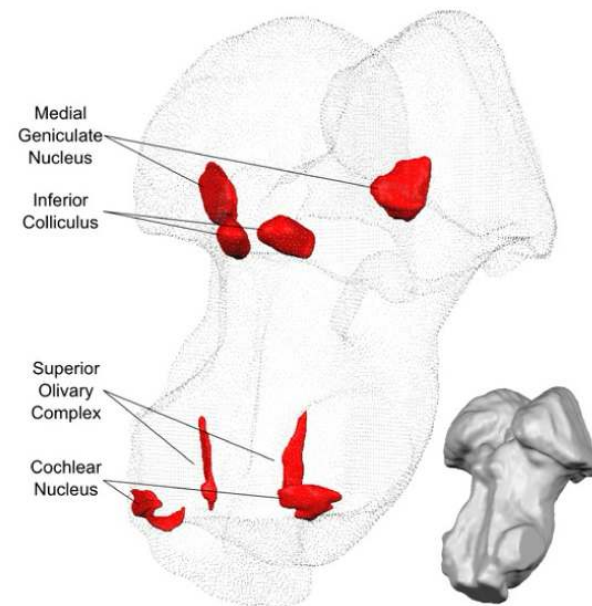
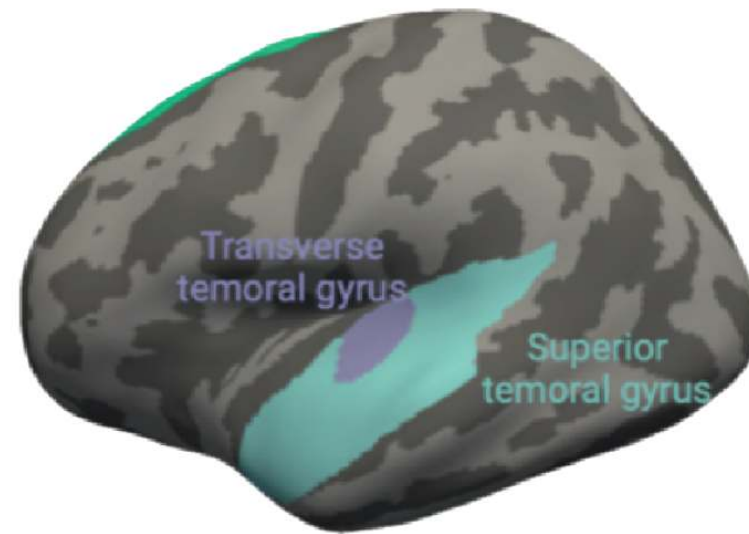


Sitek and Gulban et al. (2019 *eLife*); Sitek et al. (2022 *Frontiers Neuro*)

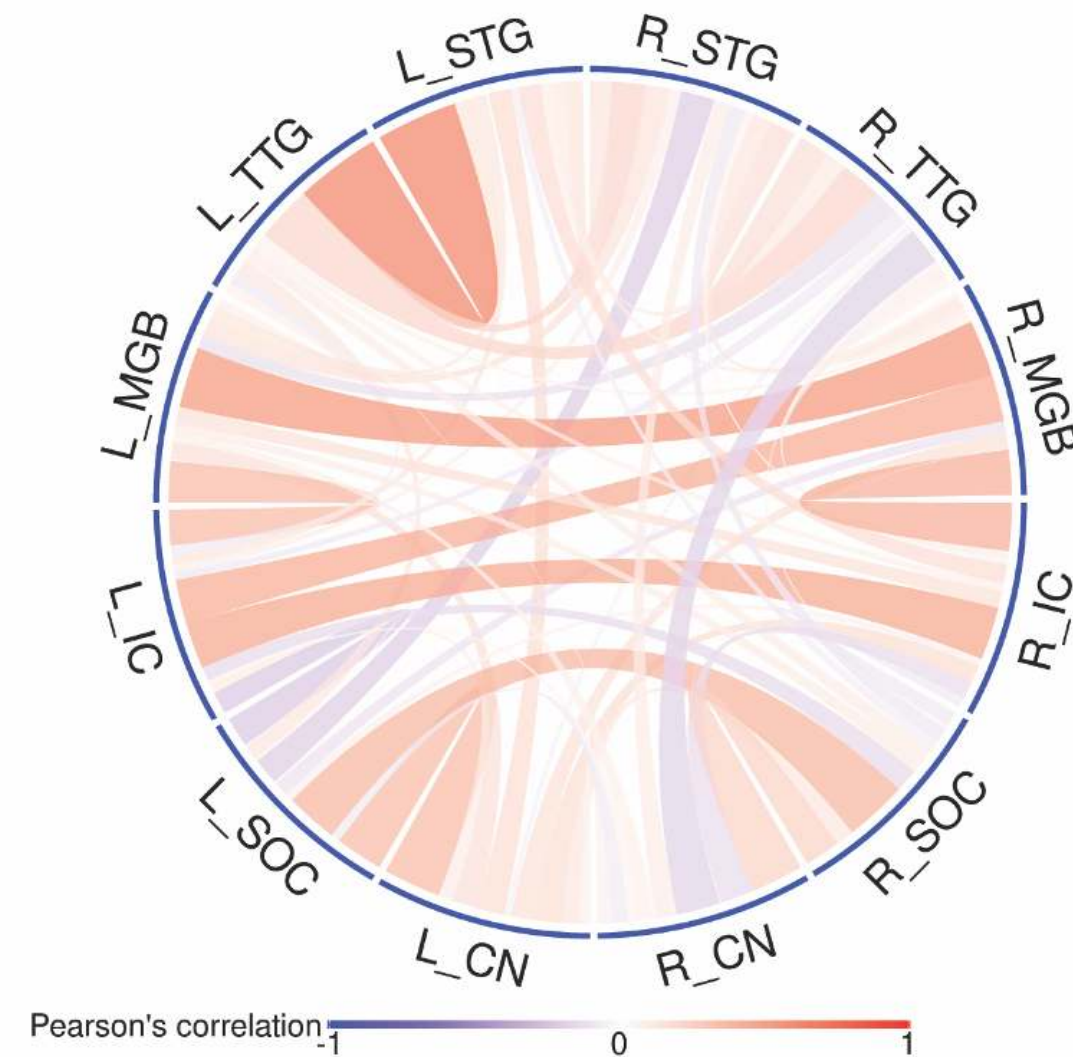


# How are the auditory structures connected functionally?

Extracting the *pathway* (partial correlations) from the *network* (simple correlations)



Simple correlation



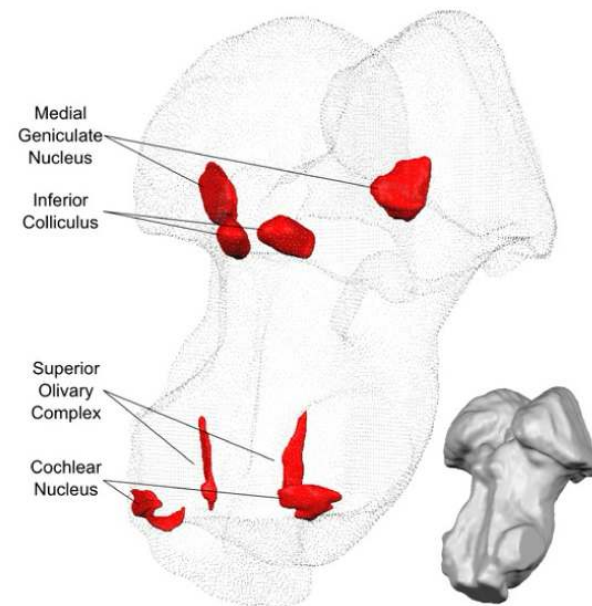
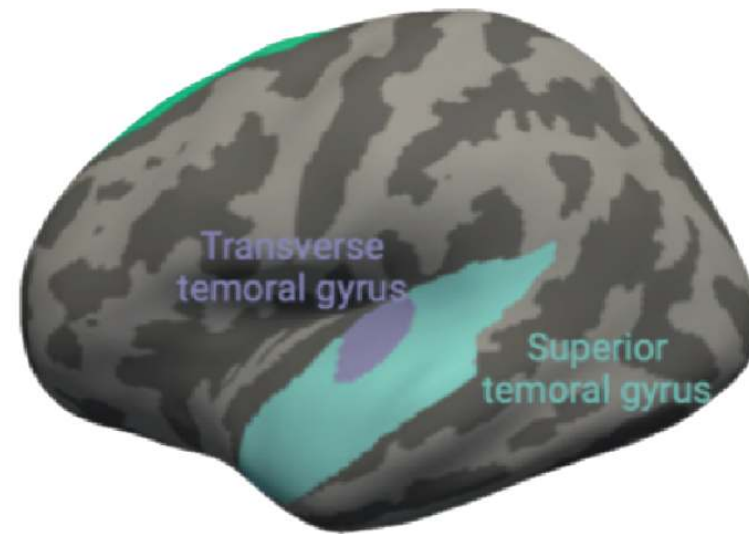
7T Human Connectome Project (106 participants) resting state **functional** MRI  
Chandra\* and Sitek\* et al. (2024 *Imaging Neuro*)



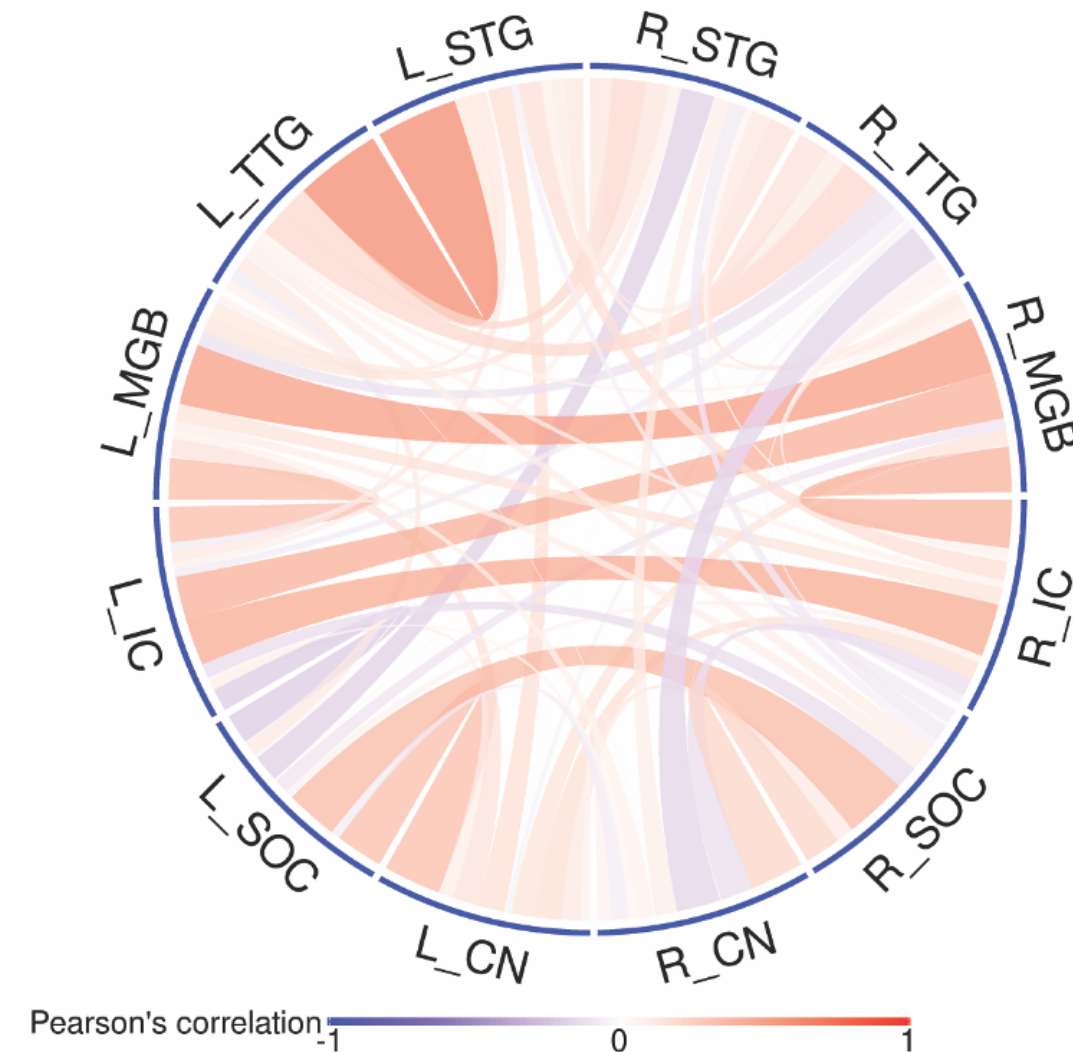


# How are the auditory structures connected functionally?

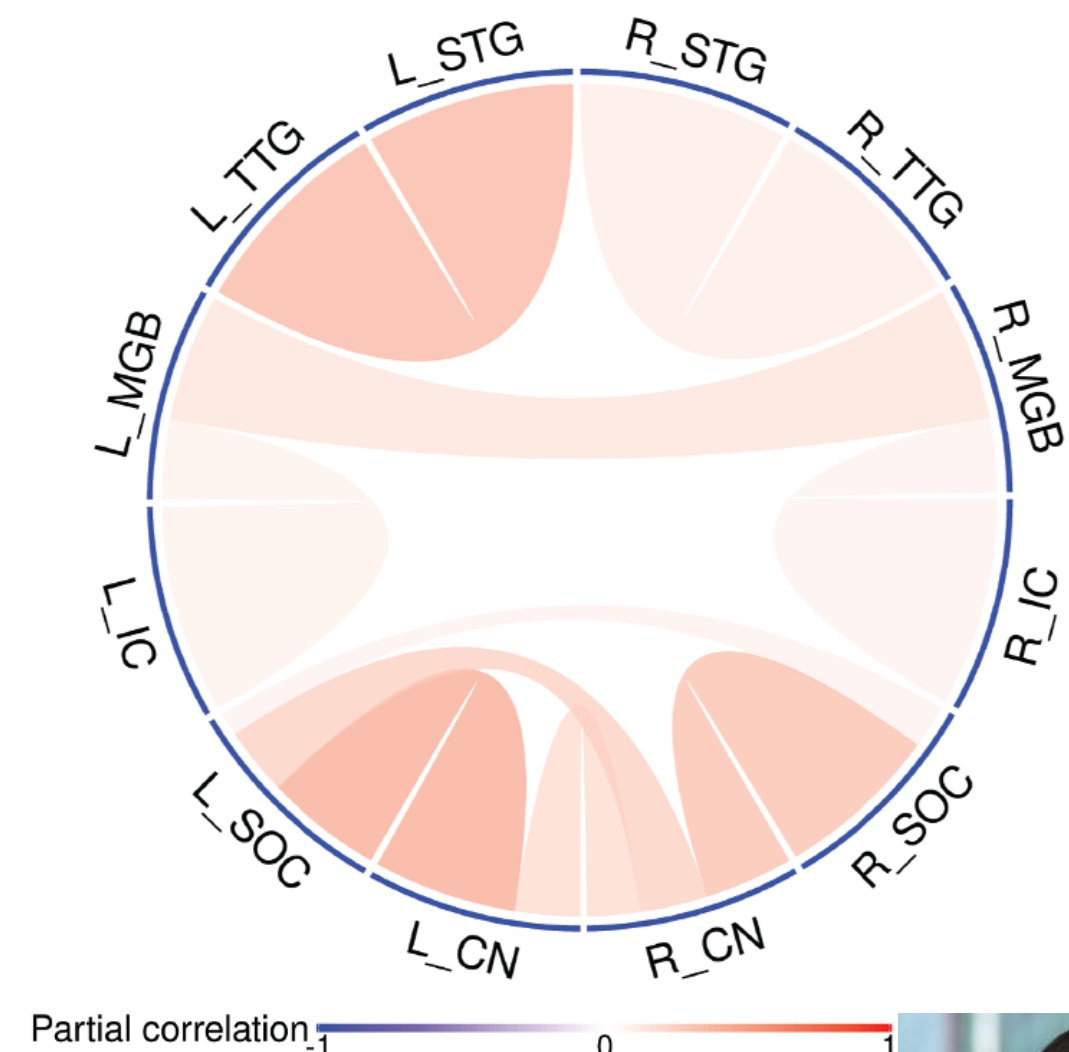
Extracting the *pathway* (partial correlations) from the *network* (simple correlations)



Simple correlation



Partial correlation



7T Human Connectome Project (106 participants) resting state **functional** MRI

Chandra\* and Sitek\* et al. (2024 *Imaging Neuro*)





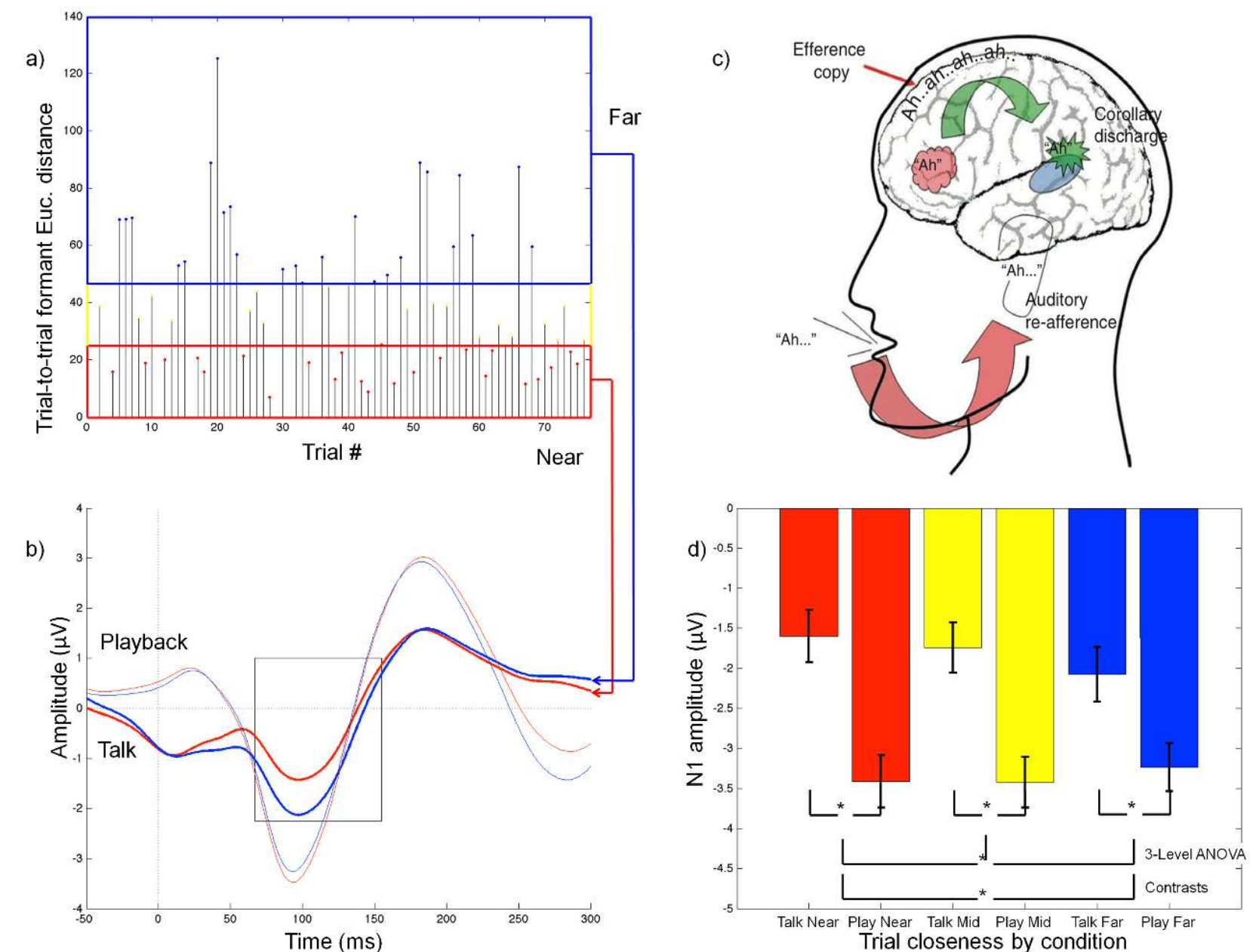
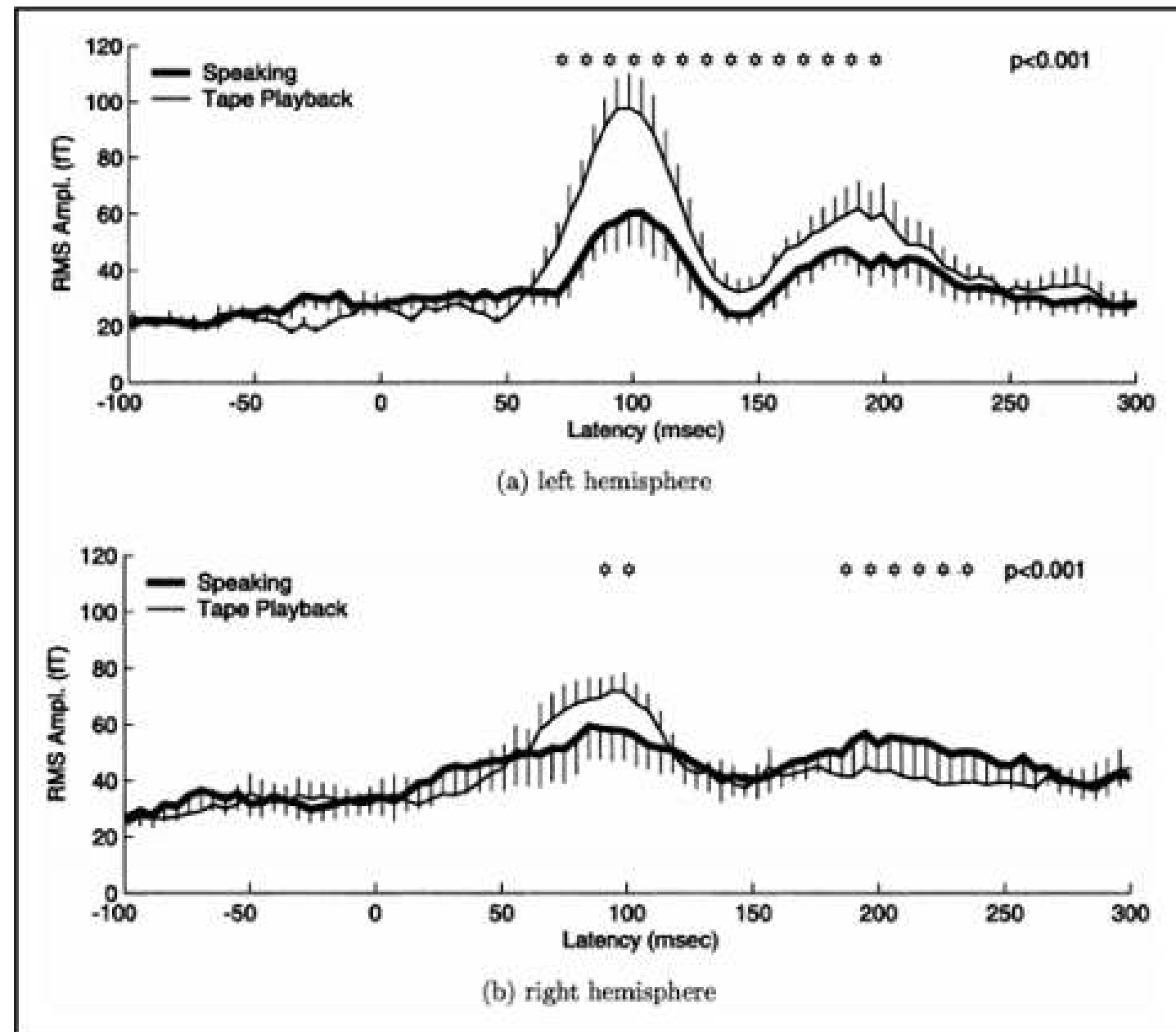


# **Auditory–motor interactions across the auditory pathway**

# Human speech and auditory feedback processing

Suppressed cortical response to own voice while speaking (Houde & Jordan 2002)

Less suppression when our own speech is *variable* (Sitek et al. 2013)



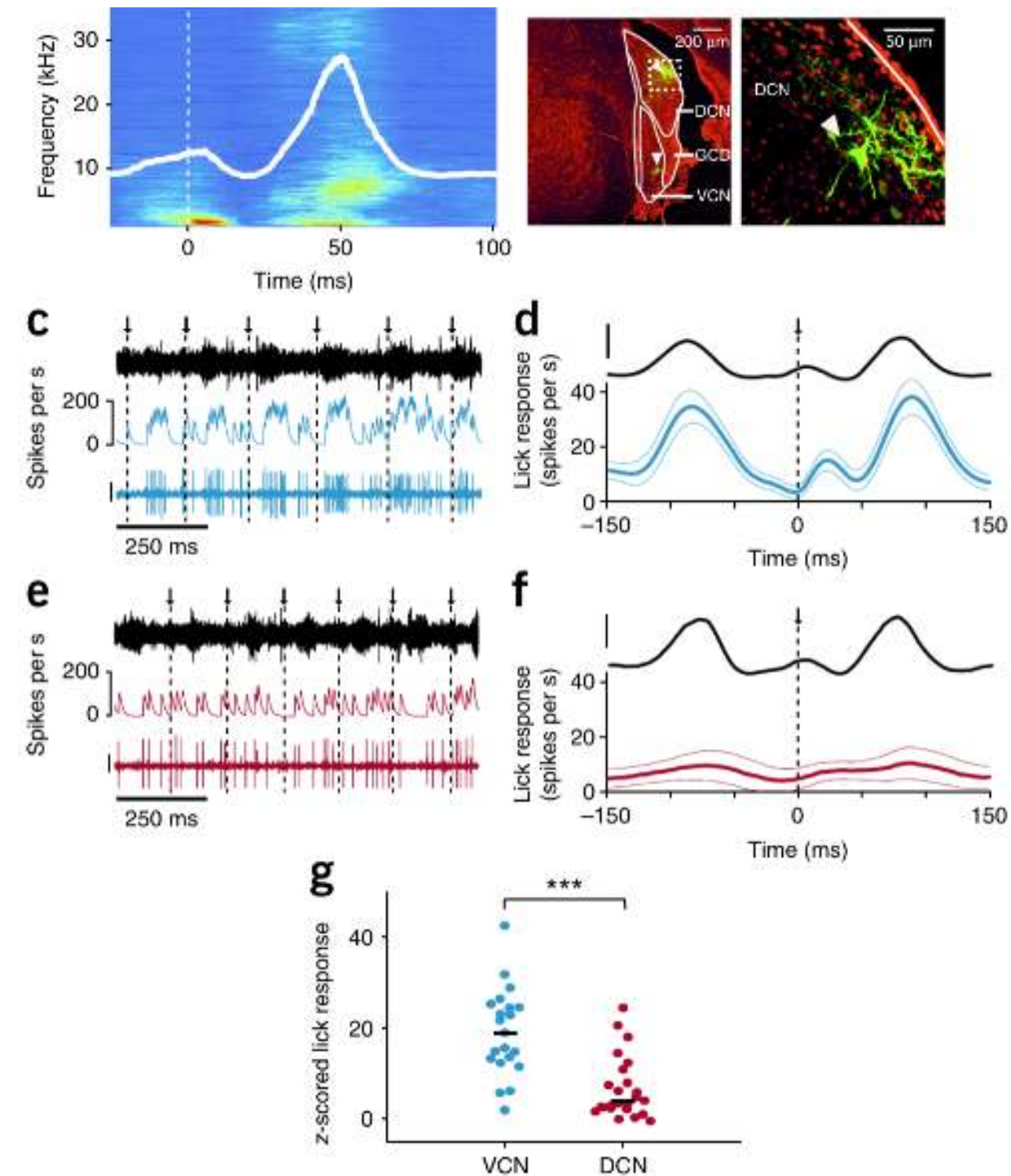
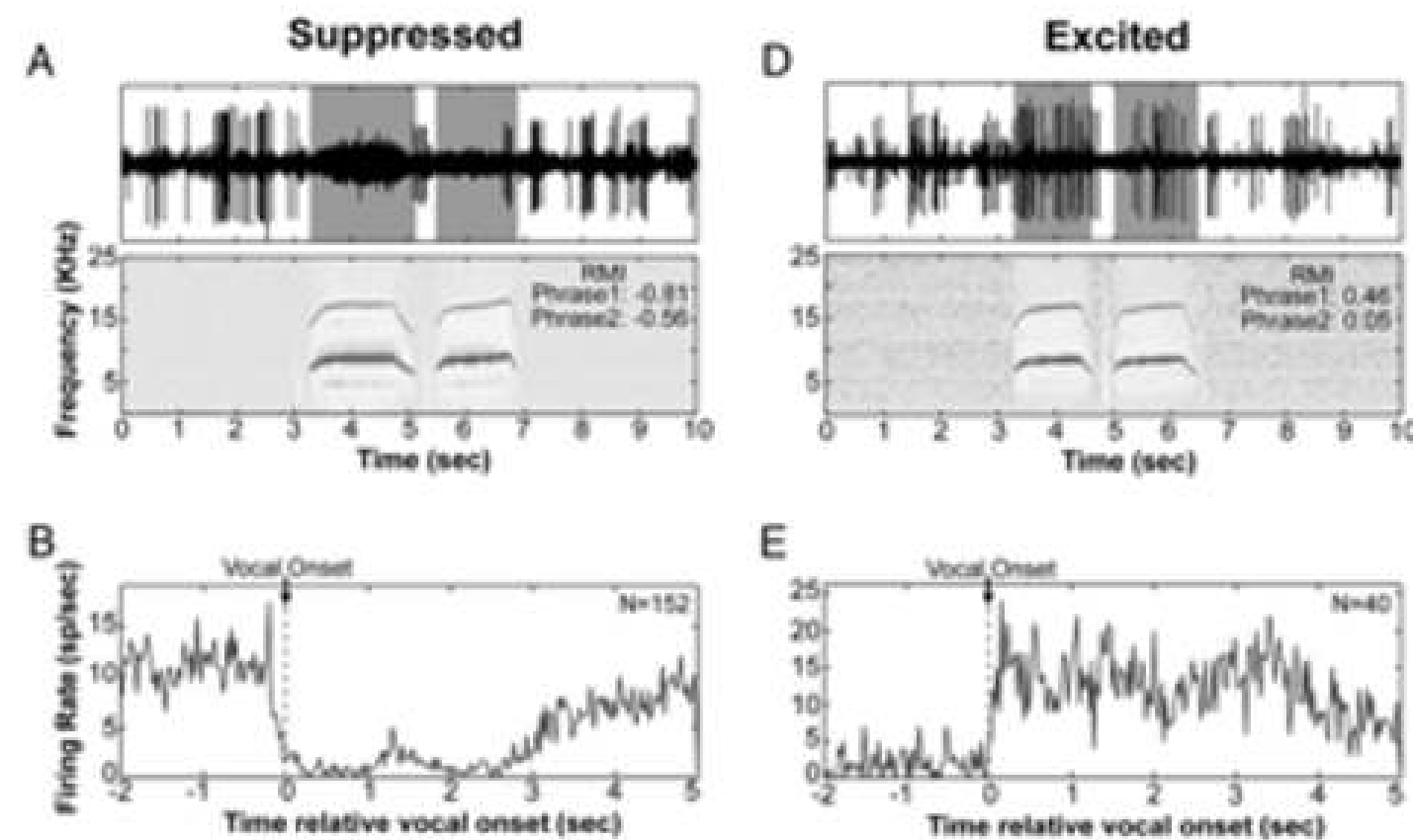
This work is actively supported by R21-DC022906 (PI: Sitek).



# Auditory feedback processing in animals

Overall suppression (but some excitation) in marmoset auditory cortex<sup>1</sup>

Decreased auditory brainstem response to own licking sound in mouse<sup>2</sup>

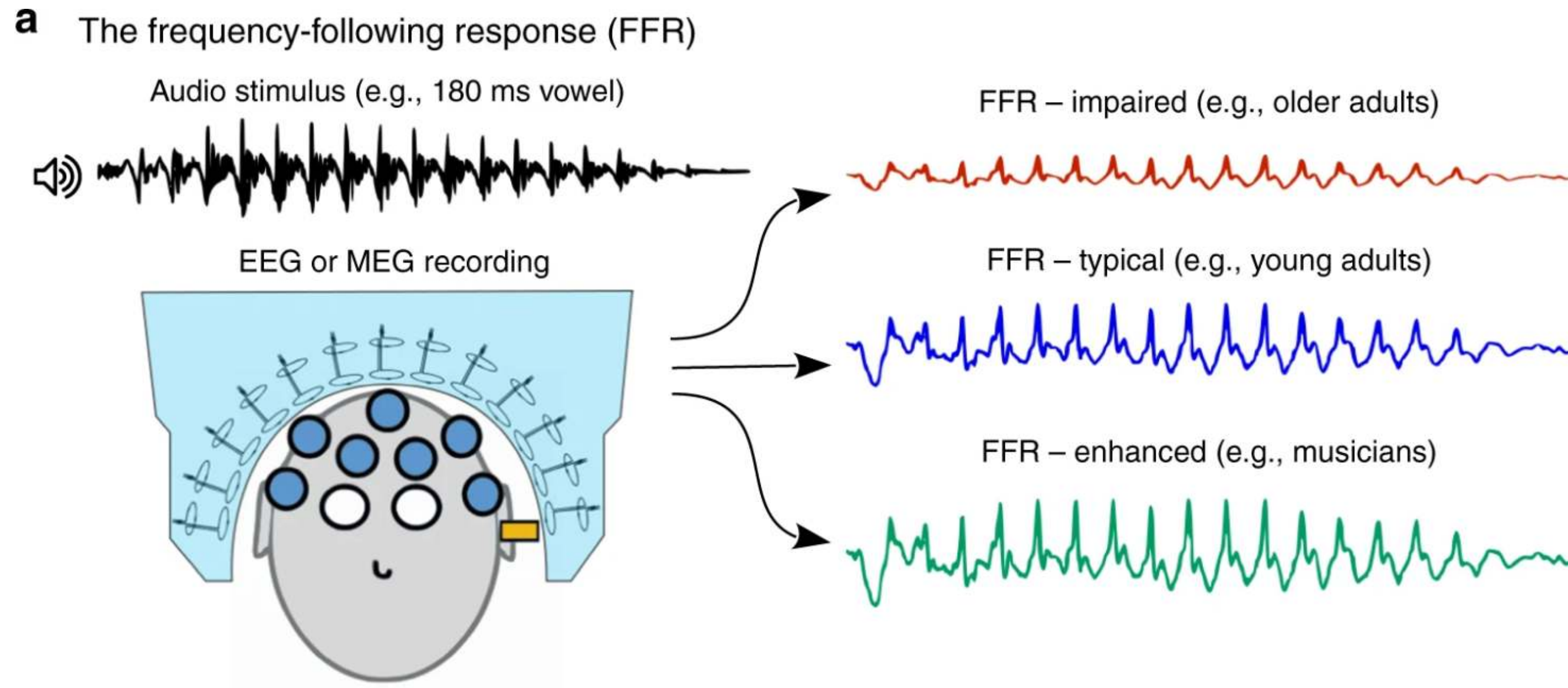


1. Eliades & Wang (2005)

2. Singla et al. (2017)

# Studying the human subcortical auditory system with EEG

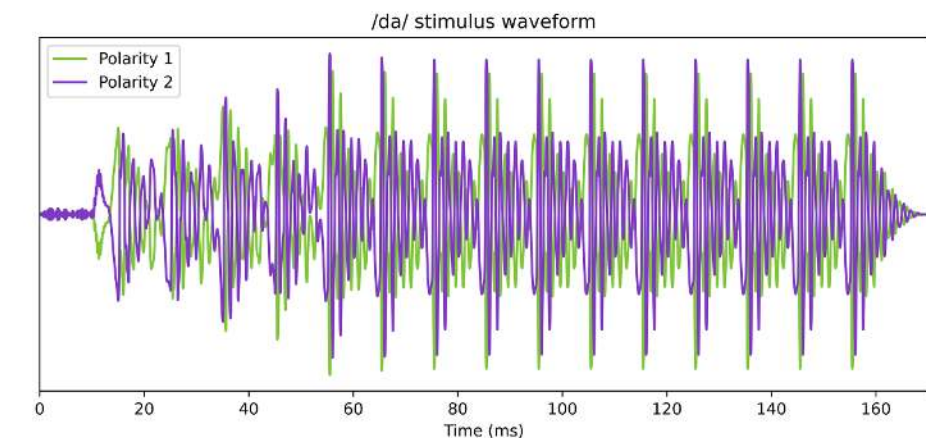
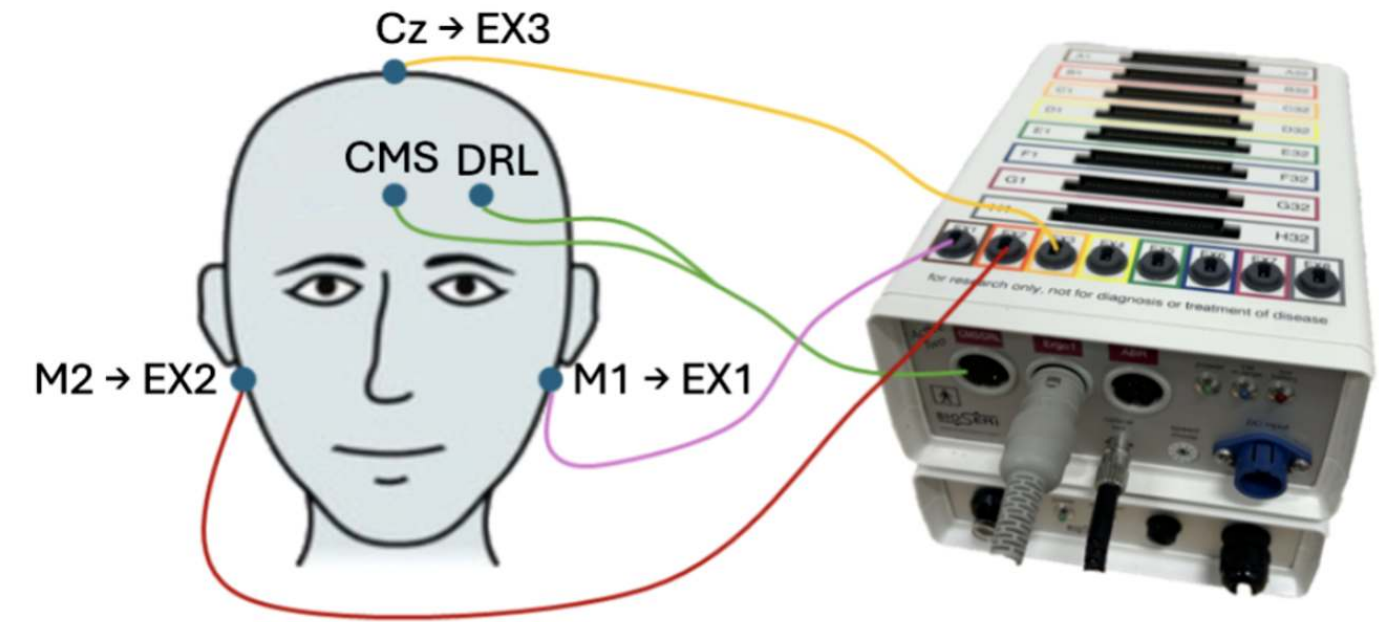
The scalp-recorded frequency-following response (FFR) shows the fidelity of incoming sound encoding



Coffey et al. (2018)

# Where are motor signals integrated? Subcortically?

- If integration is **subcortical**:
  - Expect to see **FFR** differences between *self-generated* and *passive* sound
- Recorded FFRs to /da/ stimulus
  - *Active* condition: press a button to generate a sound
  - *Passive* condition: passive presentation of the sound
- Collected data from 33 normal-hearing adults



Butler, ..., Sitek (in prep; SNL 2025)

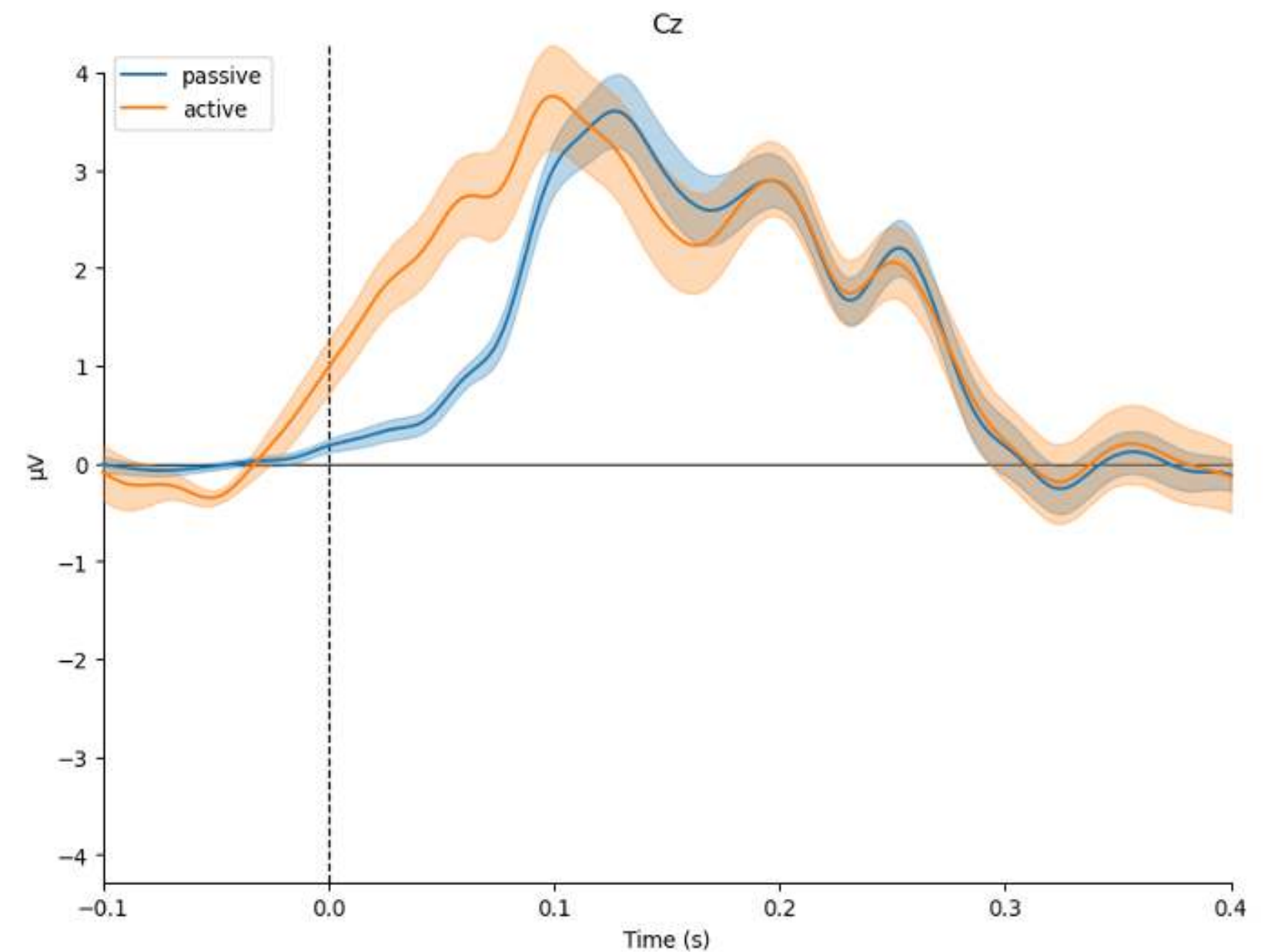
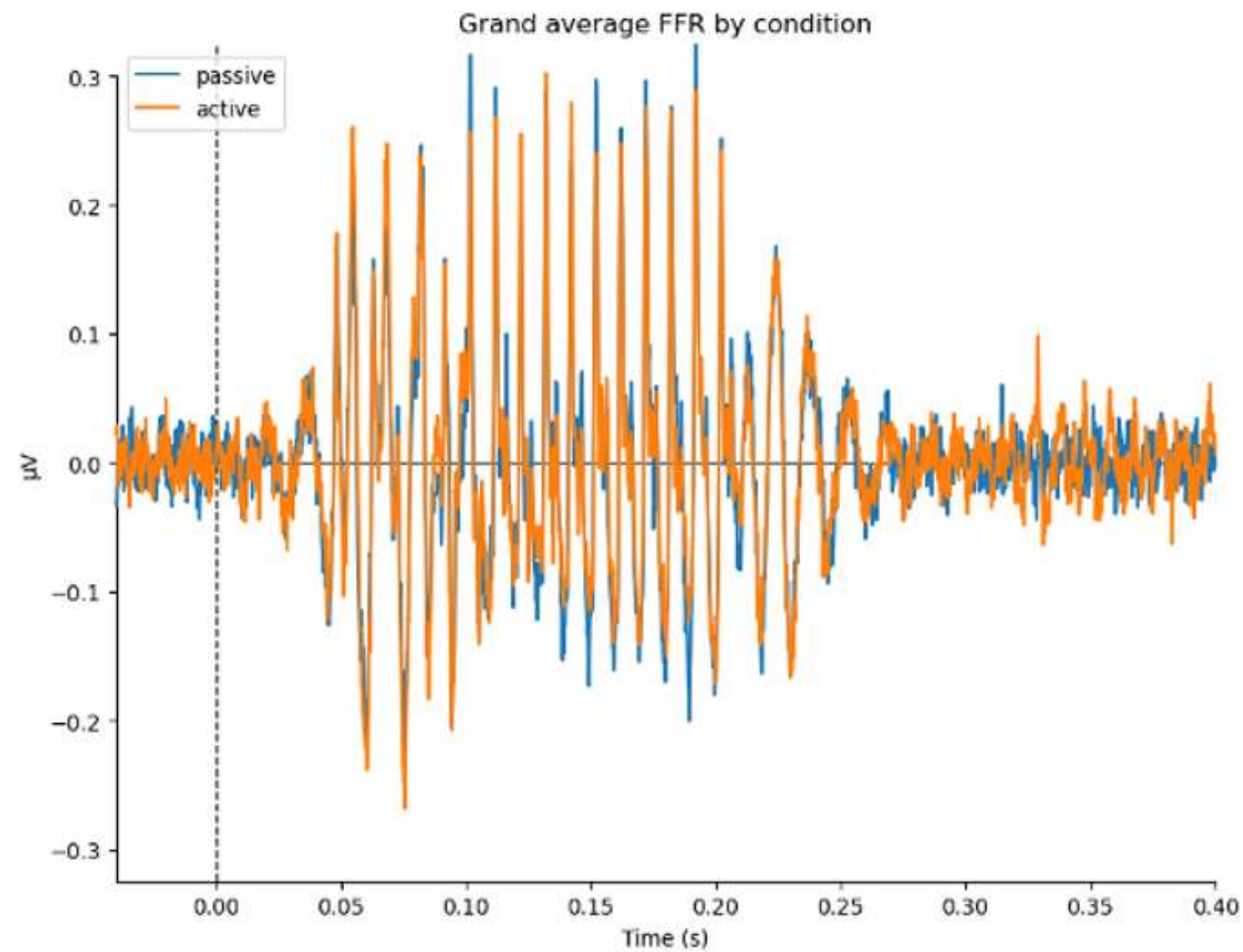




# Active and passive evoked responses

Brainstem frequency-following response

Cortical evoked response



Butler, ..., Sitek (in prep; SNL 2025)



# *Where* are motor signals integrated? Subcortically?

Brainstem frequency-following response (FFR) shows the same sound encoding when a sound is self-generated or passively presented

- No **brainstem FFR** differences, despite **cortical** evoked response differences
- *Motor signals unlikely to attenuate earliest central auditory signals*
- Next steps:
  - **Source localization** of FFR generators (ECR R21 project)
  - **Cortical layer-specific** feedback localization with 7T fMRI (planned R01)

Butler, ..., Sitek (in prep; SNL 2025)



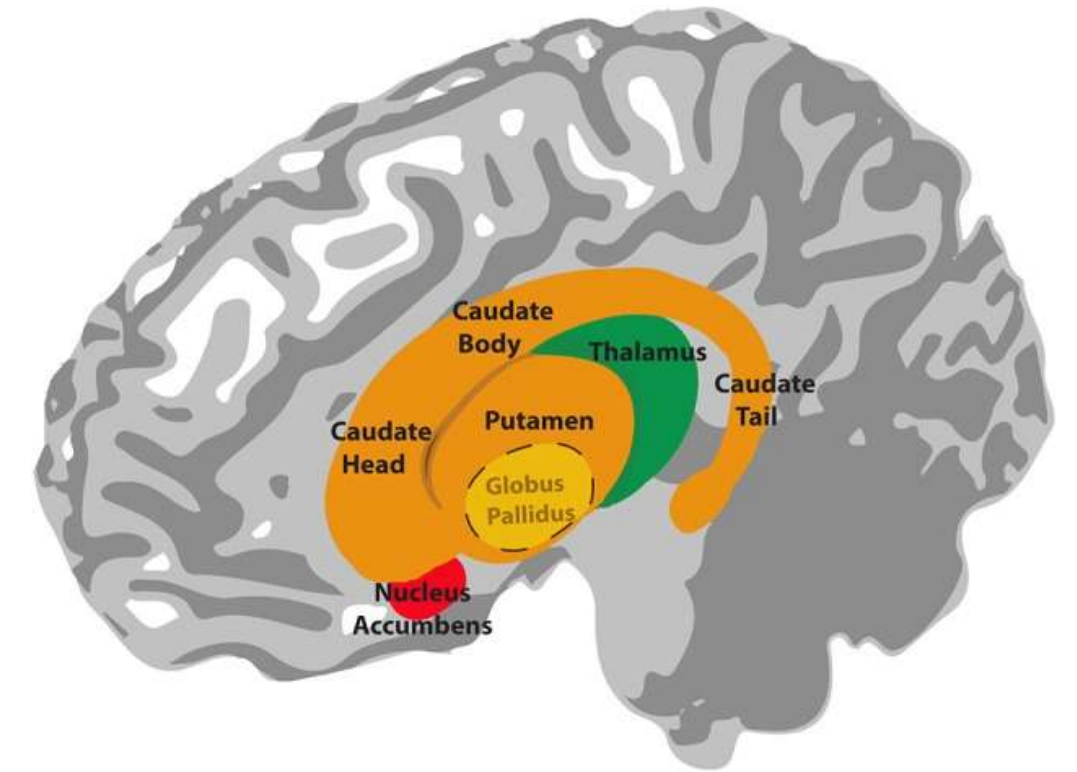




# Auditory categorization in the human brain

# How do we learn new sensory categories?

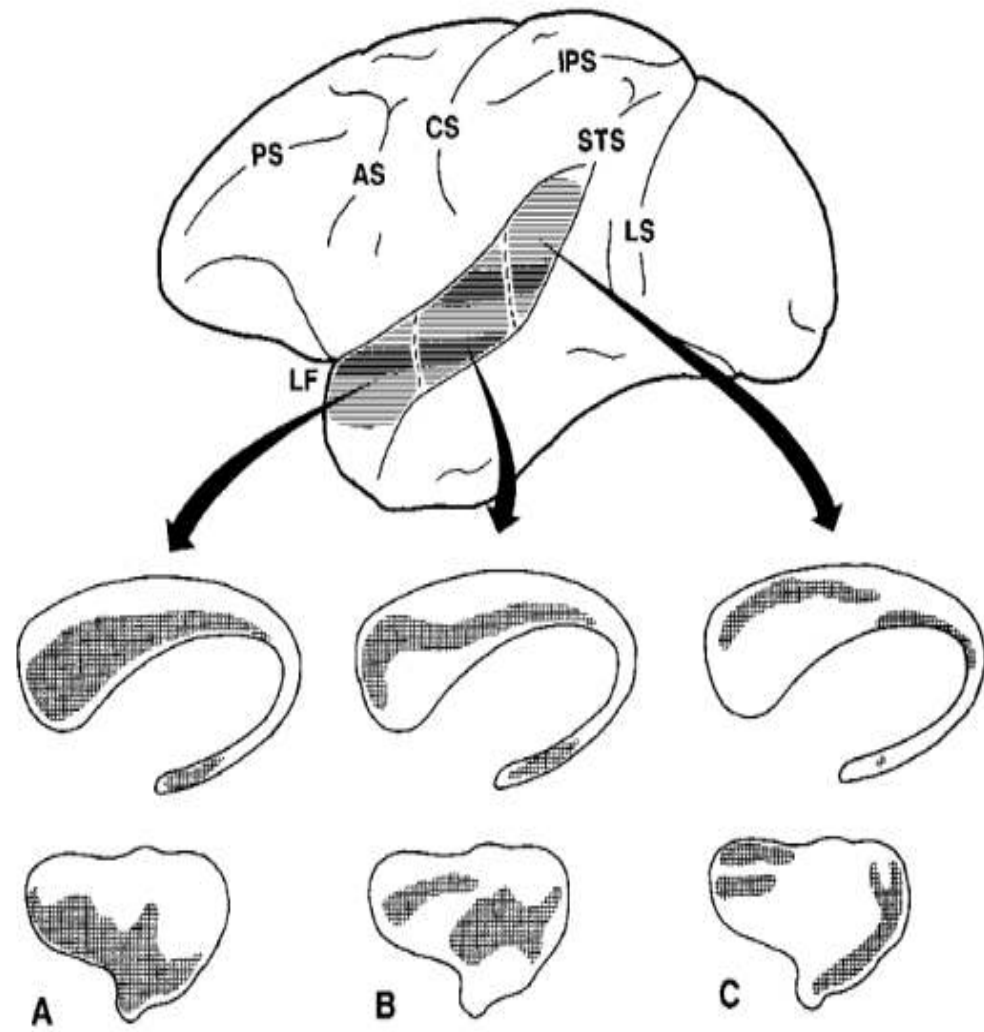
- **Dorsal striatum** contribute to learning new categories thanks to its many-to-one projections from **distributed cortical networks** that subserve:
  - motor planning
  - **task-relevant sensory processing**
  - **feedback processing**
  - decision-making
- Dorsal striatum involved in auditory category learning tasks using fMRI (Feng et al., 2019; Lim et al., 2019; Yi et al., 2016)
  - What **anatomical pathways** that connect the human auditory system to the striatum?
  - How do *subdivisions* of dorsal striatum **functionally contribute** to category learning?



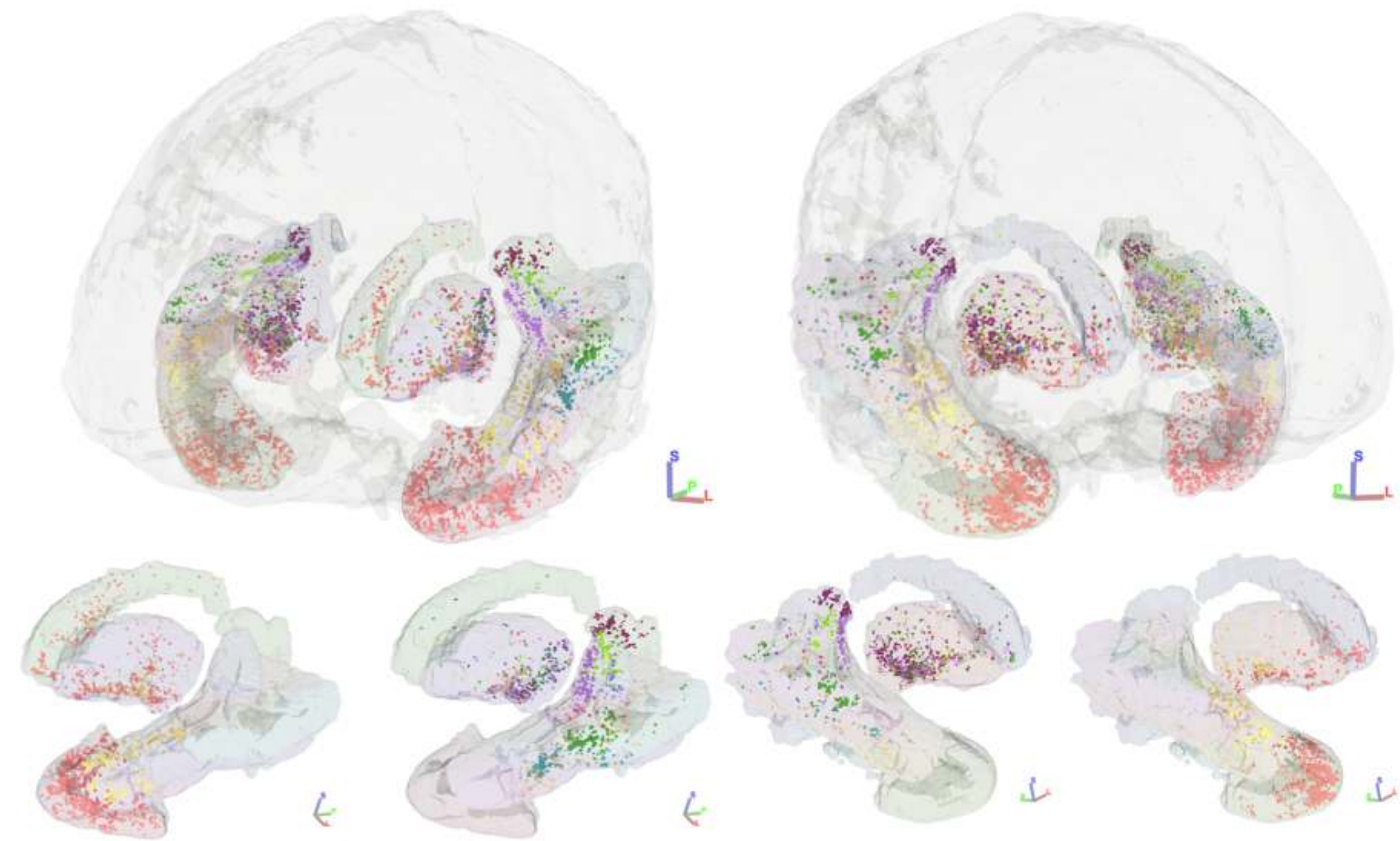
Lim et al., 2014

This work is actively supported by K01-DC019421 (PI: Sitek)

# Similar connectivity patterns in humans and macaques



Macaque (Yeterian & Pandya 1998)

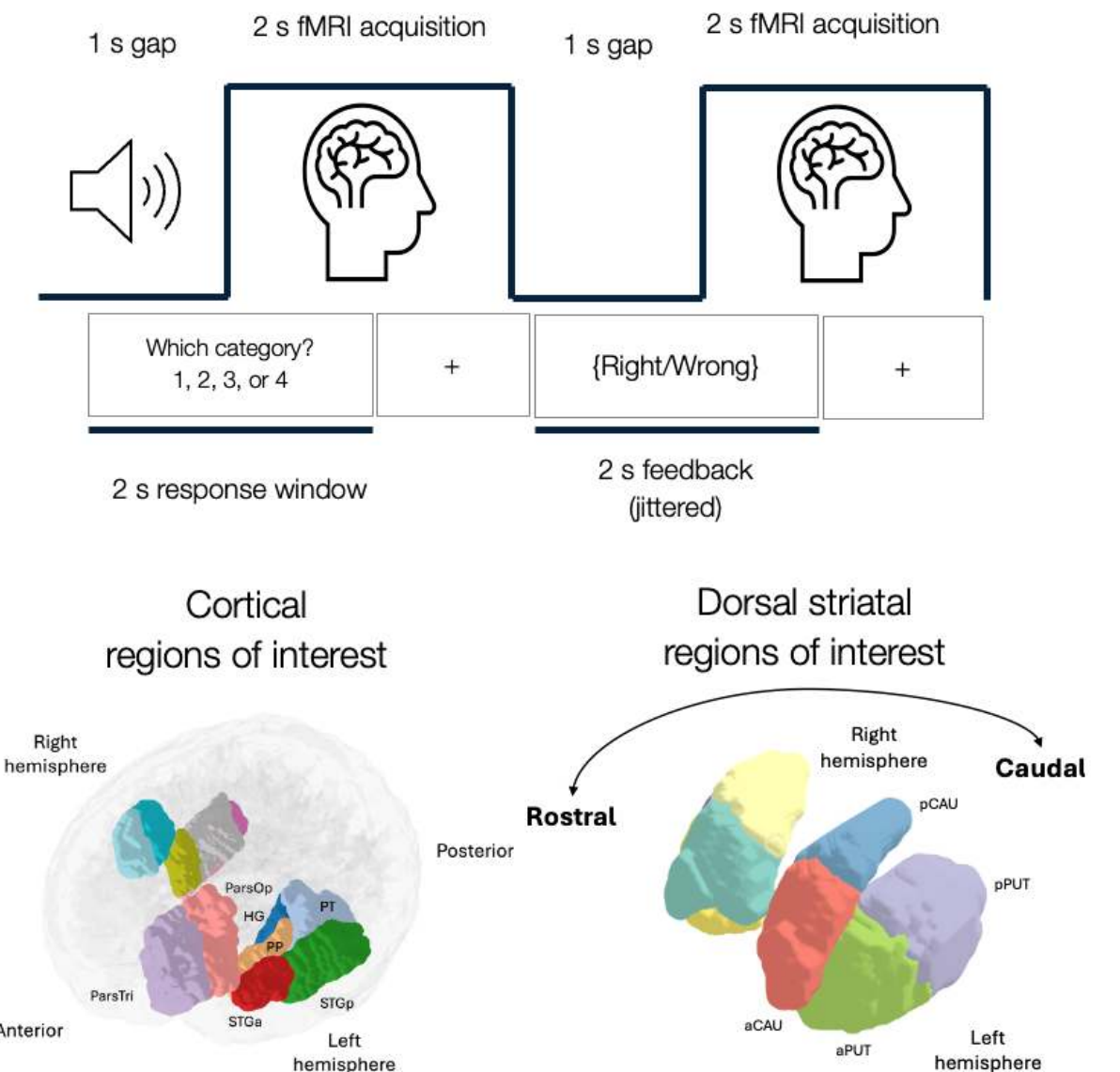
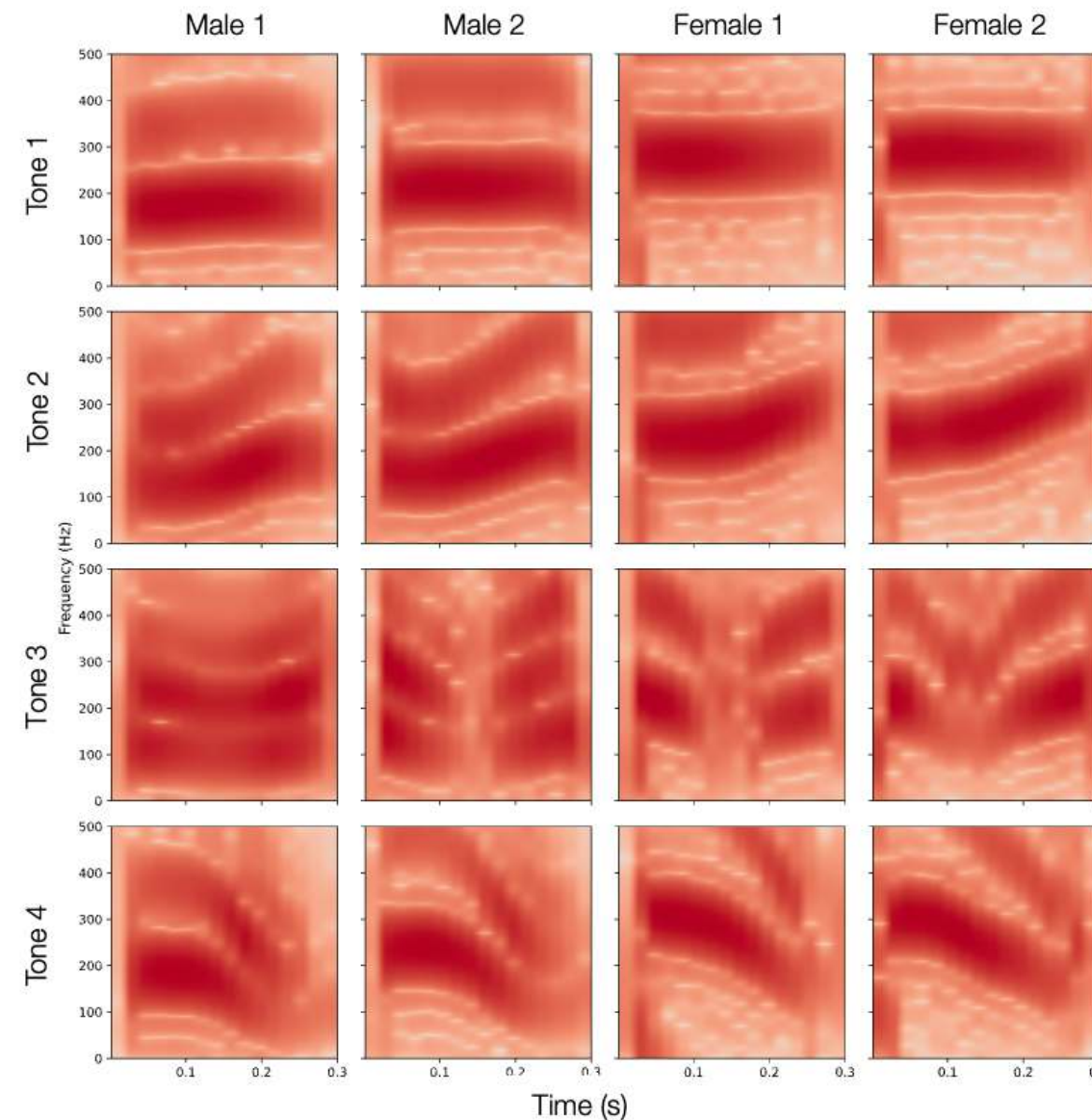


Human (Sitek et al. 2022 preprint, in revision)



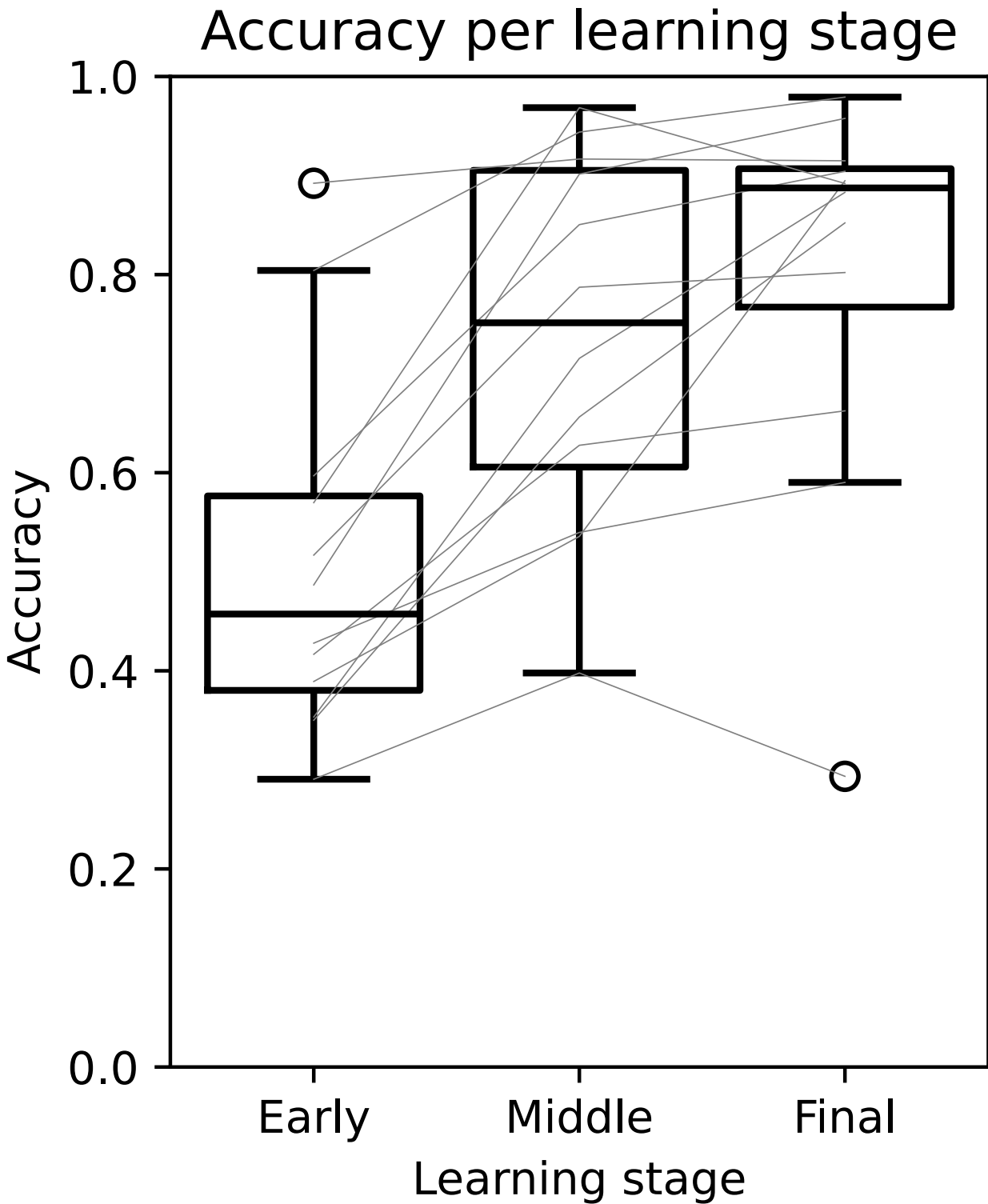
# How do humans learn new sensory categories?

- Auditory category learning task with Mandarin tone stimuli<sup>1</sup>
- 12 **non**-Mandarin-speaking participants (18–30 years old)
- 7 Tesla high resolution functional MRI (1.5 mm isotropic voxels)



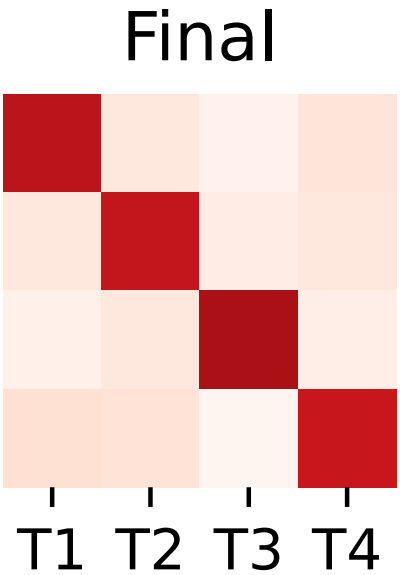
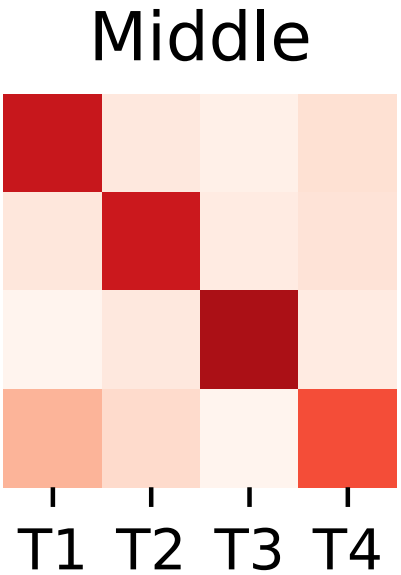
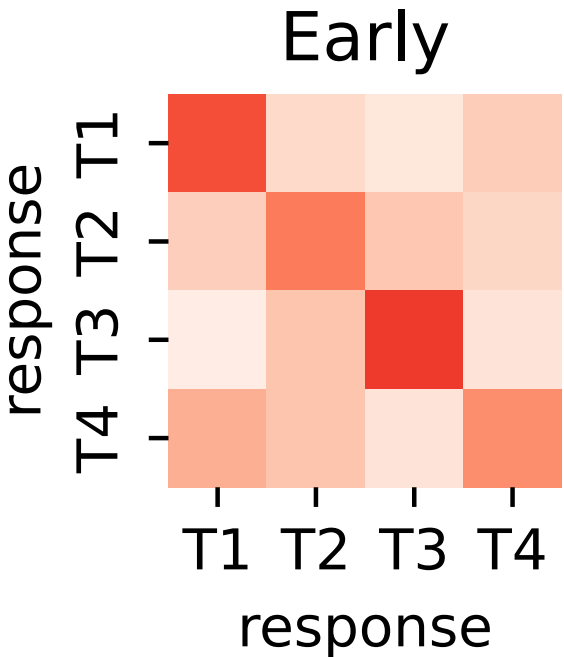
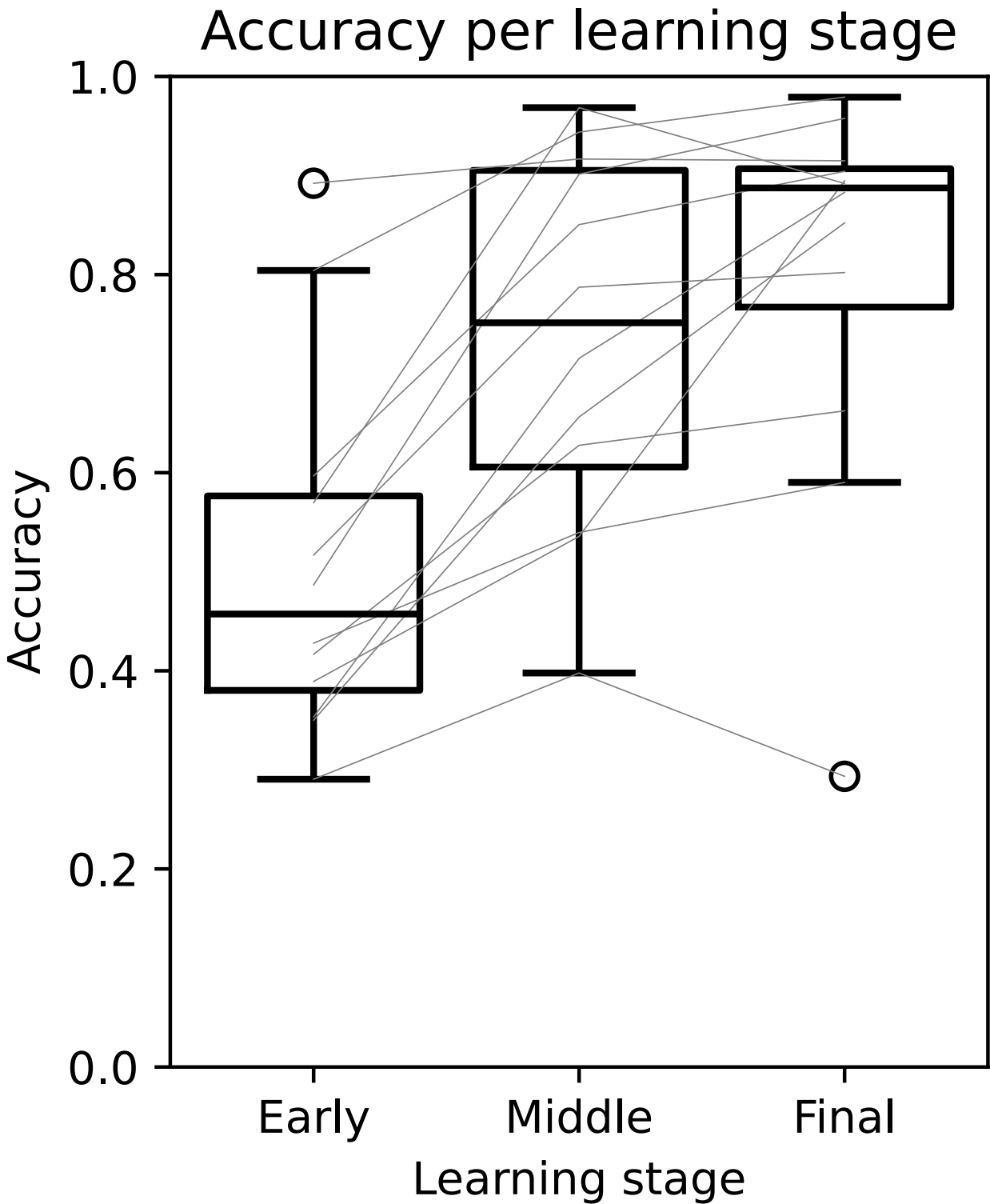
1. Yi 2016; Reetzke 2018; Feng 2019; ...

# Successful learning of new auditory categories in MRI



Sitek et al. (in revision)

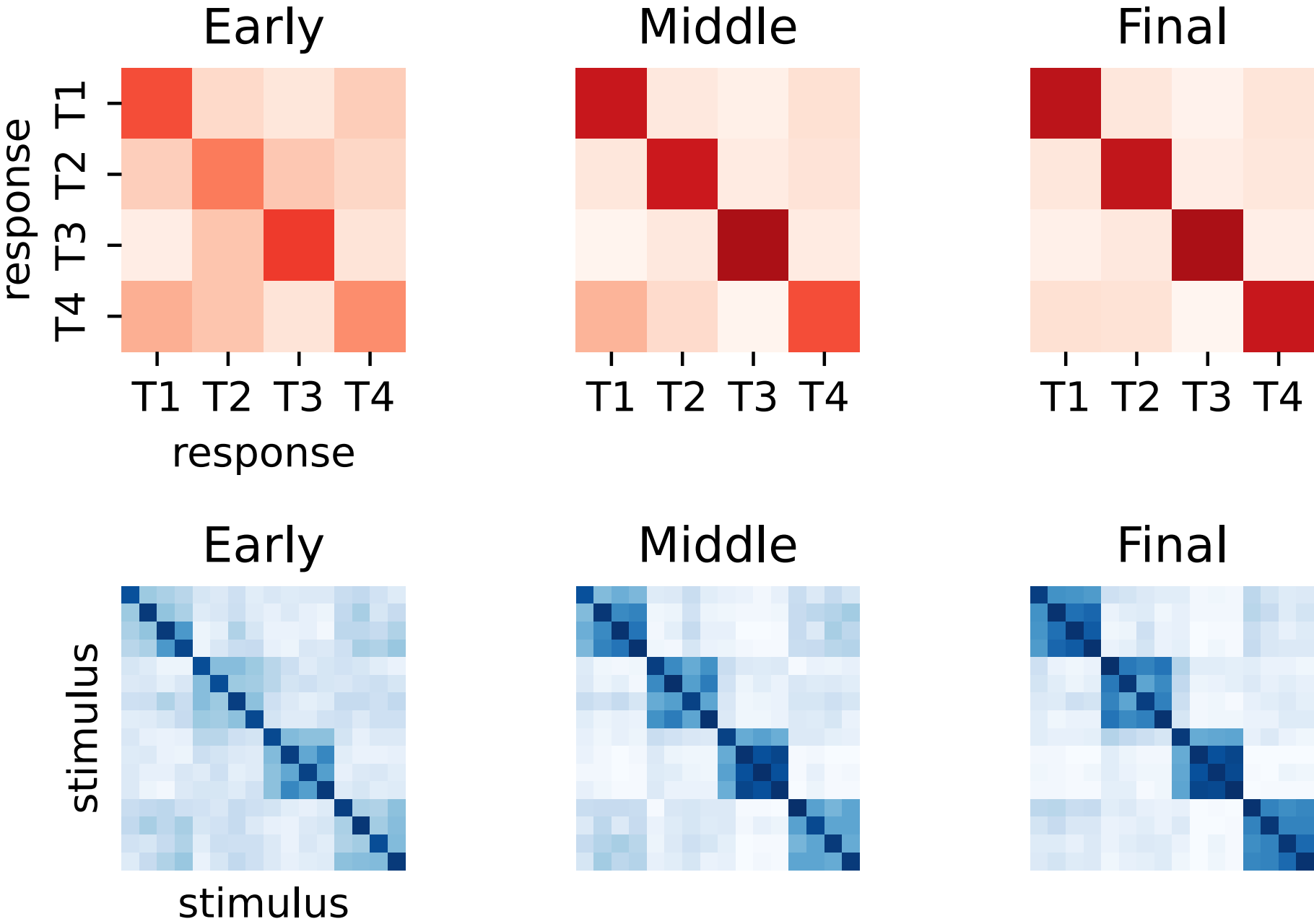
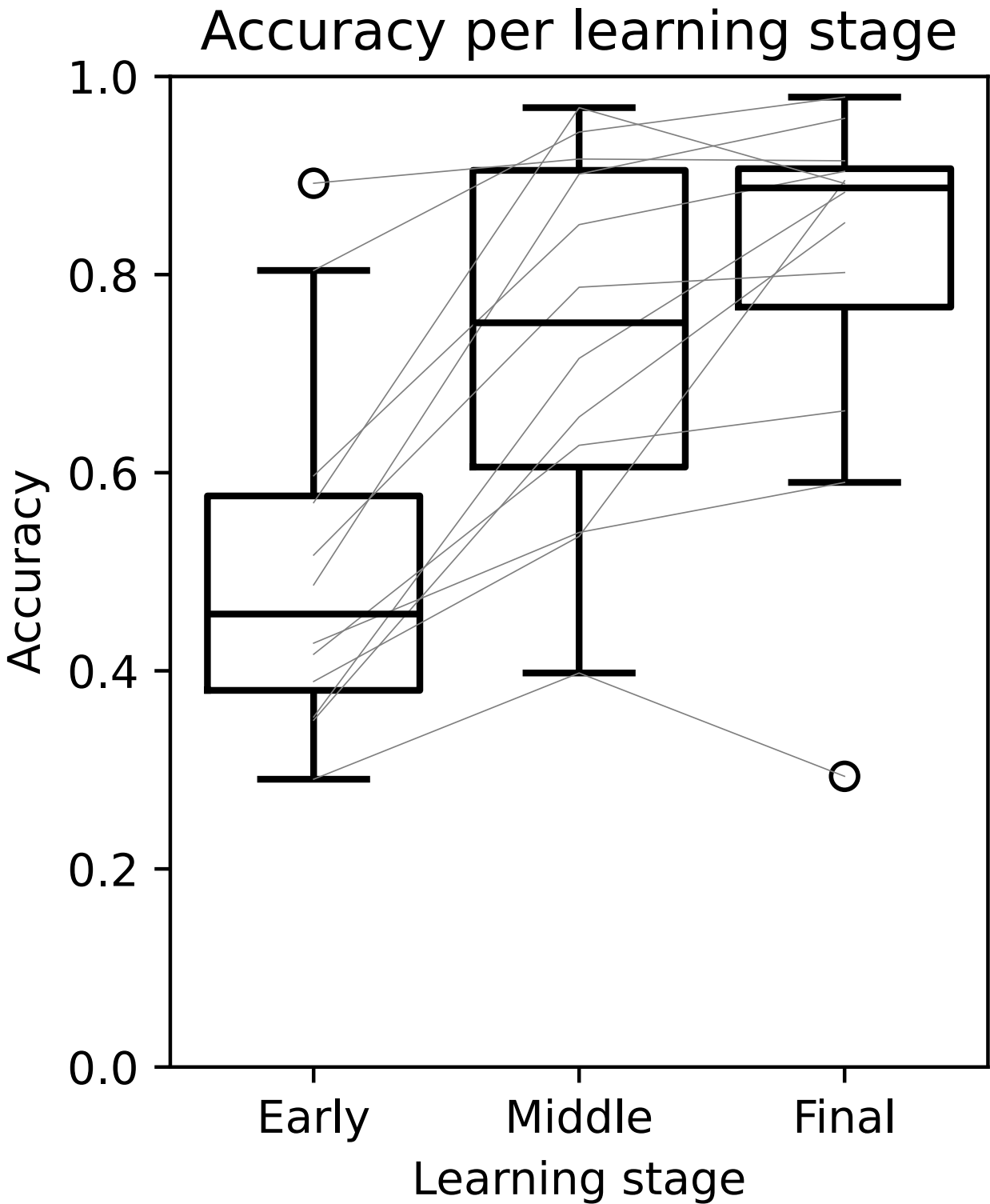
# Successful learning of new auditory categories in MRI



Sitek et al. (in revision)



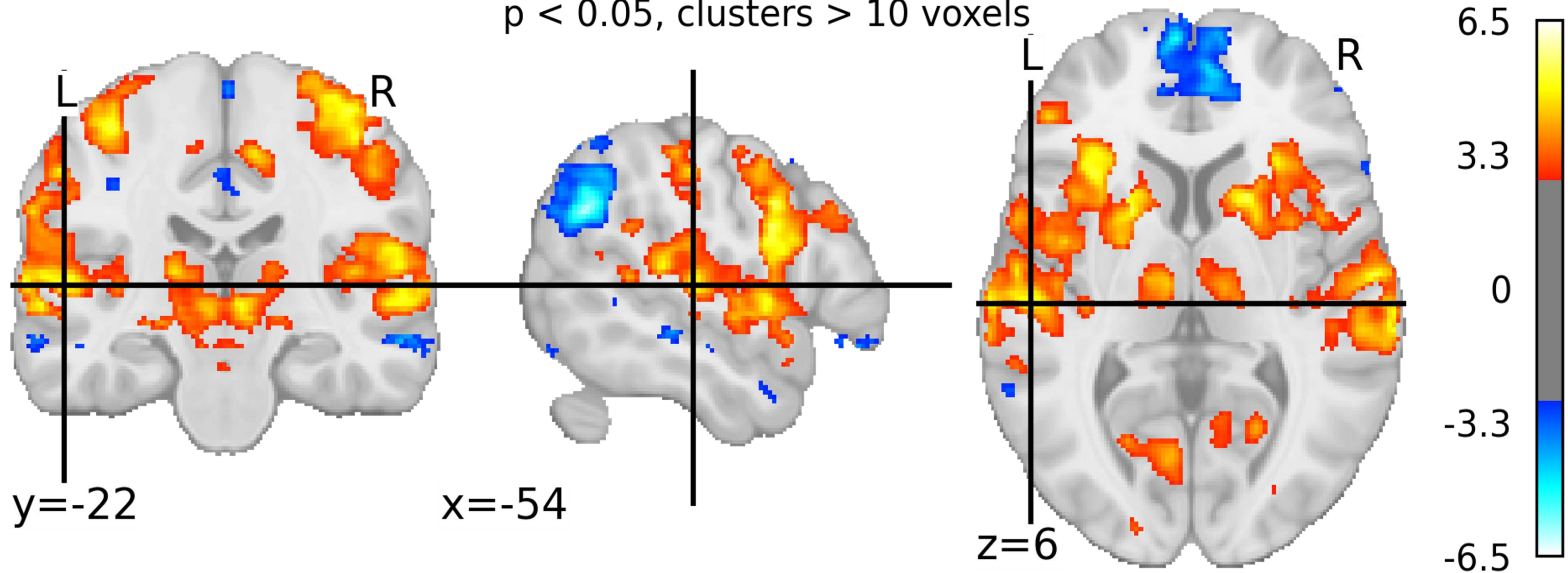
# Successful learning of new auditory categories in MRI



Sitek et al. (in revision)

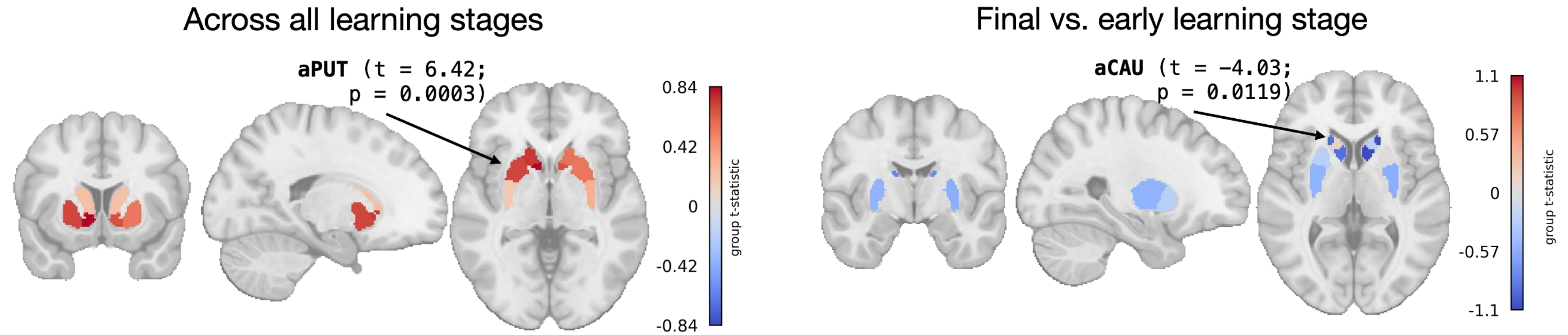
# Auditory fMRI responses to sound presentation

sound > baseline FDR-corrected z map  
 $p < 0.05$ , clusters > 10 voxels



Sitek et al. (in revision)

# fMRI responses to feedback (“correct” vs. “wrong”)



## Striatum feedback responses

- All regions showed positive responses in *early* learning stage
- But only **anterior putamen** *maintained* positive responses across all runs
- Only **anterior caudate** *significantly decreased* responses from early to final learning stages

Sitek et al. (in revision)



# Auditory categorization in the human brain

We identified putative cortico–striatal pathways for auditory decision-making

- Primary/secondary auditory cortex connects more strongly with *posterior* dorsal striatum
- Non-invasive human connectivity is similar to invasive tracing in non-human primates (Yeterian & Pandya 1998)

Dynamics of learning and categorization in dorsal striatum

When presented with feedback:

- **Anterior caudate** responds highest early on
- **Anterior putamen** responds across learning stages

**Putamen and caudate *subdivisions* have unique functional and connectivity profiles**

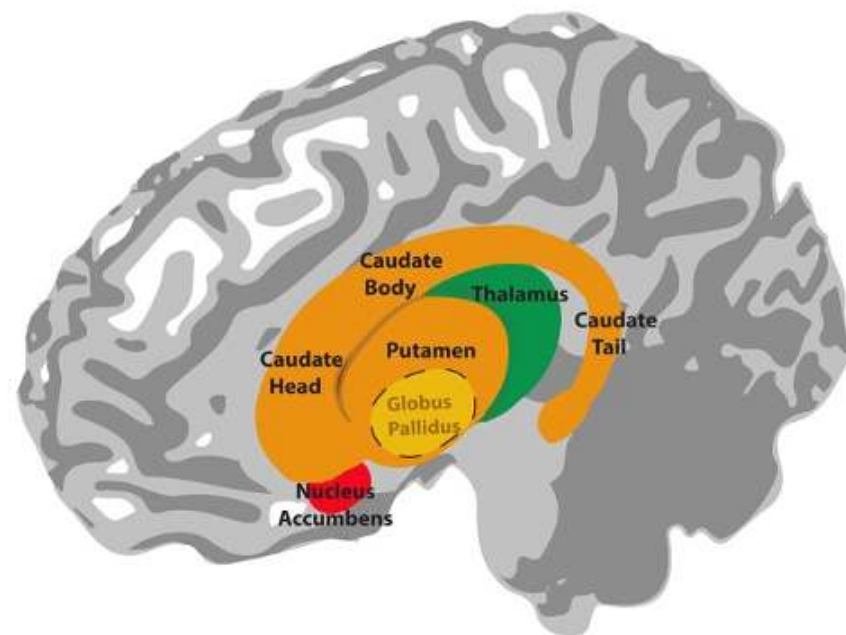
Sitek et al. (in revision)



# Where we're going next

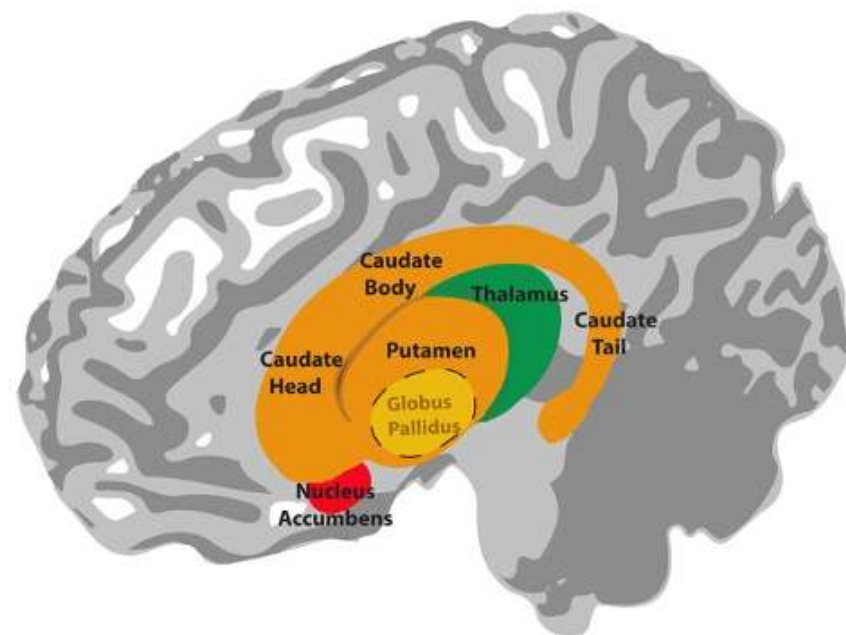


# Future research directions



***Sensory and motor  
speech learning***

# Future research directions

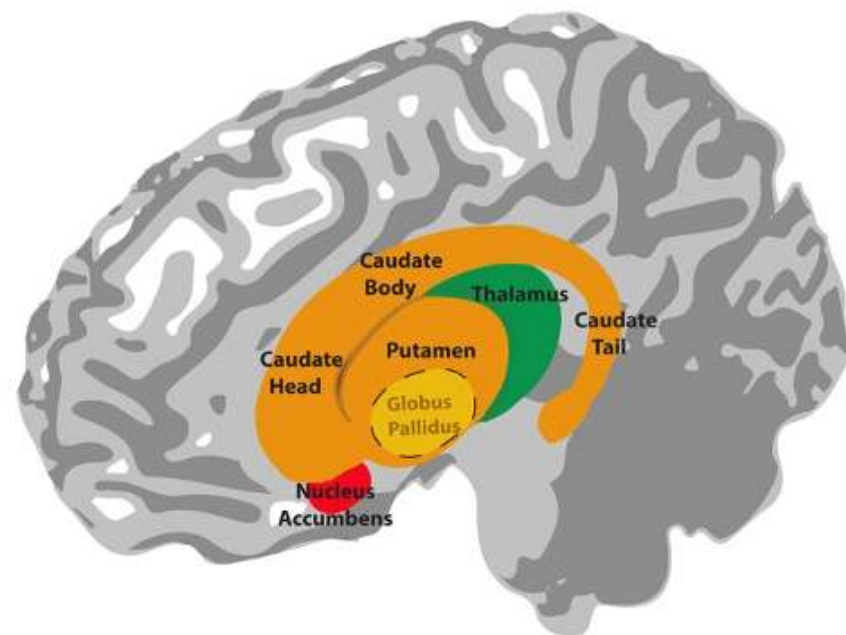


***Sensory and motor  
speech learning***



**Hearing our own  
speech**

# Future research directions



**Sensory *and* motor  
speech learning**



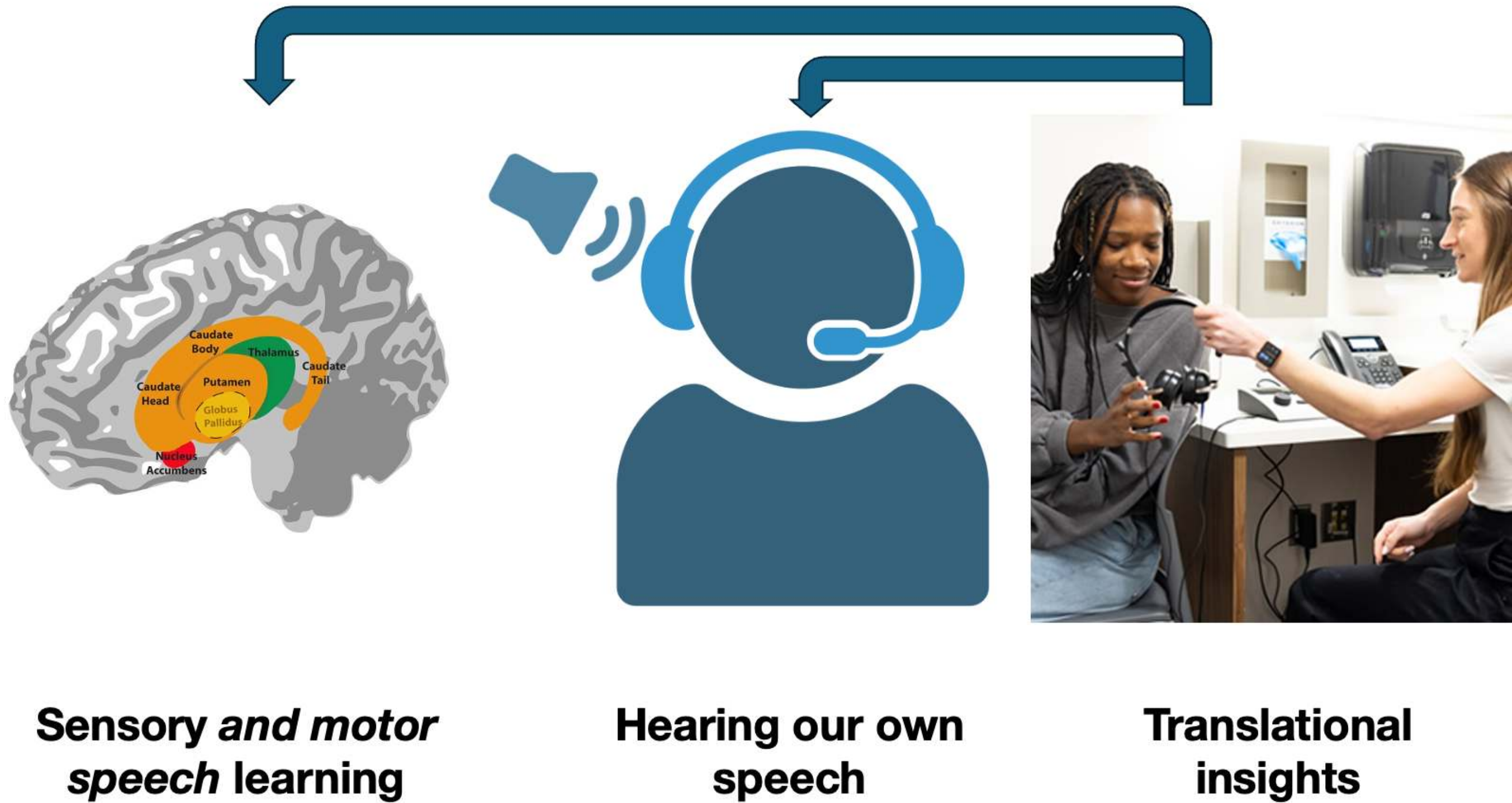
**Hearing our own  
speech**



**Translational  
insights**

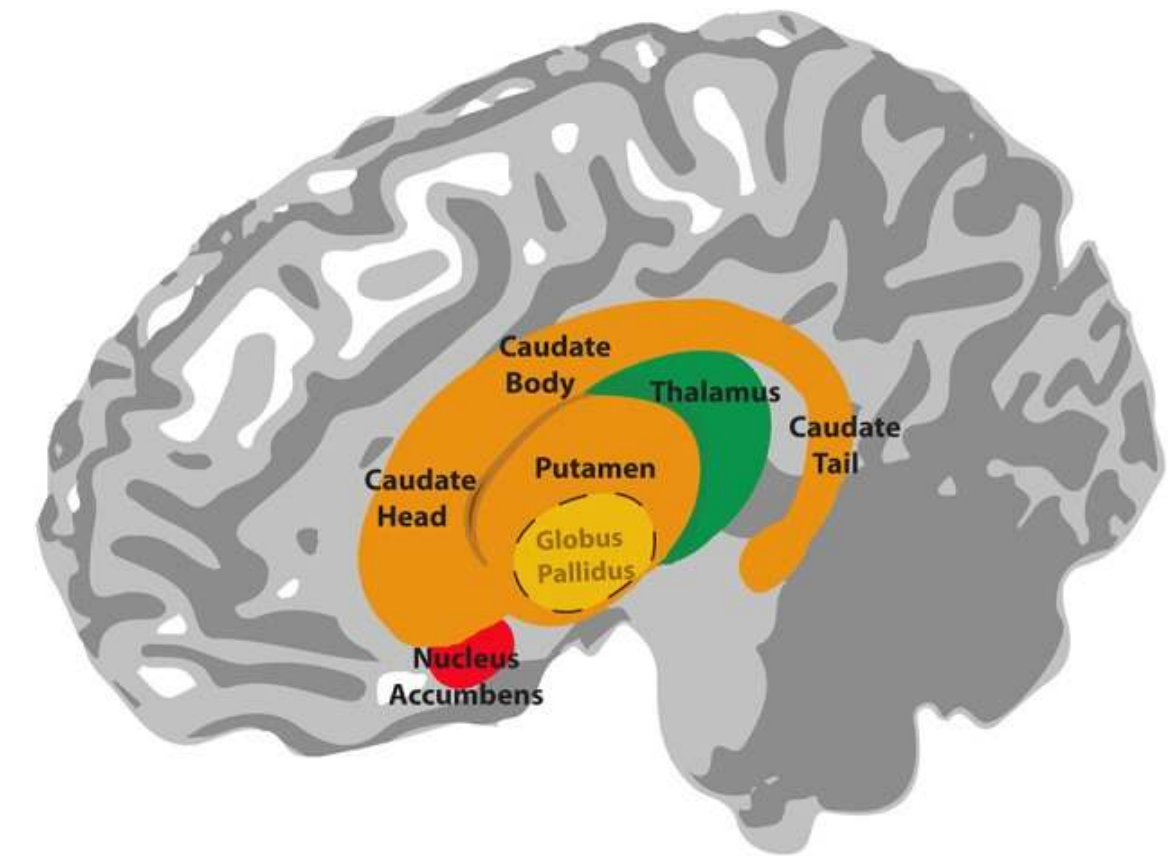


# Future research directions



# Sensory *and motor speech* learning: Shared striatal mechanisms?

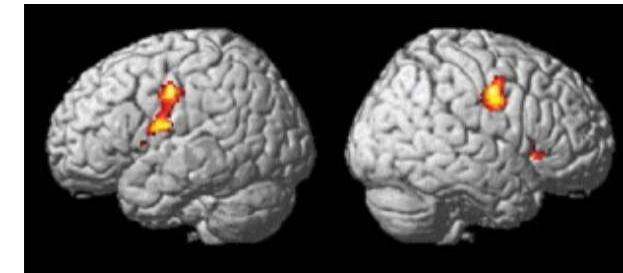
- Dorsal striatum (caudate and putamen) is critical to auditory learning (Sitek et al., in revision) *and* motor learning
- Are the neural implementations shared across perceptual and motor learning **in the context of speech and language?**
- How are shared mechanisms affected in *stuttering* or *Parkinson's*, and can they **inform therapeutic opportunities?**



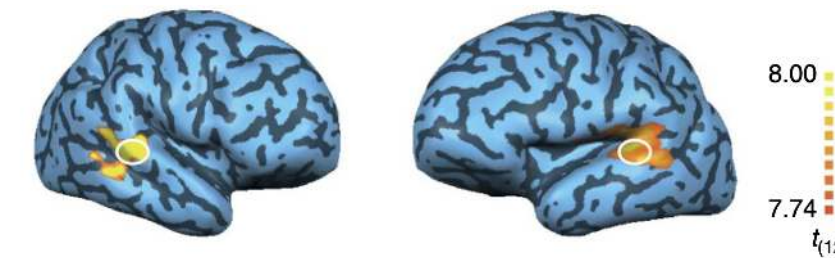
Lim et al., 2014

# Hearing our own speech: *Where* are motor signals integrated in auditory cortex?

- Auditory cortex is a hub for motor integration
  - But fMRI often shows *increased* BOLD to self-generated sounds, vs. *suppressed* M/EEG
  - Are different **laminar** responses the explanation?
- Behroozmand et al. (2015)



We can look at intra-cortical circuits with ultra-high field 7T fMRI

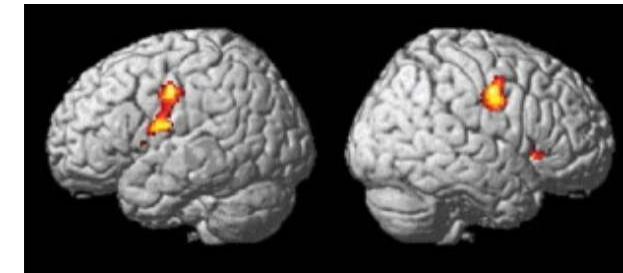


Reznik et al. (2015)



# Hearing our own speech: *Where* are motor signals integrated in auditory cortex?

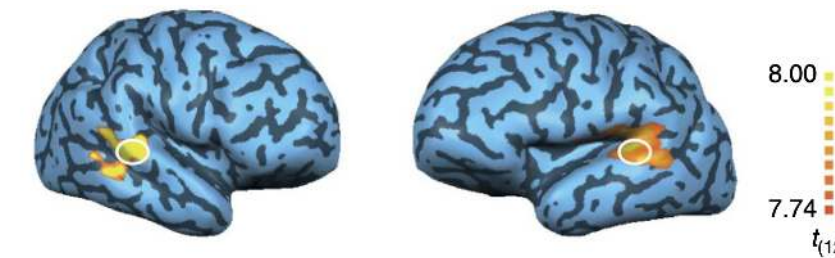
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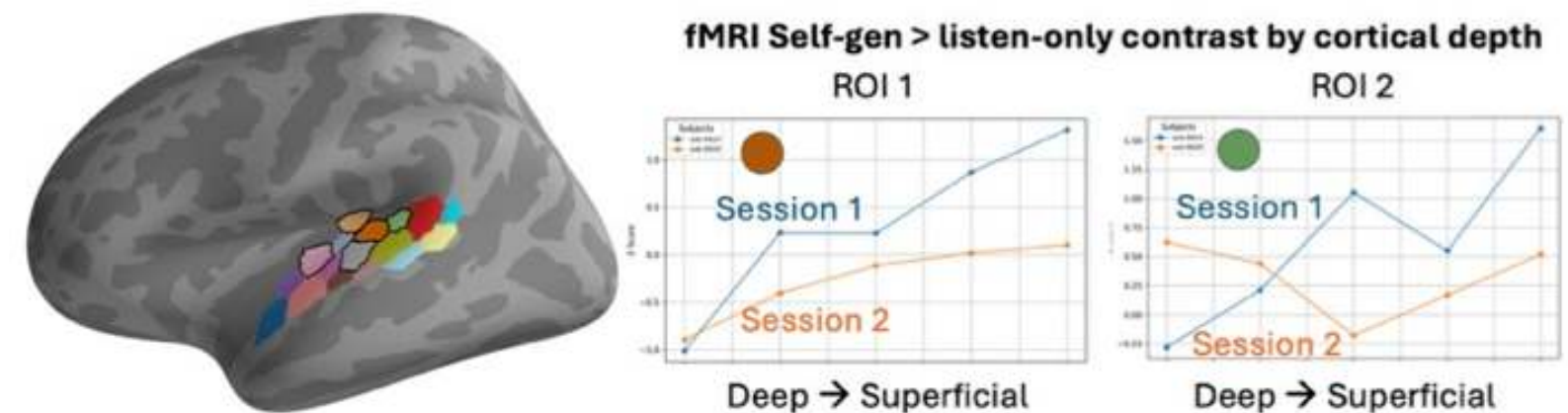
Behroozmand et al. (2015)

## We can look at intra-cortical circuits with ultra-high field 7T fMRI

- **Cortical depth profile** differences during self-generated vs. passive sound listening
- Higher increases at **superficial** cortical depths, suggesting increased **corticocortical** feedback
- Future work: Pinpointing feedback dysfunction in *stuttering*, *schizophrenia*, and other feedback processing disorders



Reznik et al. (2015)



Sitek et al. (OHBM 2025)

# Improving clinical translation at 3T with multi-echo fMRI

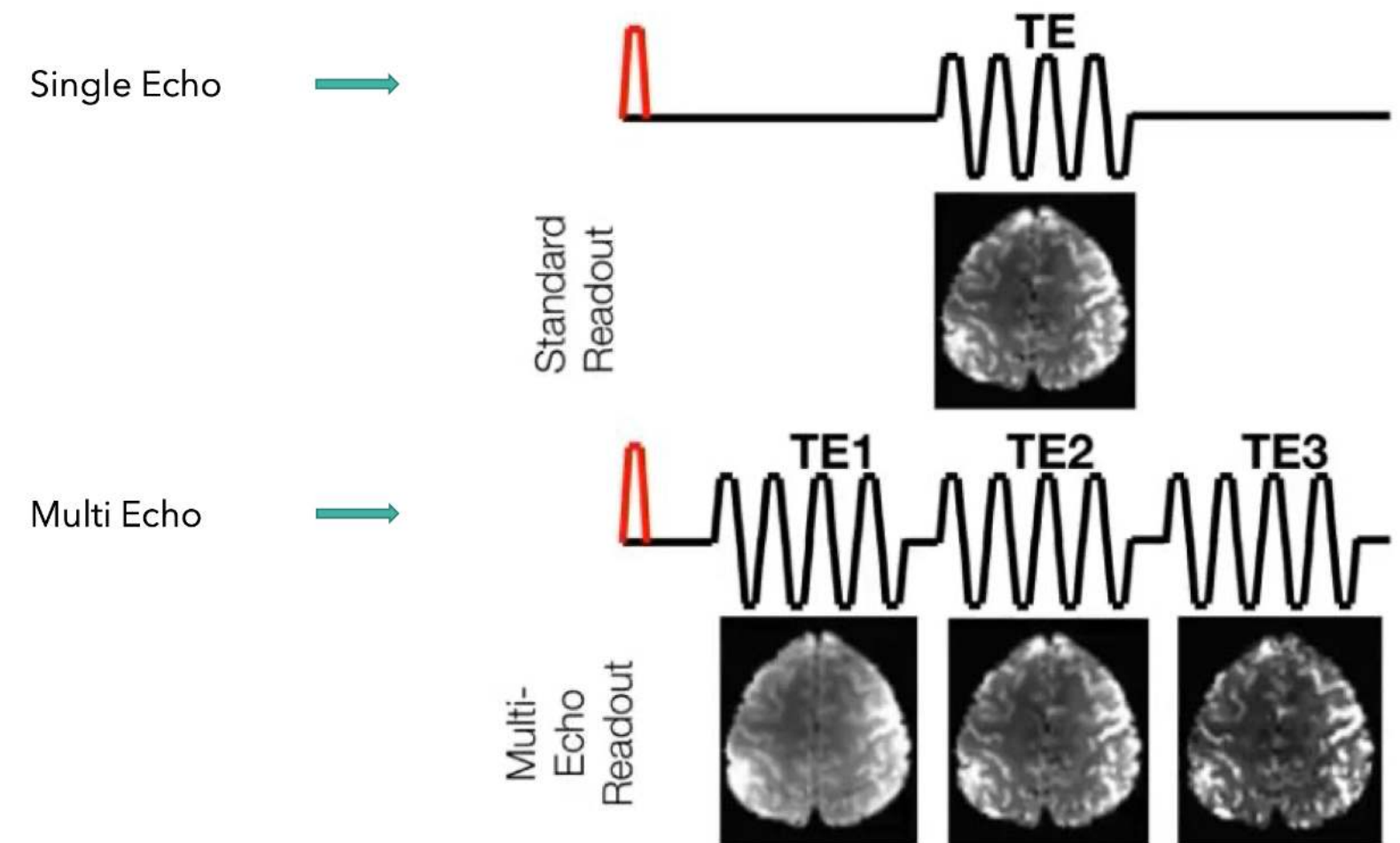
## Multi-echo fMRI improves subcortical (and cortical!) fMRI sensitivity

- Typical fMRI uses one readout (“echo”)
- But optimal echo time (TE) isn’t consistent across the brain
- Multi-echo fMRI allows *optimal combination* of different TEs across the brain

## Listening task in 3T fMRI

- 14 participants
- 10 minutes of pop-song listening
- 3 participants returned for precision-mapping session
- Multi-echo fMRI acquisition from UMinn CMRR

Medina et al. (submitted preprint); Medina, ..., Sitek (in prep.)



Dartmouth BIC

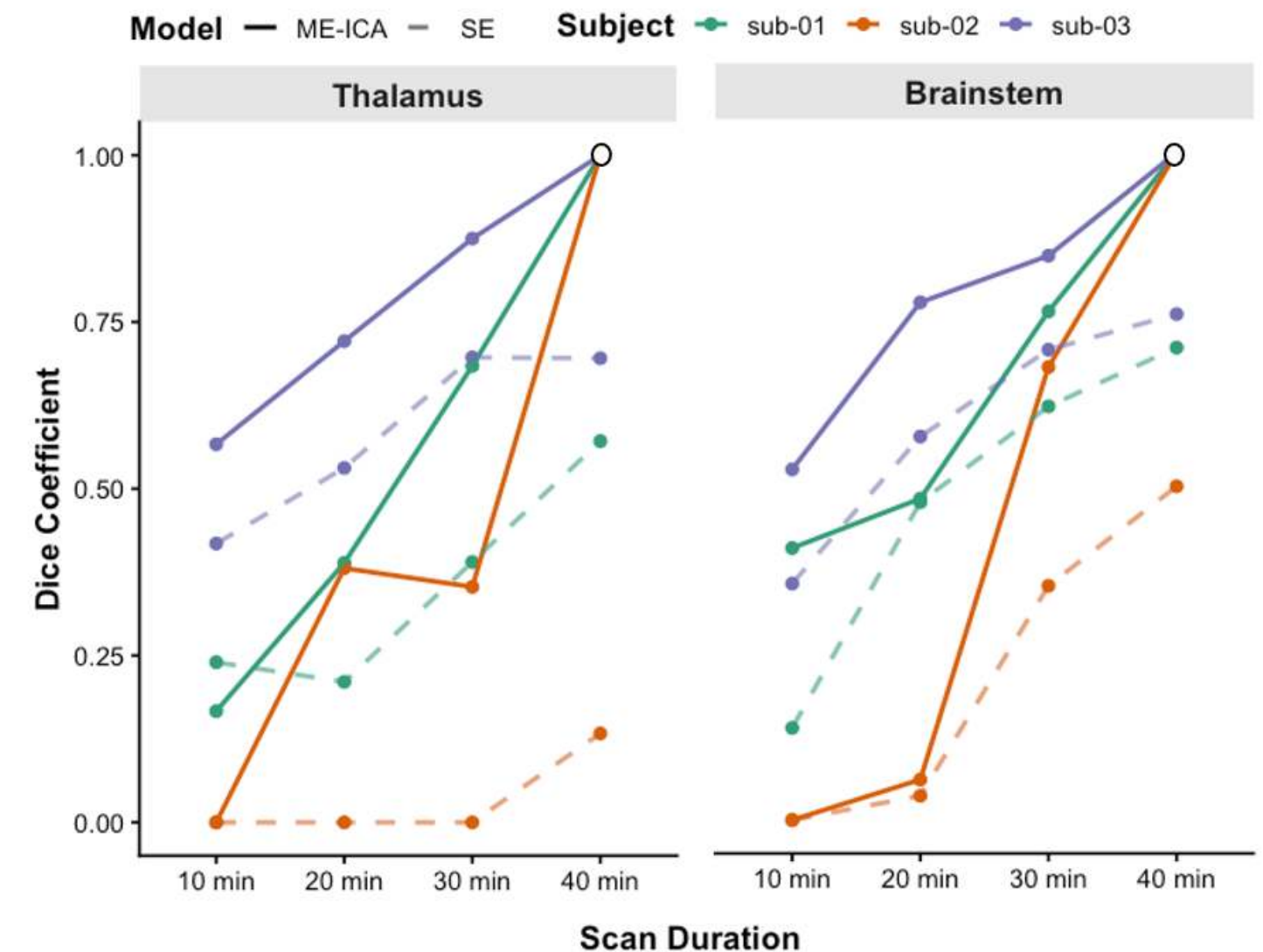
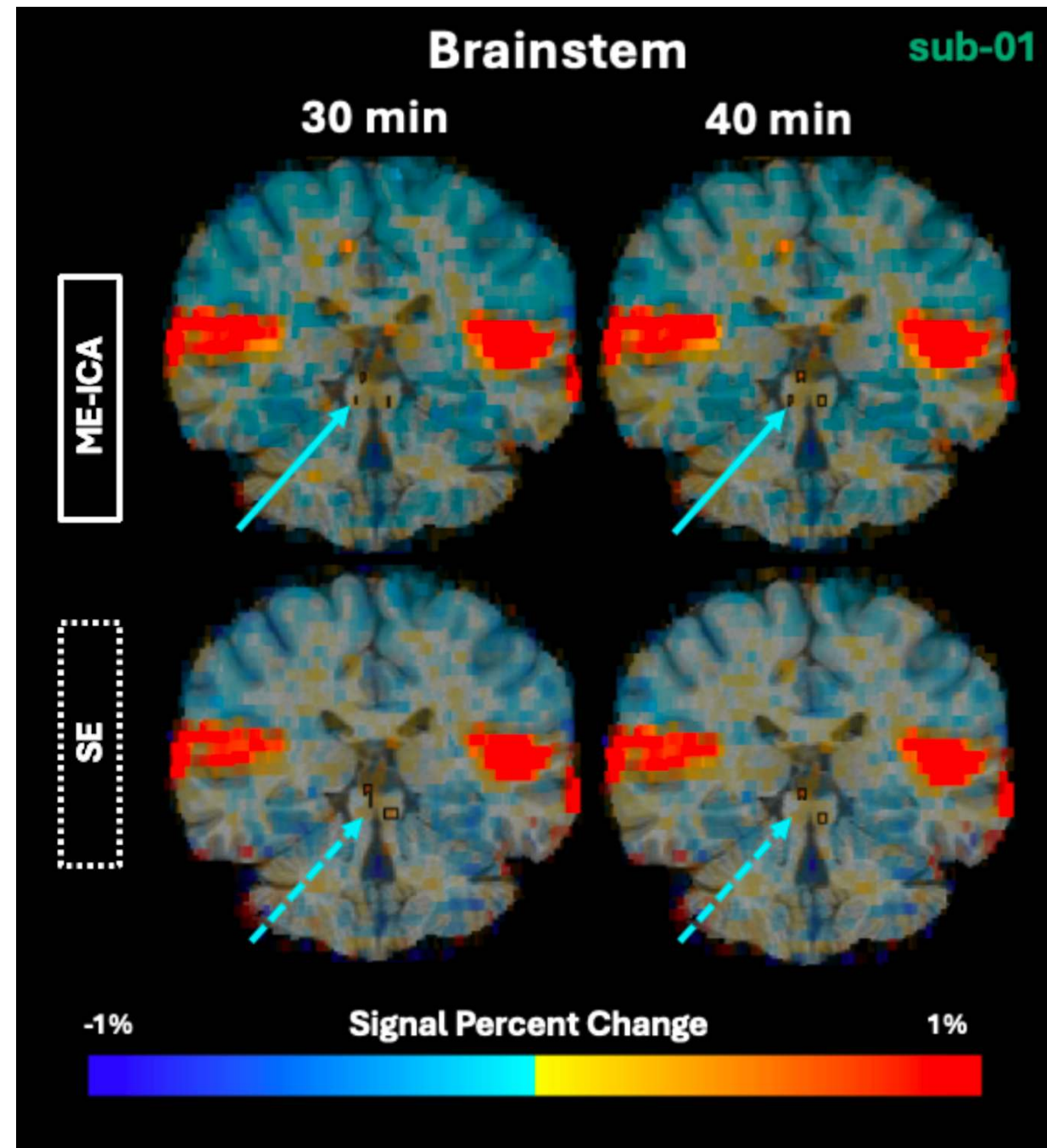




# Improved precision mapping with multi-echo fMRI

Multi-echo requires less scan time to identify auditory brainstem

Subcortical activations are stronger with multi-echo than single-echo fMRI



Medina et al. (submitted preprint); Medina, ..., Sitek (in prep.)





# Improving brainstem sensitivity at 3T with multi-echo fMRI

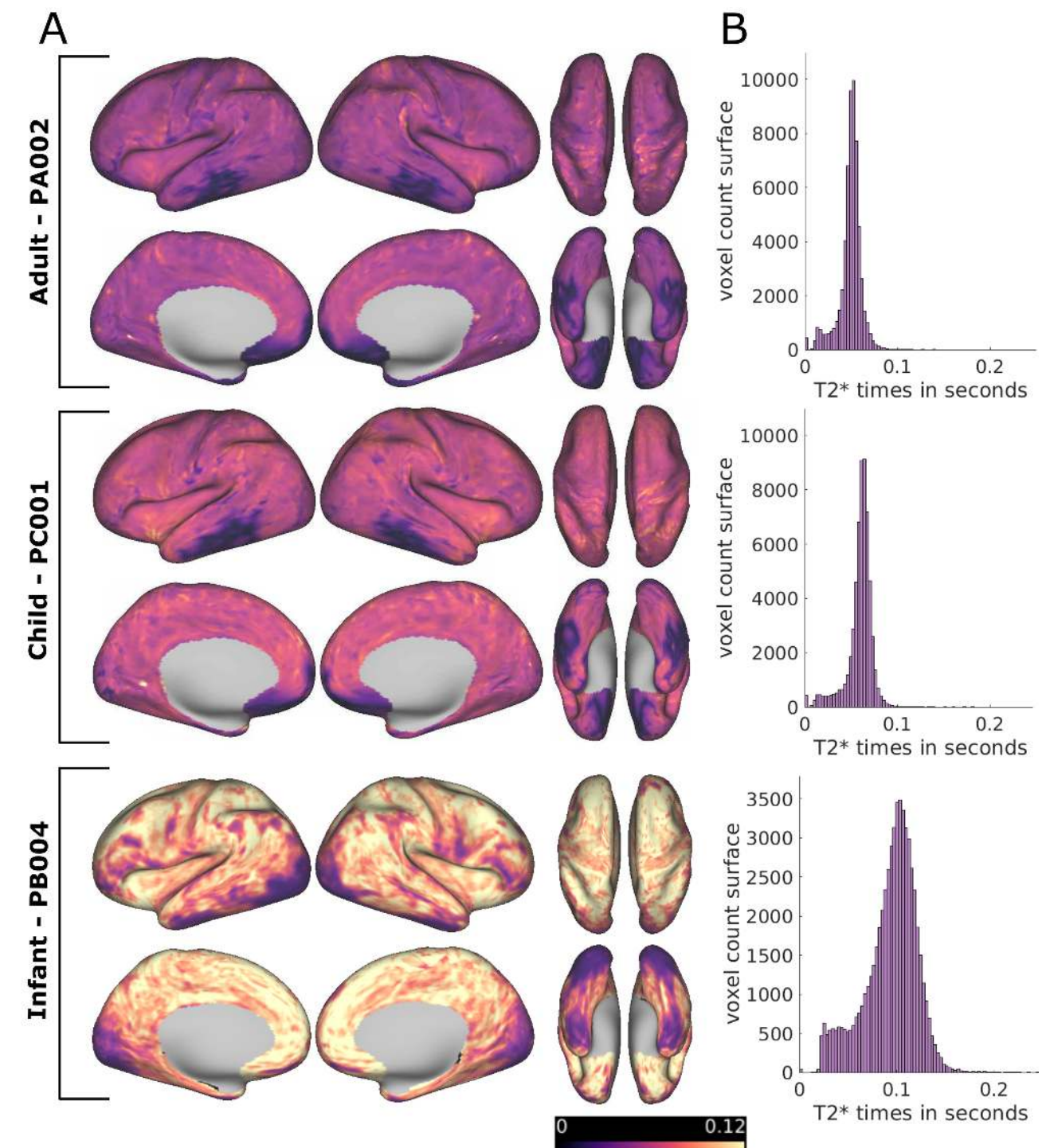
## Future applications: Auditory processing across the lifespan

### Auditory processing changes throughout adulthood

- How does speech-in-noise processing change in middle age and beyond?
- What neural patterns predict later cochlear implant success?

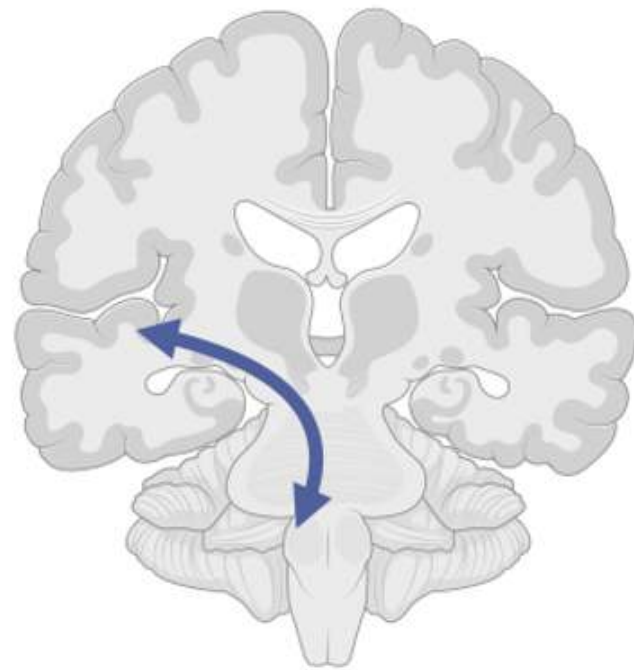
### Speech processing in the developing brain

- How do children make auditory category decisions, and how are they affected by noise?
- What compensatory neural pathways are available to children with developmental communication differences?

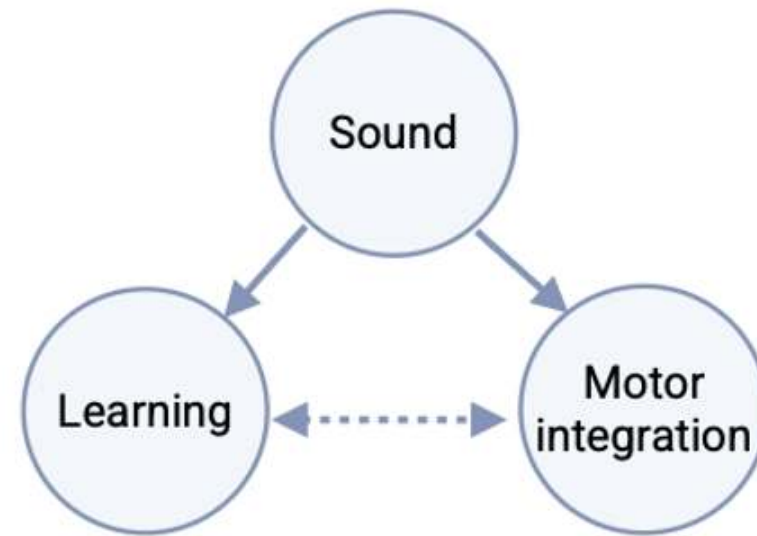


Moser et al. (2025)

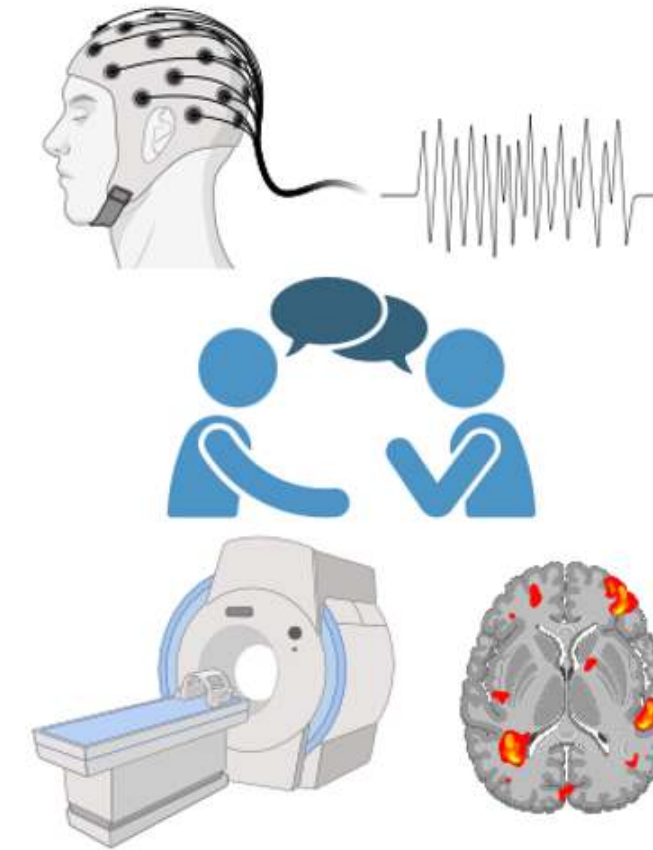
# A systems-level approach to human auditory neuroscience



**Auditory  
circuits**



**Speech and  
learning**



**Communication  
biomarkers**

# Acknowledgments

## SoundBrain Lab

- **Bharath Chandrasekaran**, Jacie McHaney, Kailyn McFarlane
- Nike Gnanateja (now UW Madison CSD)
- Casey Roark (now U New Hampshire Psychology)



**Northwestern:** Molly Bright, **Michelle Medina** (*BME PhD student*), **Gabby Butler** (*Cog Sci UG honors student*), Max Chen (*Cog Sci/Comp Sci undergrad*)

**UT Austin collaborators:** Abhra Sarkar, Noirrit Chandra (*UT Dallas*), Blake Moya (*industry*)

**Pitt:** Jay Bohland, Mandy Hampton Wray

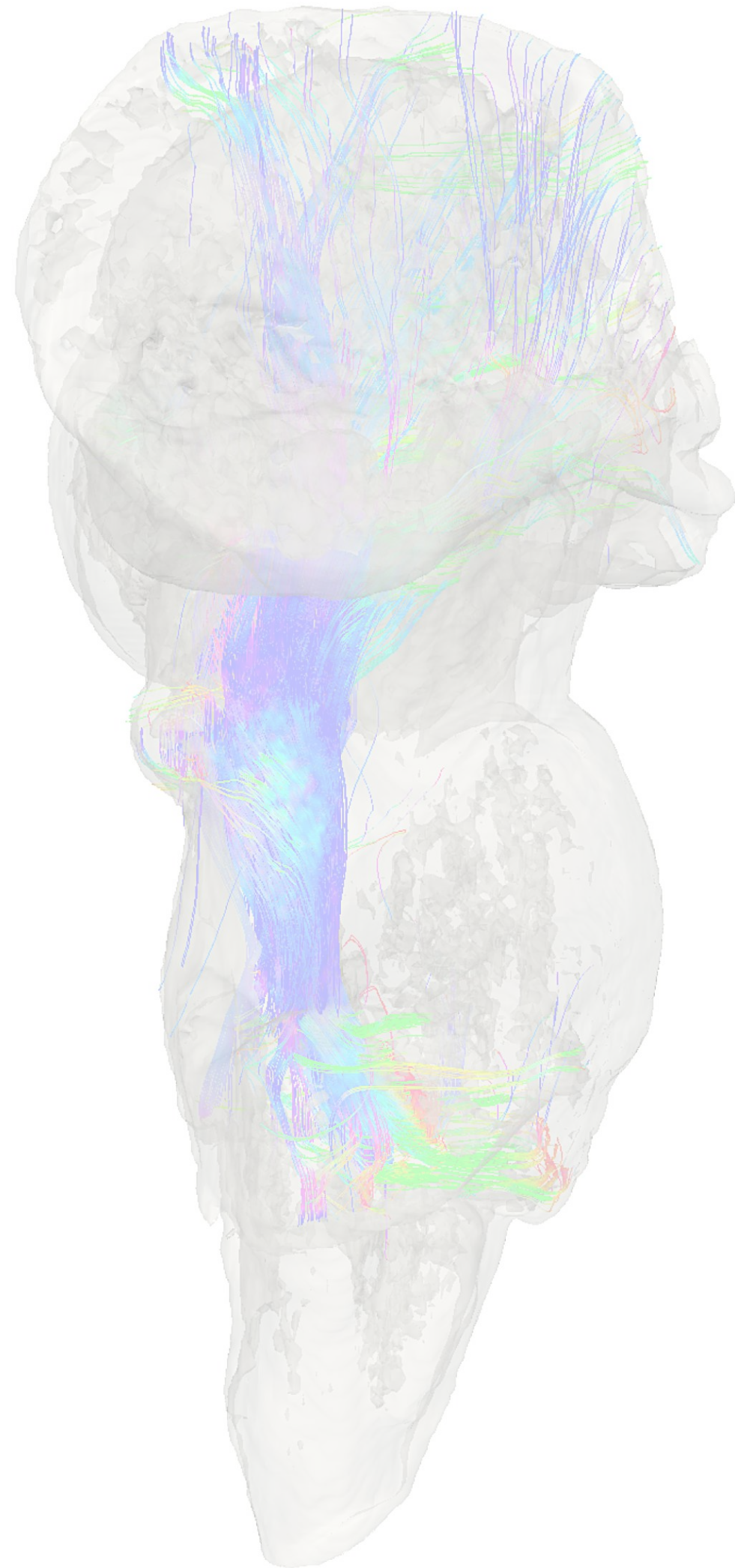
**Harvard/MIT:** Satra Ghosh (*PhD advisor*)

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- K01 DC019421 (2023–2027)
- R21 DC022906 (2025–2028)
- R01 DC020963 (2024–2027; PI: Bohland)



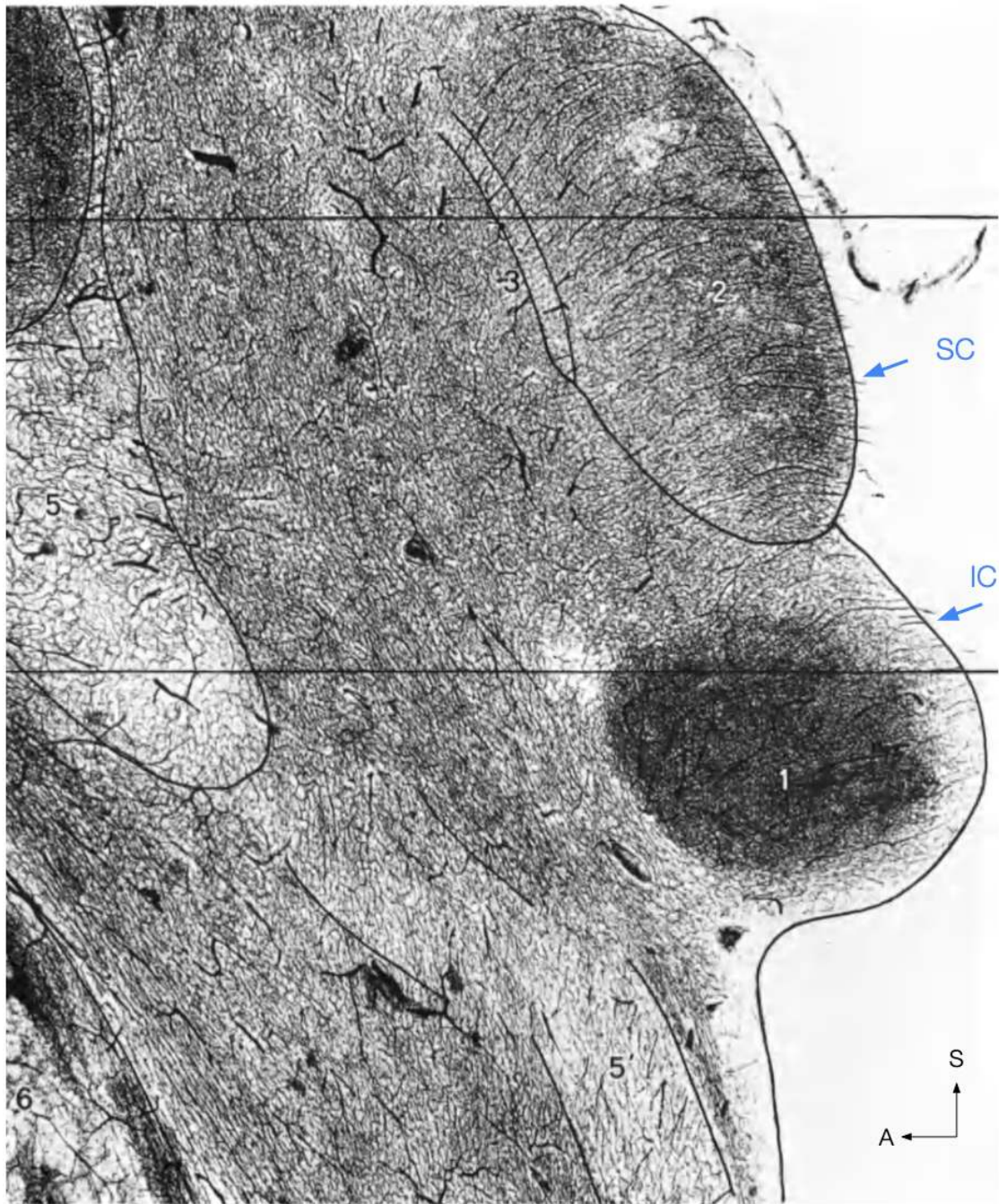
**Thank you!**



# Extra slides

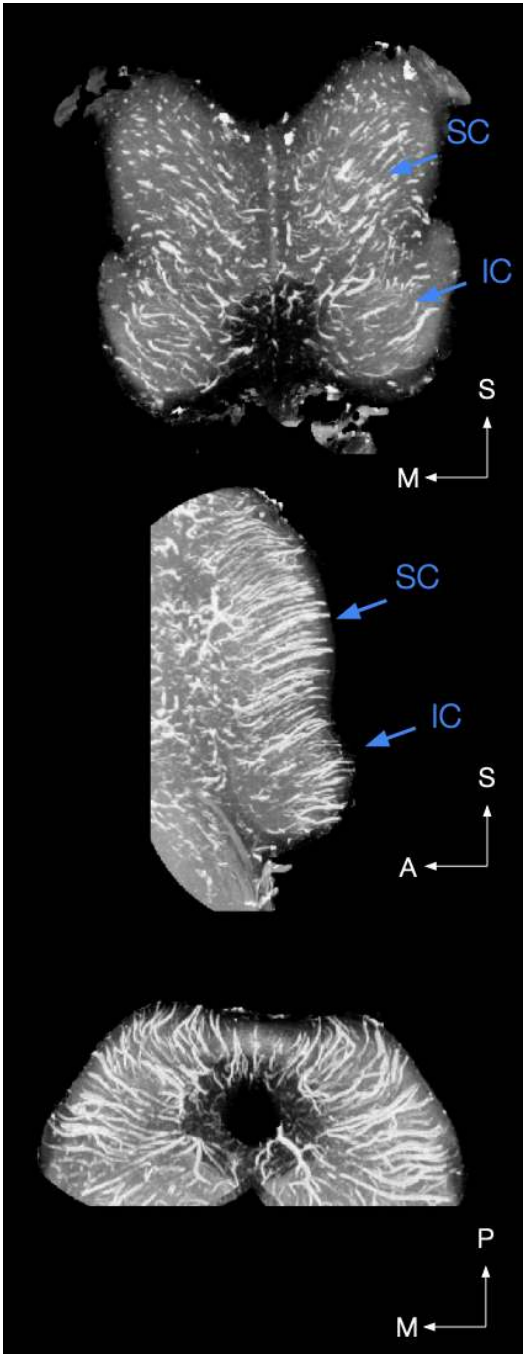


# How is the brainstem supplied oxygen?

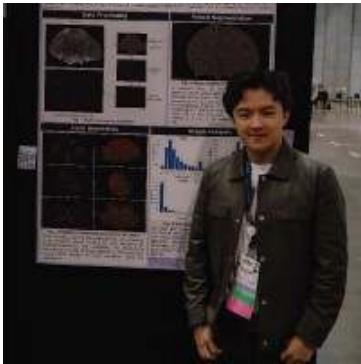
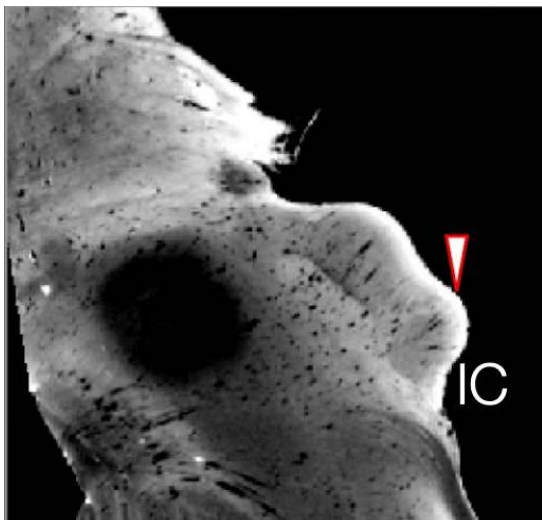


Vessel ink staining (Duvernoy 1978)

Xu et al. (ISMRM 2025); Xu, ..., S. Bollmann, *Sitek* (in prep.)

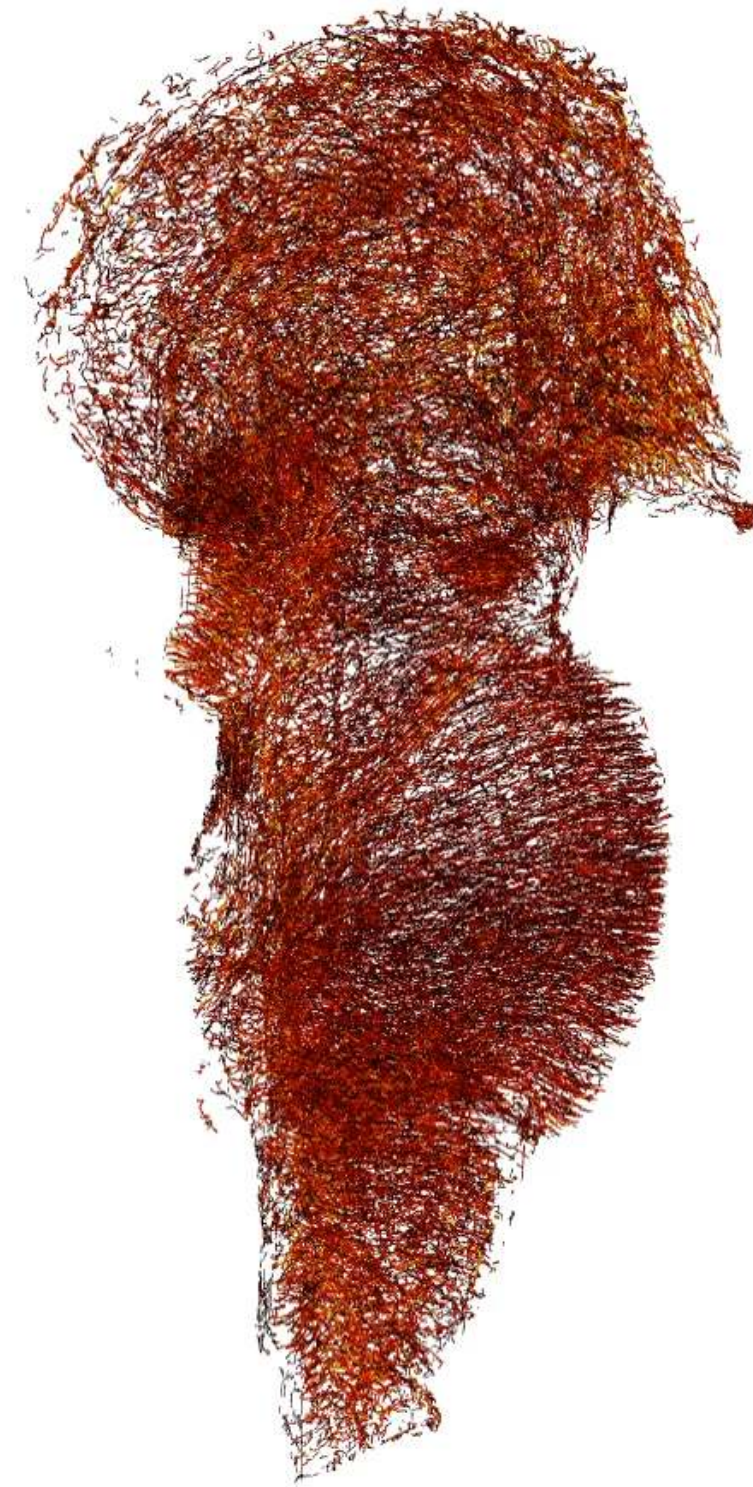


T2\*-w MRI max.  
intensity projection



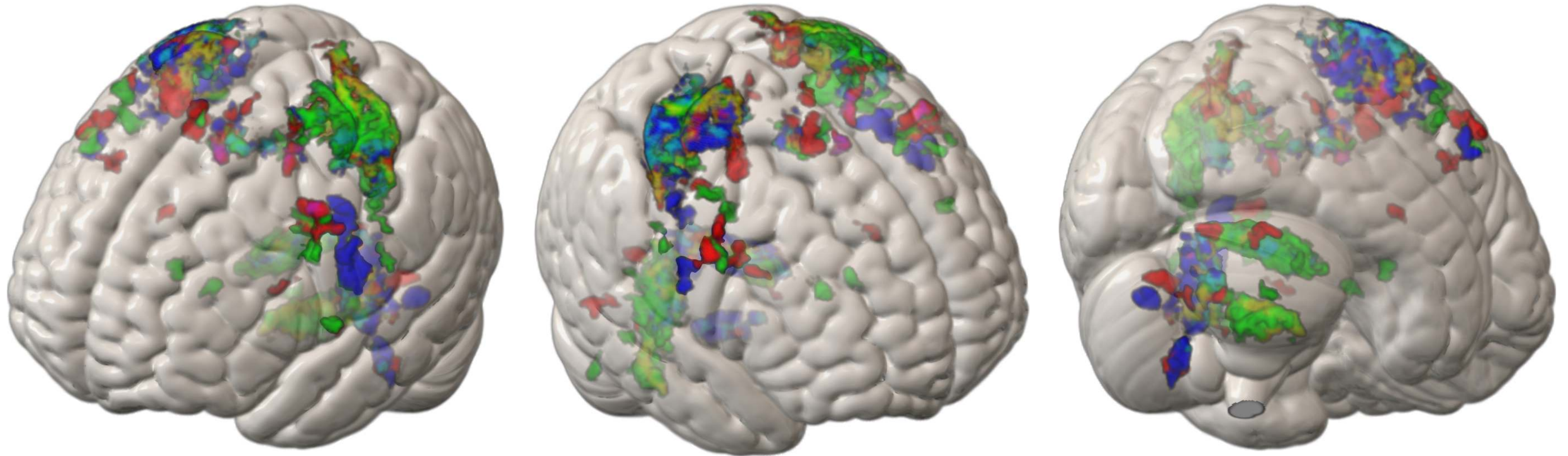


# How is the brainstem supplied oxygen?





# Finger motor representation in the human brain



Posterior mean dominance greater than 25% across voxels for the **right index (red)**, **left index (blue)**, and **middle fingers (green)**.

Moya et al. (in prep.)





# The developing auditory system

## How do we learn to hear in noisy environments?

- Adults: maturation of speech network<sup>1</sup> + compensatory networks<sup>2</sup>
- Children: fine-tuning speech network + undeveloped frontal networks
- Collected syllable-in-noise task fMRI at 3T in 35 typically developing kids
  - 5 speech-shaped noise contexts: Quiet and SNR = +8, 0, -2, and -6
  - Task: identify which of 4 syllables were presented



via The Telegraph

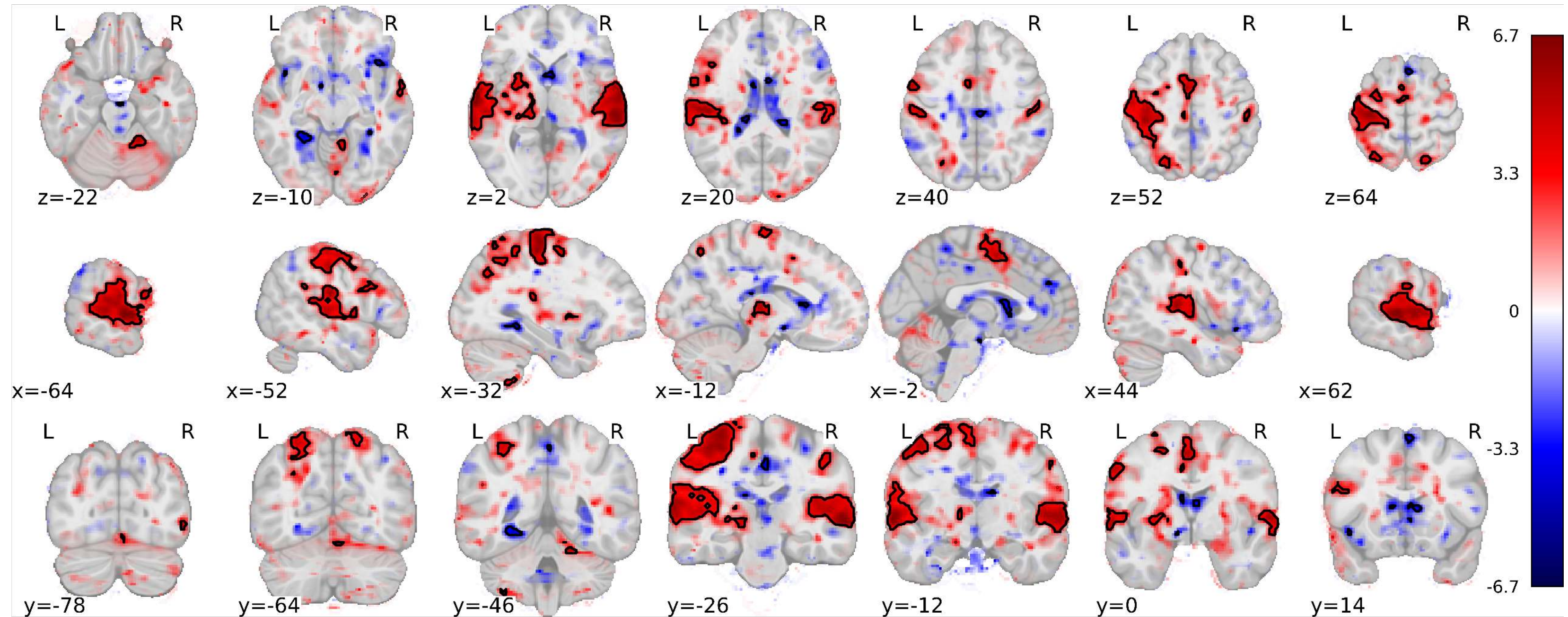
Sitek et al. (in prep.; SNL 2025)

1. Du et al. (2014)

2. Vaden et al. (2013); Eckert et al. (2016)



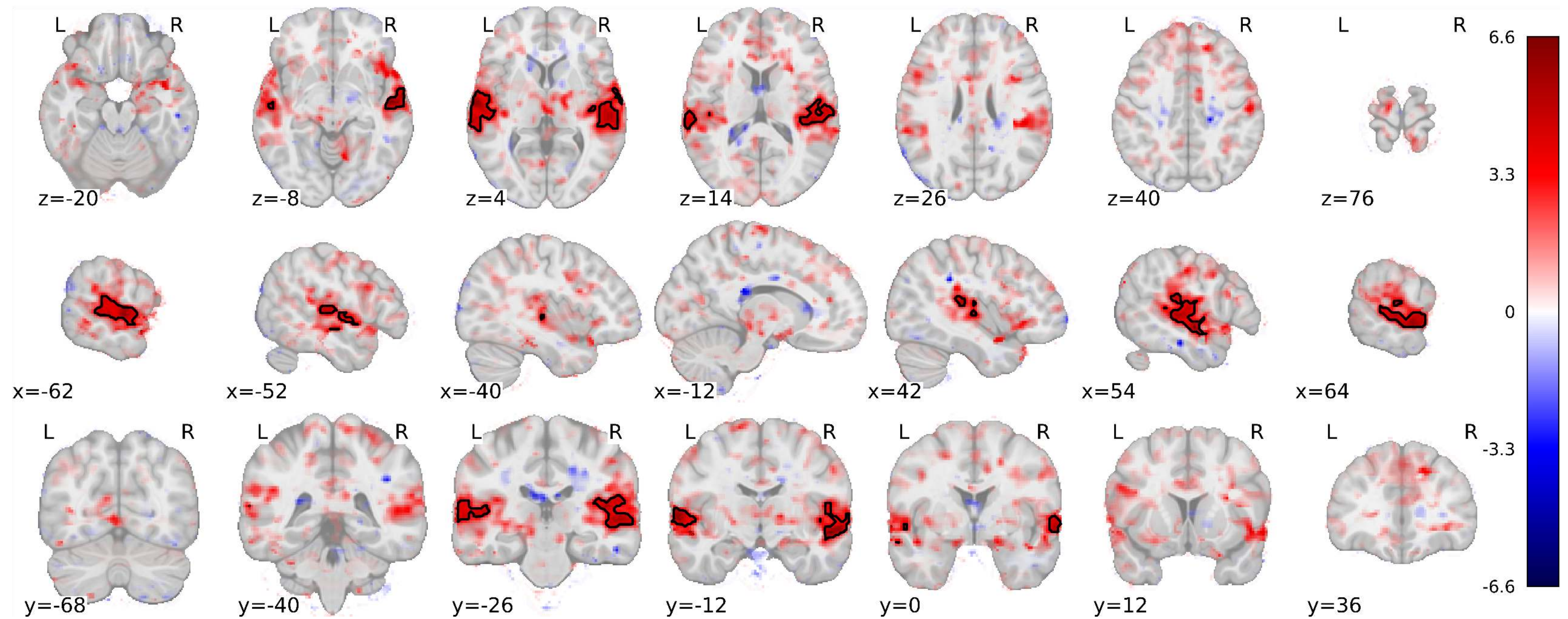
# The developing auditory system: Speech-in-quiet vs. baseline fMRI contrast



Sitek et al. (in prep.; SNL 2025)



# The developing auditory system: Speech-in-quiet vs. -6 dB SNR

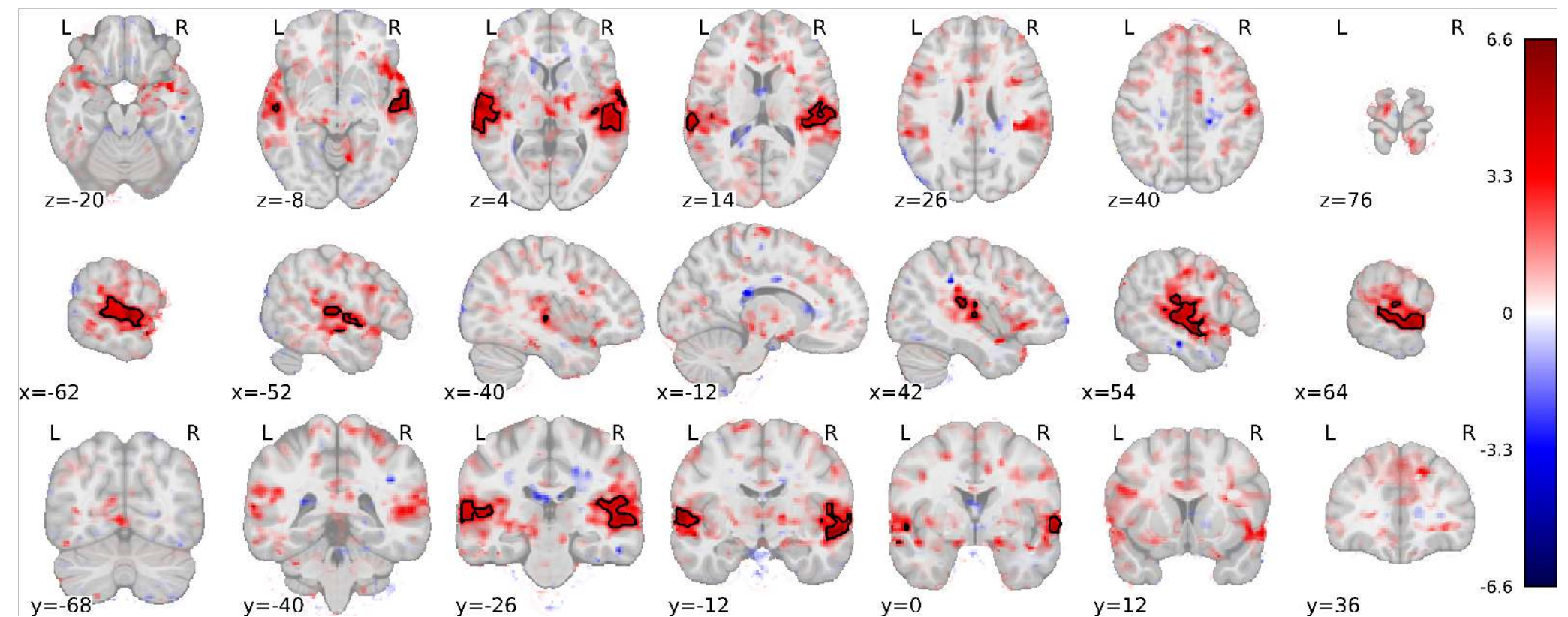


Sitek et al. (in prep.; SNL 2025)



# Systems for speech-in-noise processing are actively developing in adolescents

- Children activate broad canonical auditory and language regions more strongly in clear speech compared to noisy stimuli
- No evidence that they utilize compensatory motor and prefrontal regions seen in adults
- Investigating task-relevant *representations* in development
- Currently collecting data in **children who stutter**

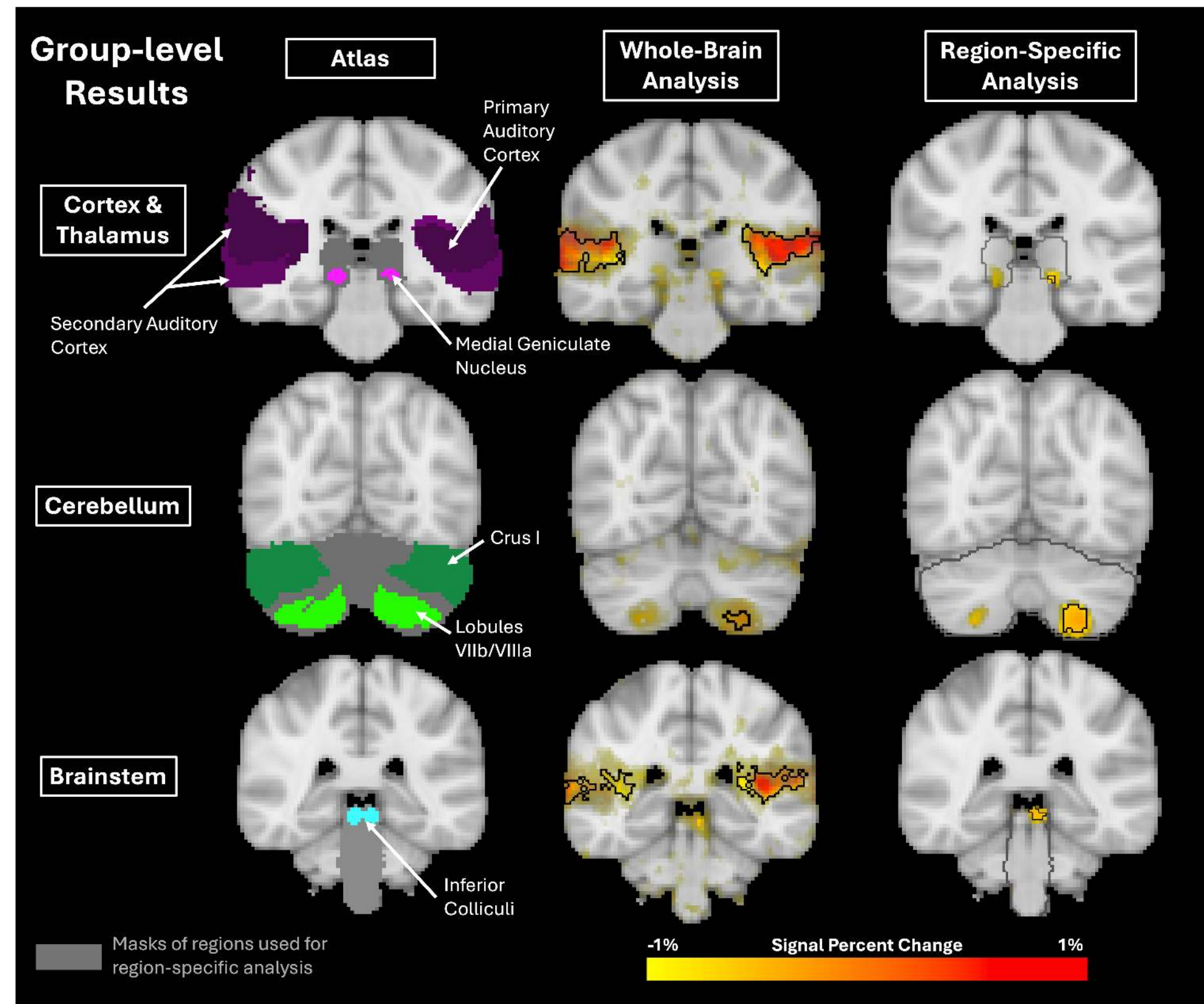


Sitek et al. (in prep.; SNL 2025);

Hampton-Wray R01DC019904



# Multi-echo fMRI improves group-level sensitivity at 3T

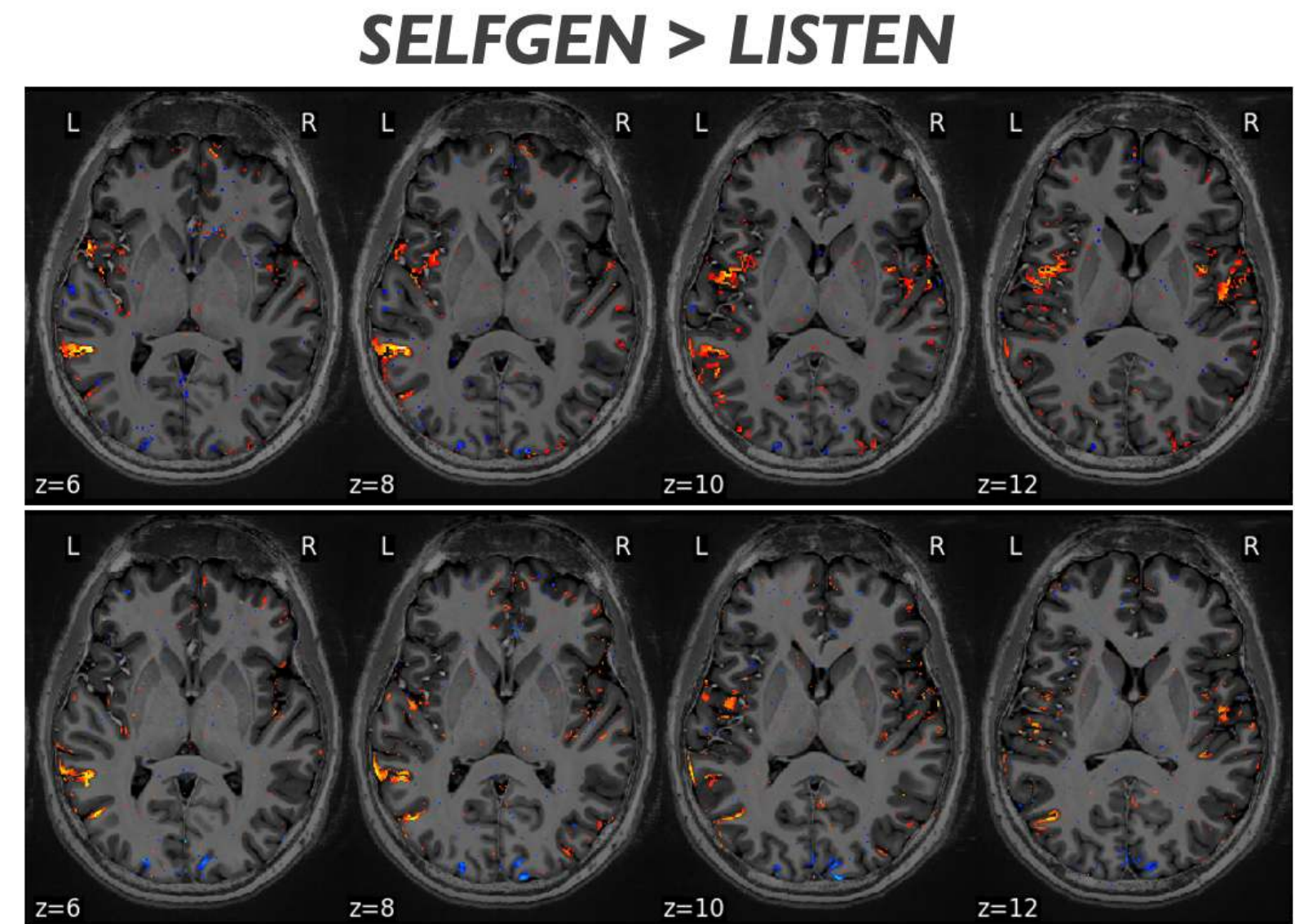
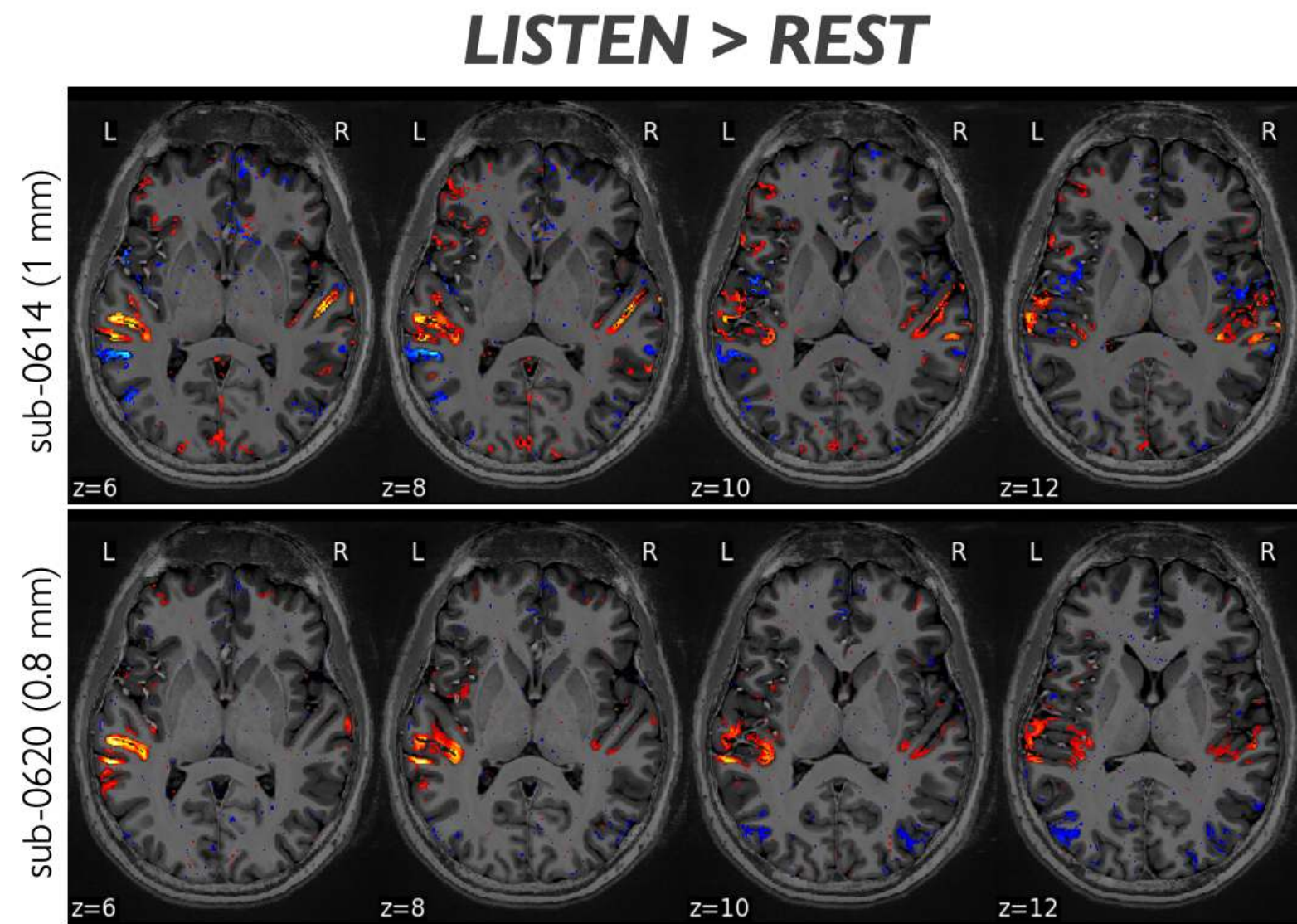


Medina et al. (submitted preprint); Medina, ..., Sitek (in prep.)





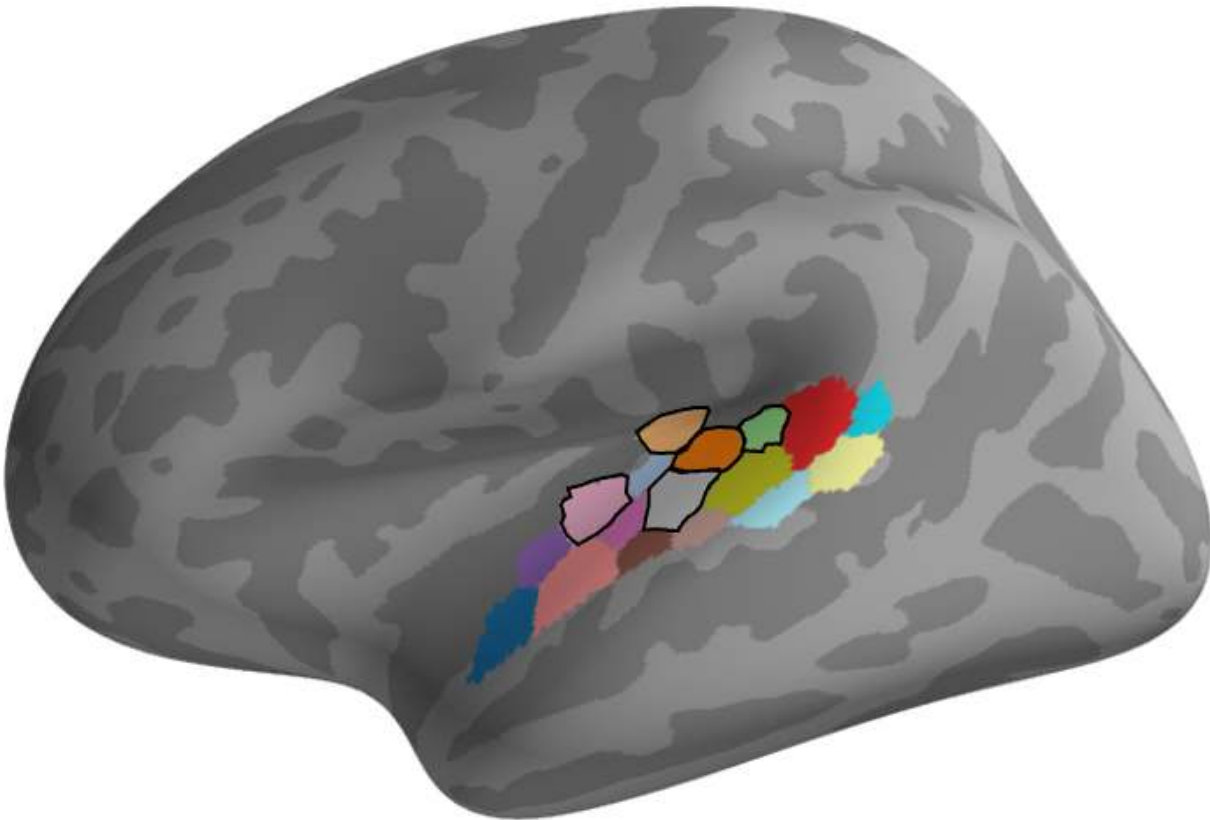
# Greater self-generated vs. passive fMRI responses in a single participant across replications



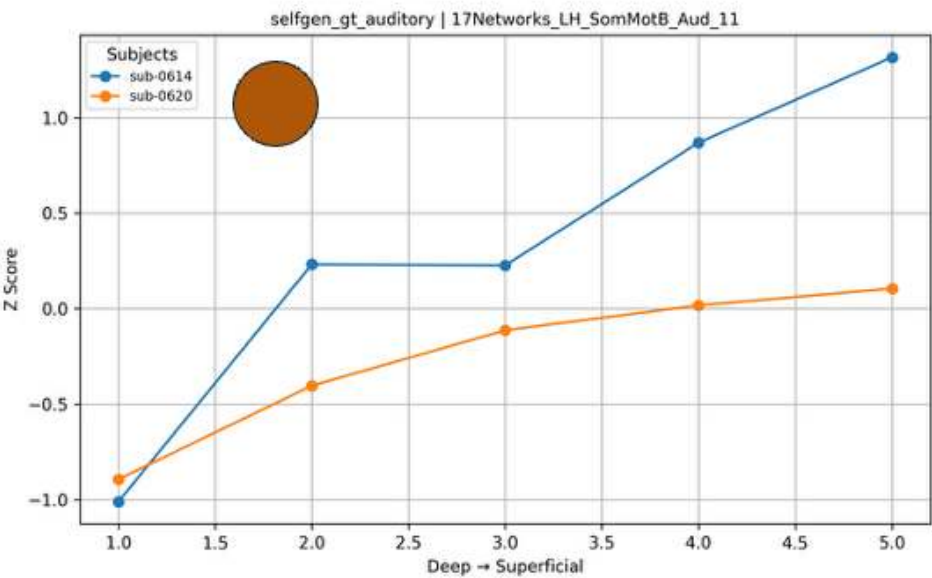
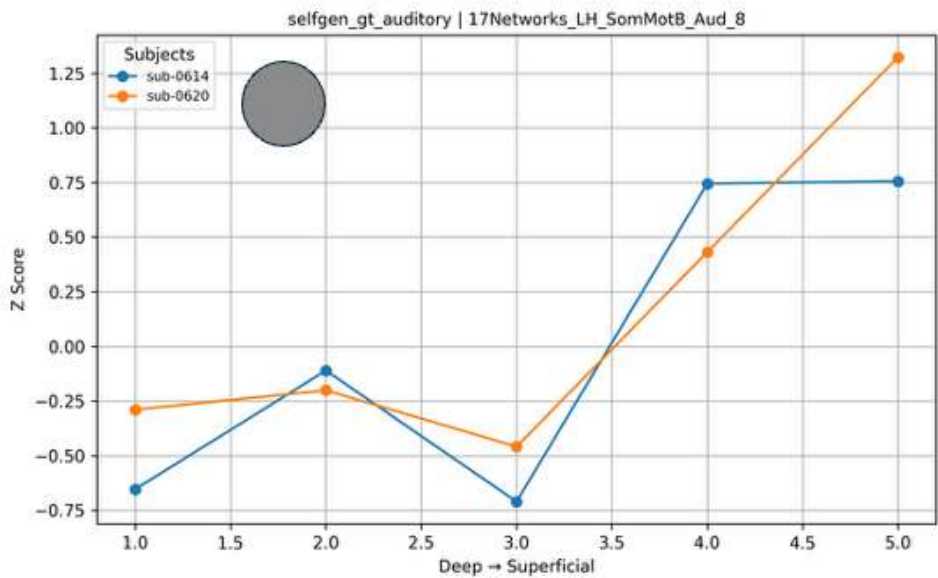
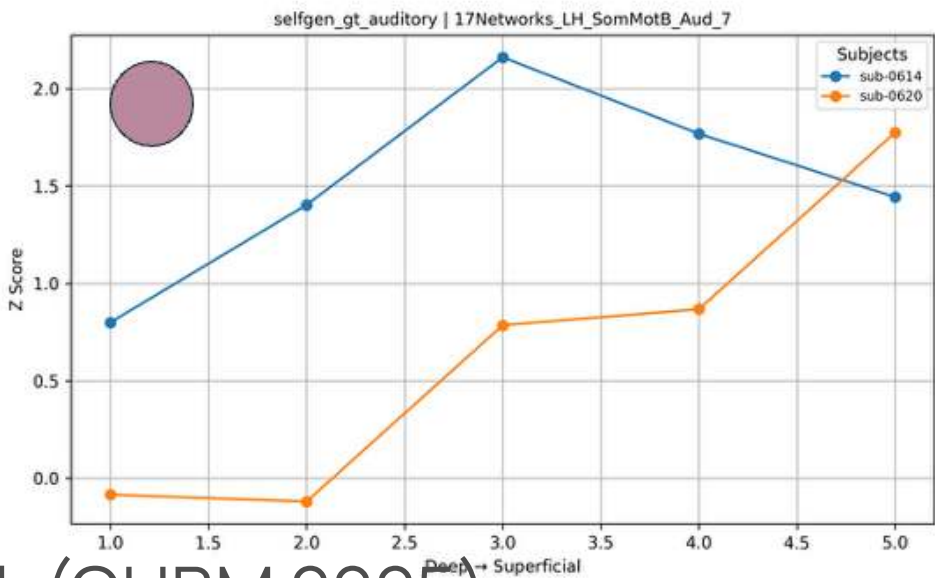
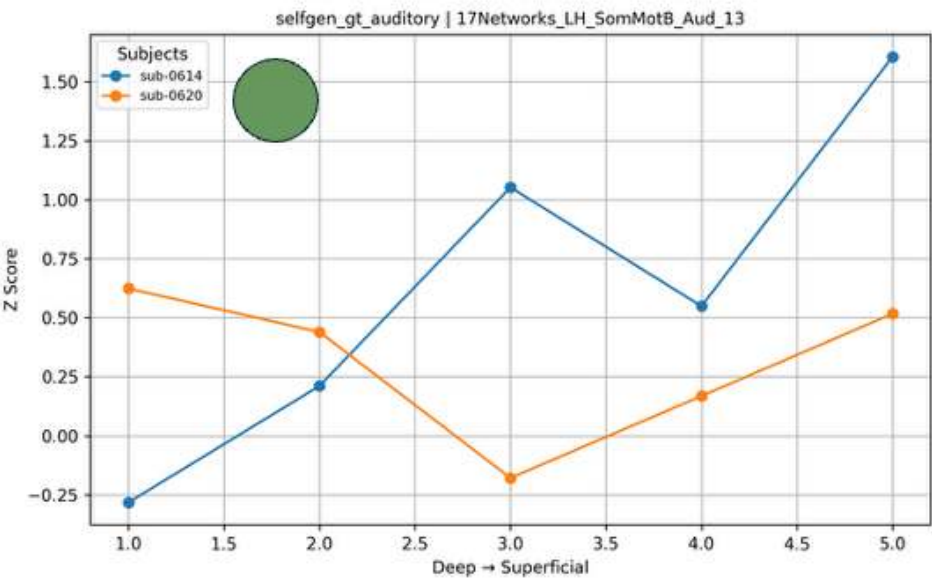
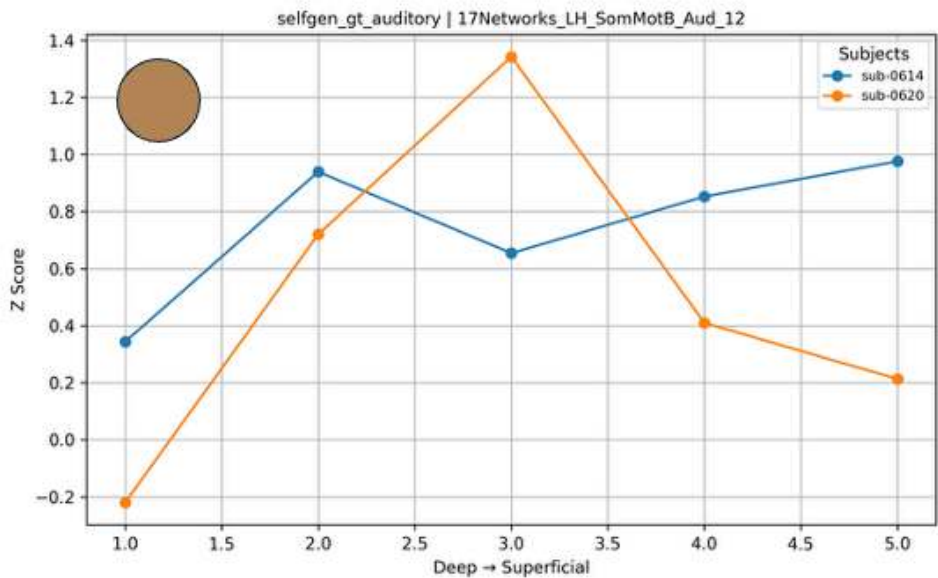
Sitek et al. (OHBM 2025)



# Stronger self-generated (vs. passive) sound responses in superficial auditory cortex



**SELFGEN > LISTEN**



Sitek et al. (OHBM 2025)